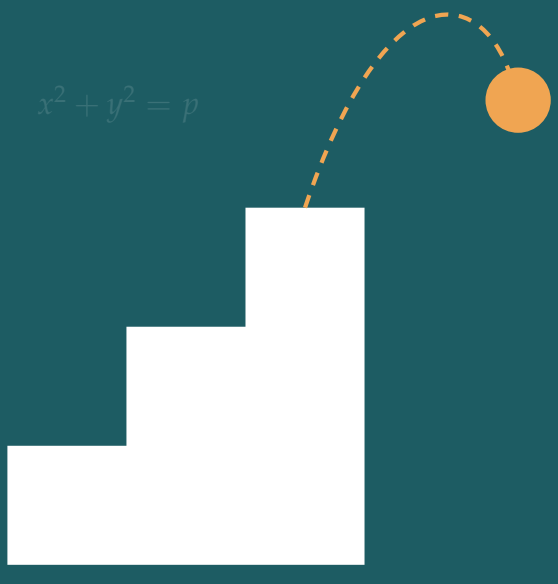
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2007 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2007 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2007 Worked Solutions

Worked solutions to all 42 questions of the 2007 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2007, Question 1

Pure • Counting Digit Arrangements

> **STEP I 2007, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/07-s1-q1

The key idea

Don't try to list balanced numbers. Instead, sort them by the common value s of the two digit sums. For each s, the first pair of digits and the last pair can be chosen independently, so the number of balanced numbers with that sum is a product of two counts. Everything then comes down to counting the ways of writing s as the sum of two digits, which is the same as counting lattice points on a diagonal of a square grid.

Solution

Write a 4-digit balanced number as $abcd$, where each digit lies in $\{0, 1, \ldots, k\}$, $a \geq 1$ (the leading digit can't be 0), and

$$a + b = c + d = s$$

for some s with $0 \leq s \leq 2k$. For each s let

- $N(s)$ be the number of ordered pairs (c, d) of digits with $c + d = s$;
- $M(s)$ be the number of ordered pairs (a, b) of digits with $a \geq 1$ and $a + b = s$.

Once s is fixed, any admissible front pair can go with any admissible back pair, so the number of balanced numbers with digit sum s is $M(s)N(s)$. Different values of s give different numbers, so

$$\text{total} = \sum_{s=0}^{2k} M(s)\, N(s).$$

Counting the pairs

$N(s)$. If $0 \leq s \leq k$, then c can be any of $0, 1, \ldots, s$, and $d = s - c$ is automatically a digit. So $N(s) = s + 1$. If $k < s \leq 2k$, then $d = s - c \leq k$ forces $c \geq s - k$, so c runs from $s - k$ to k, and $N(s) = 2k + 1 - s$.

$M(s)$. This is the same count, except that we lose the pair with $a = 0$, if it exists. That pair is $(0, s)$, and it is allowed only when $s \le k$. So

$$M(s) = \begin{cases} s, & 0 \le s \le k, \\ 2k + 1 - s, & k < s \le 2k. \end{cases}$$

In particular $M(0) = 0$: no balanced number has digit sum 0, which is right, since $a \ge 1$.

Ask yourself Why can't we just count all four-digit strings with $a + b = c + d$ and leave it there?

Because strings such as 0312 aren't 4-digit numbers. The condition $a \ge 1$ affects only the front pair, and only for small sums, which is exactly why $M(s)$ and $N(s)$ differ just when $s \le k$.

Part (i)

Here $k = 4$, and the sums run from 0 to 8:

s	0	1	2	3	4	5	6	7	8
$M(s)$	0	1	2	3	4	4	3	2	1
$N(s)$	1	2	3	4	5	4	3	2	1
$M(s)N(s)$	0	2	6	12	20	16	9	4	1

The bottom row adds up to $(2 + 6 + 12 + 20) + (16 + 9 + 4 + 1) = 40 + 30 = 70$. So seventy 4-digit balanced numbers can be made from the digits 0 to 4, as required.

Part (ii)

Now split the sum at $s = k$, using the formulae for M and N:

$$\text{total} = \sum_{s=1}^{k} s(s+1) \; + \sum_{s=k+1}^{2k} (2k + 1 - s)^2.$$

(The $s = 0$ term is zero.) In the second sum put $r = 2k + 1 - s$. As s runs from $k + 1$ to $2k$, r runs from k down to 1, so the second sum is $\sum_{r=1}^{k} r^2$. Hence

$$\begin{aligned} \text{total} &= \sum_{r=1}^{k} (r^2 + r) + \sum_{r=1}^{k} r^2 = 2\sum_{r=1}^{k} r^2 + \sum_{r=1}^{k} r \\ &= \tfrac{1}{3}k(k+1)(2k+1) + \tfrac{1}{2}k(k+1) \\ &= \tfrac{1}{6}k(k+1)\big(2(2k+1) + 3\big) \\ &= \tfrac{1}{6}k(k+1)(4k+5), \end{aligned}$$

using the given identity for $\sum r^2$ (the $r = 0$ term contributes nothing) and $\sum_{r=1}^{k} r = \frac{1}{2}k(k+1)$. This is the required result.

Checks

- Putting $k = 4$ into the formula gives $\frac{1}{6} \cdot 4 \cdot 5 \cdot 21 = 70$, agreeing with part (i).
- For $k = 1$ the formula gives 3. Directly, the 4-digit balanced numbers using only 0 and 1 are 1001, 1010 and 1111.
- A brute-force count over all strings $abcd$ with $a \geq 1$ matches the formula for every k from 1 to 15 (for example 615 when $k = 9$, the full set of decimal digits).

Common slips

- Forgetting the leading-digit condition, which gives $\sum N(s)^2 = \frac{1}{3}(k+1)(2k^2+4k+3)$ instead, which is 85 when $k = 4$.
- Getting the number of pairs wrong for $s > k$: it is $2k + 1 - s$, not $s + 1$, because both digits are capped at k.
- Counting the middle sum $s = k$ in both halves of the split, or in neither.

Beyond STEP

The count $N(s)$ is the coefficient of x^s in $(1 + x + \cdots + x^k)^2$, and the total number of balanced numbers is a coefficient of a product of such polynomials. This *generating function* method handles $2n$-digit balanced numbers too, where direct counting quickly gets out of hand. It is a central tool of undergraduate combinatorics. For more counting problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2007, Question 2

Pure • Inverse Tangent Identities

STEP I 2007, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q2

The key idea

An equation between inverse tangents becomes an equation between ordinary fractions as soon as we take the tangent of both sides and use the addition formula for $\tan(A+B)$. The price is that tan repeats every π, so $\tan(A+B)=1$ only tells us $A+B=\frac{1}{4}\pi+k\pi$; we need a word about the sizes of the angles to pin down k. After that, each part is a small piece of number theory: factorise, or use a divisibility argument.

Solution

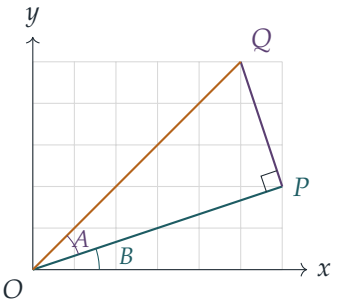
Part (i)

Since $\tan A=\frac{1}{2}$ and $\tan B=\frac{1}{3}$,

$$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{6}}=\frac{\frac{5}{6}}{\frac{5}{6}}=1.$$

Now A and B are acute, so $0<A+B<\pi$, and the only angle in $(0,\pi)$ with tangent 1 is $\frac{1}{4}\pi$. Hence $A+B=\frac{1}{4}\pi$, as required.

The picture below shows the same fact on squared paper. With $O=(0,0)$, $P=(6,2)$ and $Q=(5,5)$, the line OP makes angle $\arctan\frac{1}{3}$ with the x-axis; $PQ=(-1,3)$ is perpendicular to $OP=(6,2)$ and half as long, so angle POQ is $\arctan\frac{1}{2}$; and Q lies on the line $y=x$, at $\frac{1}{4}\pi$.



Part (i): the integers p and q

First we rule out $pq = 1$, where the addition formula would divide by zero. Then $p = q = 1$ or $p = q = -1$, giving left-hand sides $\frac{1}{2}\pi$ and $-\frac{1}{2}\pi$, neither of which is $\frac{1}{4}\pi$. So $pq \neq 1$, and taking tangents of both sides of the given equation,

$$\frac{\frac{1}{p} + \frac{1}{q}}{1 - \frac{1}{pq}} = 1.$$

Multiplying top and bottom by pq gives $\dfrac{p+q}{pq-1} = 1$, so

$$p + q = pq - 1 \quad \Longrightarrow \quad pq - p - q + 1 = 2 \quad \Longrightarrow \quad (p-1)(q-1) = 2,$$

which is the required result.

Since $p-1$ and $q-1$ are integers whose product is 2, the pair $(p-1, q-1)$ is one of

$$(1,2), \quad (2,1), \quad (-1,-2), \quad (-2,-1),$$

giving $(p,q) = (2,3)$, $(3,2)$, $(0,-1)$ or $(-1,0)$. The last two are excluded because p and q are non-zero. So $\{p,q\} = \{2,3\}$, and both orders really do satisfy the equation by the first half of part (i).

Ask yourself We only showed that a solution must satisfy $(p-1)(q-1) = 2$. Why do we need to check the candidates afterwards?

Taking tangents loses information: $(p-1)(q-1) = 2$ is equivalent to $\tan(\arctan\frac{1}{p} + \arctan\frac{1}{q}) = 1$, which also allows sums such as $-\frac{3}{4}\pi$. Here the candidate pairs either contain 0 (so $\frac{1}{p}$ is meaningless) or are 2 and 3, which we know work.

Part (ii)

Write $\theta = \arctan\frac{1}{r}$ and $\phi = \arctan\dfrac{s}{s+t}$. Taking tangents of the given equation,

$$\frac{\frac{1}{r} + \frac{s}{s+t}}{1 - \frac{s}{r(s+t)}} = 1.$$

The denominator is non-zero: since $r \geq 1$ and $t \geq 1$, we have $r(s+t) > s$. Multiplying top and bottom by $r(s+t)$,

$$\frac{(s+t)+rs}{r(s+t)-s} = 1 \quad \Longrightarrow \quad s+t+rs = rs+rt-s \quad \Longrightarrow \quad t(r-1) = 2s.$$

Ask yourself How does the condition on the highest common factor of s and t come in?

The equation $t(r-1) = 2s$ says that t divides $2s$. As $\text{hcf}(s,t) = 1$, t shares no prime factor with s, so every prime factor of t must come from the 2: t divides 2. Since t is a positive integer, the only possible values are

$$t = 1 \quad \text{or} \quad t = 2.$$

- If $t = 1$: $r - 1 = 2s$, so $r = 2s + 1$.
- If $t = 2$: $2(r-1) = 2s$, so $r = s + 1$. (Here s must be odd, since $\text{hcf}(s,2) = 1$.)

Both cases genuinely occur. Indeed $0 < \frac{1}{r} \leq 1$ and $0 < \frac{s}{s+t} < 1$, so θ and ϕ both lie in $(0, \frac{1}{4}\pi]$ and $0 < \theta + \phi < \frac{1}{2}\pi$. Reversing the algebra above, $t(r-1) = 2s$ gives $\tan(\theta+\phi) = 1$, and so $\theta + \phi = \frac{1}{4}\pi$ exactly. For example, $t = 1,\ s = 1$ gives $\arctan\frac{1}{3} + \arctan\frac{1}{2}$, and $t = 2,\ s = 1$ gives $\arctan\frac{1}{2} + \arctan\frac{1}{3}$: the identity of part (i) both ways round.

Checks

- On a calculator, $\arctan 0.5 + \arctan(1/3) = 0.4636 + 0.3218 = 0.7854 = \frac{1}{4}\pi$.
- A computer search over all non-zero p, q with $|p|, |q| \leq 200$ finds only $(2,3)$ and $(3,2)$.
- A search over $r < 400$ and $s, t < 150$ with $\text{hcf}(s,t) = 1$ finds only $t = 1,\ r = 2s+1$ and $t = 2,\ r = s+1$; for instance $\arctan\frac{1}{8} + \arctan\frac{7}{9} = \frac{1}{4}\pi$ ($s = 7,\ t = 2$) to machine precision.

Common slips

- Concluding $A + B = \frac{1}{4}\pi$ from $\tan(A+B) = 1$ without saying why $A + B$ cannot be $\frac{5}{4}\pi$.
- Forgetting the negative factorisations $(-1)(-2)$ and $(-2)(-1)$ of 2, or forgetting to discard them because they give a zero.
- Saying "t divides $2s$, so t divides 2" without quoting $\text{hcf}(s,t) = 1$: this is exactly where the hypothesis is needed.

Beyond STEP

The addition formula for tan is really multiplication of complex numbers: $\arctan \frac{1}{p}$ is the argument of $p + \mathrm{i}$, so part (i) is the statement that $(2 + \mathrm{i})(3 + \mathrm{i}) = 5 + 5\mathrm{i}$ has argument $\frac{1}{4}\pi$. Identities like Machin's $4 \arctan \frac{1}{5} - \arctan \frac{1}{239} = \frac{1}{4}\pi$, once used to compute π to hundreds of digits, come from factorising Gaussian integers in exactly this way. Needham's *Visual Complex Analysis* develops the geometry of complex multiplication beautifully, and Davenport's *The Higher Arithmetic* covers the divisibility arguments used in part (ii).

WORKED SOLUTION

STEP I 2007, Question 3

Pure • Trigonometric Integrals

STEP I 2007, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q3

The key idea

Neither $\cos^4\theta$ nor $\sin^4\theta$ is easy to integrate on its own, but their *difference* and their *sum* both are, once we use $\cos^2\theta + \sin^2\theta = 1$ to collapse them into functions of 2θ. Integrating the sum and the difference gives two simple linear equations for the two integrals we want. The same trick, one power higher, handles the sixth powers.

Solution

The identities

Factorising the difference of two squares,

$$\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta) = \cos 2\theta \cdot 1 = \cos 2\theta.$$

For the sum, we complete the square:

$$\begin{aligned}\cos^4\theta + \sin^4\theta &= (\cos^2\theta + \sin^2\theta)^2 - 2\sin^2\theta\cos^2\theta \\ &= 1 - \tfrac{1}{2}(2\sin\theta\cos\theta)^2 \\ &= 1 - \tfrac{1}{2}\sin^2 2\theta,\end{aligned}$$

using $\sin 2\theta = 2\sin\theta\cos\theta$. Both identities are proved.

The fourth powers

Write $C_4 = \int_0^{\pi/2}\cos^4\theta\,\mathrm{d}\theta$ and $S_4 = \int_0^{\pi/2}\sin^4\theta\,\mathrm{d}\theta$. Integrating the first identity,

$$C_4 - S_4 = \int_0^{\pi/2}\cos 2\theta\,\mathrm{d}\theta = \left[\tfrac{1}{2}\sin 2\theta\right]_0^{\pi/2} = 0.$$

For the second identity we need $\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta$. Using $\sin^2 2\theta = \frac{1}{2}(1 - \cos 4\theta)$,

$$\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta = \left[\tfrac{1}{2}\theta - \tfrac{1}{8}\sin 4\theta\right]_0^{\pi/2} = \frac{\pi}{4},$$

since $\sin 2\pi = 0$. Hence

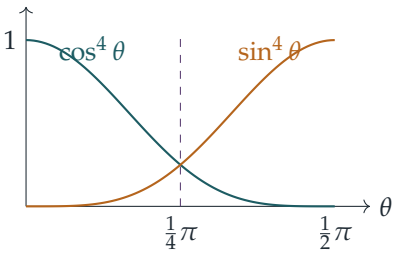
$$C_4 + S_4 = \int_0^{\pi/2} \left(1 - \tfrac{1}{2}\sin^2 2\theta\right) \mathrm{d}\theta = \frac{\pi}{2} - \frac{1}{2}\cdot\frac{\pi}{4} = \frac{3\pi}{8}.$$

Adding and subtracting the two equations,

$$\int_0^{\pi/2} \cos^4\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^4\theta \,\mathrm{d}\theta = \frac{3\pi}{16}.$$

Ask yourself Could we have seen that $C_4 = S_4$ without any calculation?

Yes. The substitution $\phi = \frac{1}{2}\pi - \theta$ turns $\cos\theta$ into $\sin\phi$ and swaps the limits 0 and $\frac{1}{2}\pi$, so $\int_0^{\pi/2} \cos^n\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^n\phi \,\mathrm{d}\phi$ for every n. The graphs of $\cos^4\theta$ and $\sin^4\theta$ are mirror images in the line $\theta = \frac{1}{4}\pi$, so they enclose equal areas:



So the first identity is really only needed for the sum; the difference integral being 0 is the symmetry in disguise.

The sixth powers

We repeat the method. With $c = \cos\theta$, $s = \sin\theta$ and $c^2 + s^2 = 1$:

$$\begin{aligned} c^6 - s^6 &= (c^2 - s^2)(c^4 + c^2s^2 + s^4) = \cos 2\theta\left(1 - \tfrac{1}{2}\sin^2 2\theta + \tfrac{1}{4}\sin^2 2\theta\right) \\ &= \cos 2\theta\left(1 - \tfrac{1}{4}\sin^2 2\theta\right), \end{aligned}$$

using the identities already proved and $c^2s^2 = \frac{1}{4}\sin^2 2\theta$. For the sum we use the cube of $c^2 + s^2$:

$$c^6 + s^6 = (c^2 + s^2)^3 - 3c^2s^2(c^2 + s^2) = 1 - \tfrac{3}{4}\sin^2 2\theta.$$

Write C_6 and S_6 for the two integrals. Since $\frac{\mathrm{d}}{\mathrm{d}\theta}\sin^3 2\theta = 6\sin^2 2\theta\cos 2\theta$,

$$C_6 - S_6 = \left[\tfrac{1}{2}\sin 2\theta - \tfrac{1}{24}\sin^3 2\theta\right]_0^{\pi/2} = 0,$$

as the symmetry argument also predicts. And, using $\int_0^{\pi/2} \sin^2 2\theta \,\mathrm{d}\theta = \frac{1}{4}\pi$ again,

$$C_6 + S_6 = \frac{\pi}{2} - \frac{3}{4}\cdot\frac{\pi}{4} = \frac{5\pi}{16}.$$

Therefore

$$\int_0^{\pi/2} \cos^6\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \sin^6\theta \,\mathrm{d}\theta = \frac{5\pi}{32}.$$

Checks

- The answers decrease as the power rises: $\frac{1}{2}\pi$ for $n = 0$, $\frac{1}{4}\pi$ for $n = 2$, then $\frac{3}{16}\pi$ and $\frac{5}{32}\pi$, as they must since $0 \le \cos\theta \le 1$ on the interval.
- Numerically, Simpson's rule gives $\int_0^{\pi/2} \cos^4\theta\,\mathrm{d}\theta \approx 0.589049$ and $\int_0^{\pi/2} \cos^6\theta\,\mathrm{d}\theta \approx 0.490874$, matching $3\pi/16$ and $5\pi/32$; the sine integrals agree with these to the same accuracy.
- Putting $\theta = \frac{1}{4}\pi$ in the sum identities gives $\frac{1}{4} + \frac{1}{4} = 1 - \frac{1}{2}$ and $\frac{1}{8} + \frac{1}{8} = 1 - \frac{3}{4}$, both true.

Common slips

- Writing $\cos^4\theta + \sin^4\theta = 1$ by analogy with $\cos^2\theta + \sin^2\theta = 1$: the cross term $2\sin^2\theta\cos^2\theta$ is lost.
- Forgetting the factor $\frac{1}{2}$ from the chain rule when integrating $\cos 2\theta$ or $\cos 4\theta$; here it doesn't matter for $\cos 2\theta$ (the integral is 0 anyway) but it does for $\sin^2 2\theta$.

Beyond STEP

These are cases of Wallis's integrals $I_n = \int_0^{\pi/2} \sin^n\theta\,\mathrm{d}\theta$, which satisfy the reduction formula $I_n = \frac{n-1}{n} I_{n-2}$ (integrate by parts). You can check that $I_4 = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$ and $I_6 = \frac{5}{6} I_4$ reproduce the answers above. Comparing I_{2n} and I_{2n+1} leads to Wallis's famous infinite product for π. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with reduction formulae, and Bartle and Sherbert, *Introduction to Real Analysis*, makes the limiting arguments rigorous.

WORKED SOLUTION

STEP I 2007, Question 4

Pure • Common Roots of Quadratics

STEP I 2007, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q4

The key idea

The first part is the classic factorisation of $x^3 + y^3 + z^3 - 3xyz$, and once you know $x + b + c$ is a factor the quadratic follows by matching coefficients. In the second part we have two quadratics sharing a root. The standard trick is to *eliminate*: combine the two equations so that one power of k cancels. Doing this in two different ways gives k and k^2 as ratios, and then $k^2 = (k)^2$ produces a relation between the coefficients alone. That relation turns out to be the first part with $x = a$.

Solution

The factorisation

Put $x = -(b + c)$ into $P(x) = x^3 - 3xbc + b^3 + c^3$:

$$\begin{aligned} P(-b-c) &= -(b+c)^3 + 3bc(b+c) + b^3 + c^3 \\ &= -(b^3 + 3b^2c + 3bc^2 + c^3) + 3b^2c + 3bc^2 + b^3 + c^3 = 0, \end{aligned}$$

so by the factor theorem $x + b + c$ is a factor. Matching coefficients (the x^2 term of P is zero, and the constant term is $b^3 + c^3$) gives

$$x^3 - 3xbc + b^3 + c^3 = (x + b + c)\,\mathrm{Q}(x), \qquad \mathrm{Q}(x) = x^2 + b^2 + c^2 - xb - bc - cx.$$

Expanding the right-hand side confirms it: the x^2 terms are $x^2(b + c) - x^2(b + c) = 0$, the x terms collect to $-3bcx$, and the constants are $b^3 + c^3$.

For the squares, each of x^2, b^2, c^2 appears twice in $2\mathrm{Q}(x)$ and each cross term appears as $-2xb$ and so on, so

$$2\mathrm{Q}(x) = (x - b)^2 + (b - c)^2 + (c - x)^2.$$

The common root

Since k is a root of both equations,

$$ak^2 + bk + c = 0, \tag{1}$$
$$bk^2 + ck + a = 0. \tag{2}$$

To remove k^2, take $b \times (1) - a \times (2)$:

$$(b^2 - ac)k + bc - a^2 = 0, \qquad \text{so} \qquad (ac - b^2)\,k = bc - a^2.$$

To remove the k terms instead, take $c \times (1) - b \times (2)$:

$$(ac - b^2)k^2 + c^2 - ab = 0, \qquad \text{so} \qquad (ac - b^2)\,k^2 = ab - c^2.$$

This is the similar expression involving k^2.

Now use $k^2 = k \cdot k$. Since $ac \neq b^2$ we may divide by $ac - b^2$, giving $k = \dfrac{bc - a^2}{ac - b^2}$, and so

$$(ac - b^2)\,\frac{(bc - a^2)^2}{(ac - b^2)^2} = ab - c^2 \quad \Longrightarrow \quad (ac - b^2)(ab - c^2) = (bc - a^2)^2,$$

after multiplying through by $ac - b^2$. This is the required result.

Expanding both sides,

$$a^2bc - ac^3 - ab^3 + b^2c^2 = b^2c^2 - 2a^2bc + a^4,$$
$$\text{so} \quad 0 = a^4 + ab^3 + ac^3 - 3a^2bc = a\left(a^3 - 3abc + b^3 + c^3\right).$$

Since $a \neq 0$, we get $a^3 - 3abc + b^3 + c^3 = 0$, as required.

Ask yourself Where have we seen $a^3 - 3abc + b^3 + c^3$ before?

It is the first part with $x = a$. So

$$(a + b + c)\,\mathrm{Q}(a) = 0, \qquad 2\mathrm{Q}(a) = (a - b)^2 + (b - c)^2 + (c - a)^2.$$

Hence either $a + b + c = 0$ or $\mathrm{Q}(a) = 0$.

Case $\mathrm{Q}(a) = 0$**.** Because a, b, c are real, the three squares are non-negative, and their sum is zero only if each is zero. So $a = b = c$, and the two equations are both $a(y^2 + y + 1) = 0$: they are identical.

Case $a + b + c = 0$**.** Substitute $c = -a - b$ into our formula for k:

$$bc - a^2 = -ab - b^2 - a^2, \qquad ac - b^2 = -a^2 - ab - b^2,$$

so $bc - a^2 = ac - b^2$, and since this is non-zero, $k = 1$. (Indeed $y = 1$ satisfies both equations, as each then reads $a + b + c = 0$.)

So either $k = 1$ or the two equations are identical, which completes the question.

Notice that in the identical case $a = b = c$ gives $ac = b^2$, which the hypotheses rule out. So under the stated conditions the common root is in fact always $k = 1$; the "identical" alternative is what happens if you drop the condition $ac \neq b^2$, and then the common roots are the complex cube roots of unity.

Checks

Take $a = 1$, $b = 2$, $c = -3$, so $a + b + c = 0$. The equations are $y^2 + 2y - 3 = (y - 1)(y + 3) = 0$ and $2y^2 - 3y + 1 = (2y - 1)(y - 1) = 0$, which share $k = 1$. Here $ac - b^2 = -7$, $bc - a^2 = -7$ and $ab - c^2 = -7$, so $(ac - b^2)k = bc - a^2$, $(ac - b^2)k^2 = ab - c^2$ and $(-7)(-7) = (-7)^2$ all hold, and $a^3 - 3abc + b^3 + c^3 = 1 + 18 + 8 - 27 = 0$. We also tested both identities of the first part, and the expansion $(ac - b^2)(ab - c^2) - (bc - a^2)^2 = -a(a^3 - 3abc + b^3 + c^3)$, in exact arithmetic on a thousand random triples.

Common slips

- Eliminating the constant term $(a \times (1) - c \times (2))$ gives $k\big((a^2 - bc)k + ab - c^2\big) = 0$, which still needs $k \neq 0$ to be useful. The combination $c \times (1) - b \times (2)$ avoids this.
- Dividing by $ac - b^2$ without saying that the question guarantees it is non-zero.
- Concluding $a = b = c$ from $\mathrm{Q}(a) = 0$ without using that a, b, c are real: the sum-of-squares argument fails for complex numbers.

Beyond STEP

The condition for two polynomials to have a common root is given in general by the *resultant*, a determinant built from their coefficients; the relation $(ac - b^2)(ab - c^2) = (bc - a^2)^2$ is exactly the resultant of these two quadratics set to zero. The expression $a^3 + b^3 + c^3 - 3abc$ is also the determinant of a circulant matrix, and it factorises completely as $(a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$ with ω a complex cube root of unity. For more on polynomials and their roots, try Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP I 2007, Question 5

Pure • Geometry of Polyhedra

STEP I 2007, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q5

The key idea

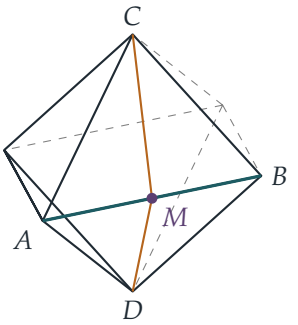
A regular octahedron sits very neatly in coordinates: its six vertices can be the points one unit along each of the positive and negative axes. With that model, both parts are just vectors. The angle between two faces is measured in a plane perpendicular to their common edge, so we use the two lines that run from the midpoint of the edge across each face to the opposite vertex. The face centres are simple averages of three vertices.

Solution

Every regular octahedron is a scaled copy of any other, and neither an angle nor a ratio of volumes changes under scaling. So we may work with the octahedron whose vertices are

$$(\pm 1, 0, 0), \qquad (0, \pm 1, 0), \qquad (0, 0, \pm 1).$$

Any two of these that are not opposite are $\sqrt{2}$ apart, which gives 12 edges, and the faces are the 8 triangles with one vertex on each axis, one for each choice of signs. Every face is equilateral with side $\sqrt{2}$, so this really is a regular octahedron.



Part (i)

Take two faces that share an edge; by symmetry every edge looks the same, so one will do. Let $A = (1,0,0)$, $B = (0,1,0)$, $C = (0,0,1)$ and $D = (0,0,-1)$, so the faces ABC and ABD meet along AB. Let $M = (\frac{1}{2}, \frac{1}{2}, 0)$ be the midpoint of AB.

Ask yourself How do we measure the angle between two planes that meet along a line?

We take a point on the common line and draw, in each plane, the line through it perpendicular to the common line; the angle between those two lines is the angle between the faces. Since each face is equilateral, the median MC of triangle ABC is perpendicular to AB, and so is the median MD of triangle ABD. So the angle we want is $\angle CMD$.

Now

$$\overrightarrow{MC} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\right), \qquad \overrightarrow{MD} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, -1\right),$$

and both are perpendicular to $\overrightarrow{AB} = (-1,1,0)$, as they should be. Each has length squared $\frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$, and

$$\overrightarrow{MC} \cdot \overrightarrow{MD} = \tfrac{1}{4} + \tfrac{1}{4} - 1 = -\tfrac{1}{2}.$$

Hence

$$\cos \angle CMD = \frac{-\frac{1}{2}}{\frac{3}{2}} = -\frac{1}{3},$$

and the angle between the faces is $\arccos\left(-\frac{1}{3}\right)$, about $109.5°$, as required. The cosine is negative, so the angle is obtuse: the octahedron is less sharp along its edges than a cube.

Without coordinates the same calculation is the cosine rule in triangle CMD. With edge length a, the medians are $MC = MD = \frac{\sqrt{3}}{2}a$ and C, D are opposite vertices: the four points A, C, $(-1,0,0)$, D form a square of side a, so $CD = \sqrt{2}\,a$. Then

$$2a^2 = \tfrac{3}{4}a^2 + \tfrac{3}{4}a^2 - 2 \cdot \tfrac{3}{4}a^2 \cos \angle CMD \quad \Longrightarrow \quad \cos \angle CMD = -\tfrac{1}{3}.$$

Part (ii)

The centre of a face is the average of its three vertices. A face has one vertex $(\pm 1, 0, 0)$, one $(0, \pm 1, 0)$ and one $(0, 0, \pm 1)$, so its centre is

$$\left(\pm\tfrac{1}{3}, \pm\tfrac{1}{3}, \pm\tfrac{1}{3}\right),$$

and the eight choices of signs give the eight centres. These are exactly the vertices of a cube with side $\frac{2}{3}$, so

$$V_{\text{cube}} = \left(\tfrac{2}{3}\right)^3 = \tfrac{8}{27}.$$

The octahedron is two square-based pyramids glued together along the square with vertices $(\pm1, 0, 0)$, $(0, \pm1, 0)$. That square has diagonals of length 2, so its area is $\frac{1}{2} \cdot 2 \cdot 2 = 2$, and each pyramid has height 1. Therefore

$$V_{\text{oct}} = 2 \times \tfrac{1}{3} \times 2 \times 1 = \tfrac{4}{3}.$$

The ratio is

$$\frac{V_{\text{oct}}}{V_{\text{cube}}} = \frac{4/3}{8/27} = \frac{9}{2},$$

so the octahedron's volume is to the cube's as $9 : 2$.

Checks

- $\arccos(-\frac{1}{3}) \approx 109.47°$. Repeating the midpoint calculation numerically for all 12 edges of the coordinate model gives $\cos = -\frac{1}{3}$ every time.
- The octahedron is $|x| + |y| + |z| \leq 1$. A Monte Carlo estimate of its volume from 10^6 random points in $[-1, 1]^3$ gave 1.330, against the exact $\frac{4}{3}$.
- The cube is inside the octahedron (each centre satisfies $|x| + |y| + |z| = 1$), so the ratio must be greater than 1. And $\frac{9}{2}$ is.

Common slips

- Taking the angle between the outward normals, $(1, 1, 1)$ and $(1, 1, -1)$, which gives $\arccos \frac{1}{3}$. The angle between the faces themselves is the supplement of that.
- Measuring the angle along lines that are not perpendicular to the common edge, such as two edges CA and DA. That angle is $90°$ here, and it isn't the angle between the faces.
- Thinking the cube's side is the distance between the centres of two opposite faces. That distance, $\frac{2}{\sqrt{3}}$, is a space diagonal of the cube.

Beyond STEP

The cube and the octahedron are *dual* polyhedra: joining the face centres of either gives the other, with faces and vertices swapping roles. Duality is the first sign that the symmetry groups of the two solids are the same group, of order 48, which you'll meet in a first course on groups. For the symmetry groups of the Platonic solids, try Armstrong, *Groups and Symmetry*.

WORKED SOLUTION

STEP I 2007, Question 6

Pure • Diophantine Equations

STEP I 2007, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q6

The key idea

Both parts begin with a difference of powers, and $x-y$ is a factor of every one of them. Cancelling that factor (allowed because $d \neq 0$) leaves a symmetric expression in x and y. That, together with $x-y=d$, pins down x and y. To get integer solutions we then choose d so that the formulas come out as whole numbers. In part (ii) that means making a square root rational.

Solution

Part (i)

Factorising the left-hand side,

$$(x-y)(x+y) = (x-y)^3.$$

Since $x-y=d\neq 0$, we may divide by $x-y$ to get $x+y=(x-y)^2=d^2$. Adding and subtracting this and $x-y=d$ gives

$$x=\frac{d^2+d}{2}=\frac{d(d+1)}{2}, \qquad y=\frac{d^2-d}{2}=\frac{d(d-1)}{2}.$$

For the integer pair, put $x=\sqrt{m}$ and $y=\sqrt{n}$. Then $x^2-y^2=m-n$, and the required condition is exactly $x^2-y^2=(x-y)^3$. With $d=\sqrt{m}-\sqrt{n}$, which is nonzero because $m>n$, we get

$$m=\left(\frac{d(d+1)}{2}\right)^2, \qquad n=\left(\frac{d(d-1)}{2}\right)^2.$$

If d is a positive integer then $d(d\pm 1)$ is a product of consecutive integers, so it is even. Then $\sqrt{m}$ and $\sqrt{n}$ are integers, and so are m and n. Also $m>n$ because $d>0$. We need $n>100$, that is $\frac{1}{2}d(d-1)>10$. The value $d=5$ gives $\frac{1}{2}d(d-1)=10$, which is not enough, so we take $d=6$:

$$\sqrt{m}=21, \quad \sqrt{n}=15, \qquad m=441, \quad n=225.$$

Check: $m - n = 216 = 6^3 = (21 - 15)^3$. (Other answers, such as $d = 7$ with $m = 784$, $n = 441$, are equally valid.)

Part (ii)

Factorise again: $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Dividing by $x - y = d \neq 0$ gives

$$x^2 + xy + y^2 = d^3.$$

We also have $(x - y)^2 = d^2$, that is, $x^2 - 2xy + y^2 = d^2$. Subtracting this from the previous equation gives $3xy = d^3 - d^2$, as required.

Now substitute $y = x - d$:

$$3x(x - d) = d^3 - d^2 \quad \Longrightarrow \quad 3x^2 - 3dx - (d^3 - d^2) = 0.$$

By the quadratic formula,

$$x = \frac{3d \pm \sqrt{9d^2 + 12(d^3 - d^2)}}{6} = \frac{3d \pm \sqrt{3d^2(4d - 1)}}{6}.$$

Now $\sqrt{3d^2(4d-1)} = |d|\sqrt{3(4d-1)}$, and the $\pm$ already covers both signs, so we may replace $|d|$ by d. Multiplying by 2,

$$2x = d \pm \frac{d\sqrt{3(4d-1)}}{3} = d \pm d\sqrt{\frac{4d-1}{3}},$$

which is the required result.

Ask yourself How do we choose d so that x and y are integers?

If $m = x$ and $n = y$ are integers, then $d = m - n$ is a nonzero integer. For x to be rational, $\sqrt{(4d-1)/3}$ must be rational. Write it as k. Then $k^2 = (4d-1)/3$ is an integer, and the square root of an integer is rational only if the integer is a perfect square, so k is a whole number. Hence we need

$$4d - 1 = 3k^2 \quad \text{for some integer } k \geq 0.$$

The left-hand side is odd, so k is odd. Then $2x = d(1 \pm k)$ is even, so x is an integer, and so is $y = x - d$.

- $k = 1$ gives $d = 1$, so $x = 1$, $y = 0$ or $x = 0$, $y = -1$. Neither pair consists of two positive integers.
- $k = 3$ gives $d = 7$, so $2x = 7 \pm 21$. The plus sign gives $x = 14$, $y = 7$.

So $m = 14$, $n = 7$ works:

$$m^3 - n^3 = 2744 - 343 = 2401 = 7^4 = (m - n)^4.$$

The order matters: with $m = 7$, $n = 14$ the left-hand side is -2401 but the right-hand side is $+2401$.

Checks

- Part (i): $21^2 - 15^2 = 441 - 225 = 216$ and $(21-15)^3 = 216$.
- Part (ii): with $d = 7$, $3xy = 3 \cdot 14 \cdot 7 = 294$ and $d^3 - d^2 = 343 - 49 = 294$.
- A computer search over $1 \le m, n < 200$ finds exactly $(14, 7)$, $(57, 38)$ and $(148, 111)$. These come from $k = 3, 5, 7$ (so $d = 7, 19, 37$), matching the formula. The search also confirms that $(441, 225)$ is the smallest solution in part (i) with $n > 100$.

Common slips

- Forgetting that the cancellation needs $x - y \neq 0$. This is exactly why the question assumes $d \neq 0$.
- In part (i), taking $d = 5$ gives $n = 100$, which fails the strict inequality $n > 100$.
- Writing $\sqrt{d^2} = d$ without comment. It equals $|d|$, and the $\pm$ is what lets us drop the modulus.

Beyond STEP

Part (ii) solves the Diophantine equation $m^3 - n^3 = (m-n)^4$ completely: with $d = (3k^2+1)/4$ for odd $k \ge 3$, the solutions are $m = \frac{1}{2}d(k+1)$ and $n = \frac{1}{2}d(k-1)$. Requiring a square root to be rational is the standard way to turn such an equation into the much simpler condition $4d - 1 = 3k^2$. Equations like this, and the arithmetic of quadratic forms behind them, are treated in Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 2007, Question 7

Pure • Distance Between Lines

STEP I 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q7

The key idea

Write down a general point on each line, subtract, and square the length: D^2 is then a quadratic in the two parameters. The trick is to write that quadratic as a sum of squares, because a sum of squares is smallest when each square is as small as it can be. In part (ii) the same method works, and the way the squares fall apart for different values of k tells us the geometry.

Solution

Part (i)

A general point of L_1 is $P = (1+2\lambda,\ 2\lambda,\ 2-3\lambda)$ and a general point of L_2 is $Q = (4+\mu,\ -2+2\mu,\ 9-2\mu)$. So

$$\overrightarrow{PQ} = \begin{pmatrix} 3+\mu-2\lambda \\ -2+2\mu-2\lambda \\ 7-2\mu+3\lambda \end{pmatrix}$$

and

$$D^2 = (3+\mu-2\lambda)^2 + (-2+2\mu-2\lambda)^2 + (7-2\mu+3\lambda)^2.$$

Collecting terms coefficient by coefficient:

$$\mu^2: \quad 1+4+4=9, \qquad \lambda^2: \quad 4+4+9=17, \qquad \lambda\mu: \quad -4-8-12=-24,$$
$$\mu: \quad 6-8-28=-30, \qquad \lambda: \quad -12+8+42=38, \qquad \text{const}: \quad 9+4+49=62,$$

so $D^2 = 9\mu^2 + 17\lambda^2 - 24\lambda\mu - 30\mu + 38\lambda + 62$.

Now expand the proposed form:

$$\begin{aligned}(3\mu-4\lambda-5)^2 + (\lambda-1)^2 + 36 &= \left(9\mu^2+16\lambda^2+25-24\lambda\mu-30\mu+40\lambda\right) \\ &\quad + \left(\lambda^2-2\lambda+1\right) + 36 \\ &= 9\mu^2+17\lambda^2-24\lambda\mu-30\mu+38\lambda+62.\end{aligned}$$

The two agree, so

$$D^2 = (3\mu - 4\lambda - 5)^2 + (\lambda - 1)^2 + 36,$$

as required.

Ask yourself Why can we make both squares zero at once?

Each square is at least 0, so $D^2 \geq 36$ for every choice of λ and μ. Equality needs $\lambda = 1$ and $3\mu = 4\lambda + 5 = 9$, that is $\mu = 3$. These two conditions are compatible (the first fixes λ, and then the second fixes μ), so $D^2 = 36$ really is attained. Hence the minimum distance between the lines is

$$D_{\min} = 6,$$

attained at

$$\lambda = 1 : \ P = (3,\ 2,\ -1) \quad \text{on } L_1, \qquad \mu = 3 : \ Q = (7,\ 4,\ 3) \quad \text{on } L_2.$$

As a check, $\overrightarrow{PQ} = (4, 2, 4)$ has length $\sqrt{16 + 4 + 16} = 6$, and it is perpendicular to both direction vectors: $(4, 2, 4) \cdot (2, 2, -3) = 8 + 4 - 12 = 0$ and $(4, 2, 4) \cdot (1, 2, -2) = 4 + 4 - 8 = 0$. That is exactly what we expect of the shortest segment joining two lines.

Part (ii)

A general point of L_3 is $P = (2,\ 3 + \alpha,\ 5)$ and a general point of L_4 is $Q = \big(3 + 4k\beta,\ 3 + (1 - k)\beta,\ -2 - 3k\beta\big)$. So

$$\overrightarrow{PQ} = \begin{pmatrix} 1 + 4k\beta \\ (1 - k)\beta - \alpha \\ -7 - 3k\beta \end{pmatrix}$$

and

$$\begin{aligned} D^2 &= (1 + 4k\beta)^2 + \big((1 - k)\beta - \alpha\big)^2 + (7 + 3k\beta)^2 \\ &= \big((1 - k)\beta - \alpha\big)^2 + 25k^2\beta^2 + 50k\beta + 50 \\ &= \big((1 - k)\beta - \alpha\big)^2 + 25(k\beta + 1)^2 + 25, \end{aligned}$$

since $(1 + 4k\beta)^2 + (7 + 3k\beta)^2 = 1 + 8k\beta + 16k^2\beta^2 + 49 + 42k\beta + 9k^2\beta^2$.

Whatever β is, we can choose $\alpha = (1 - k)\beta$ to make the first square zero, because α appears nowhere else. So the question is how small $25(k\beta + 1)^2$ can be made, and this is where the two cases separate.

Ask yourself What changes when $k = 0$?

If $k \neq 0$ we can take $\beta = -1/k$, which makes $k\beta + 1 = 0$. If $k = 0$ the term $25(k\beta + 1)^2$ is stuck at 25 whatever β is.

Case $k \neq 0$**.** Every square is non-negative, so $D^2 \geq 25$, with equality at $\beta = -1/k$, $\alpha = (1 - k)\beta = (k - 1)/k$. The minimum distance is

$$D_{\min} = 5,$$

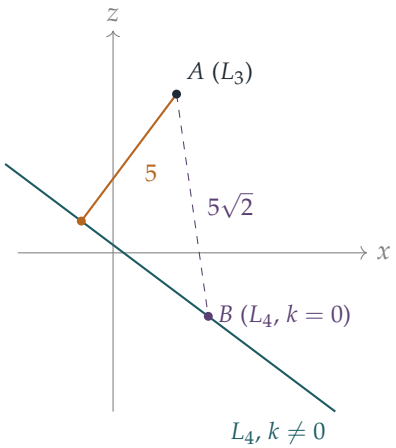
whatever the (non-zero) value of k. The closest points are $P = \left(2,\ 3+\frac{k-1}{k},\ 5\right)$ and $Q = \left(-1,\ 3+\frac{k-1}{k},\ 1\right)$, with $\overrightarrow{PQ} = (-3, 0, -4)$ of length 5. Since $D_{\min} > 0$ the lines never meet, and their directions $(0,1,0)$ and $(4k, 1-k, -3k)$ are not parallel when $k \neq 0$ (the first and third components of the second are not both zero). So the lines are *skew*, and 5 is the length of their common perpendicular, which is unique.

Case $k = 0$. Now L_4 has direction $(0,1,0)$, the same as L_3: the lines are *parallel*. Here $D^2 = (\beta - \alpha)^2 + 50$, minimised whenever $\alpha = \beta$, so

$$D_{\min} = \sqrt{50} = 5\sqrt{2},$$

and it is attained by infinitely many pairs of points: any point of one line together with the point of the other directly opposite it (same y-coordinate). This is just the constant distance between two parallel lines.

The geometry. Both lines are easiest to see by looking straight down the y-axis, that is, projecting onto the xz-plane. Since L_3 runs parallel to the y-axis, it projects to the single point $A = (2,5)$, and distances perpendicular to L_3 are unchanged by the projection. For $k \neq 0$, L_4 projects to the line through $(3,-2)$ with direction $(4,-3)$ (the $1-k$ component disappears), and the distance from A to that line is $\left|1\cdot(-3) - (-7)\cdot 4\right|/5 = 5$, independent of k. For $k = 0$, L_4 projects to the single point $B = (3,-2)$, and the distance is $AB = \sqrt{1+49} = 5\sqrt{2}$.



So the answer is 5 when $k \neq 0$ (skew lines, one shortest segment, which is perpendicular to both) and $5\sqrt{2}$ when $k = 0$ (parallel lines, the distance is the same all along).

Checks

In part (i), substituting $\lambda = 1$, $\mu = 3$ gives points 6 apart, and $\overrightarrow{PQ}$ is perpendicular to both lines. A numerical search over a grid of (λ, μ) found nothing below $D^2 = 36$. In part (ii), minimising D^2 numerically for $k = 0.001,\ 0.3,\ 1,\ -2,\ 5$ gave 25 each time, and 50 for $k = 0$. Notice that $D_{\min}$ jumps from 5 to $5\sqrt{2}$ at $k = 0$. For tiny k the closest points are a long way out, near $\beta = -1/k$, so the limit doesn't have to be continuous.

Common slips

- Forgetting to check that both squares can vanish *simultaneously*. Here they can, but that has to be said, not assumed.
- Dividing by k when $k = 0$: the choice $\beta = -1/k$ is only available for $k \neq 0$, which is exactly why the two cases arise.
- Calling the $k \neq 0$ lines "intersecting". They are not parallel, but $D_{\min} = 5 > 0$, so they are skew.

Beyond STEP

Completing the square in several variables is the start of the theory of quadratic forms: writing D^2 as a sum of squares is diagonalising a symmetric matrix, and the minimum is a least-squares problem. The perpendicularity we checked at the end is the "normal equations" of least squares, and the general formula $|(\mathbf{b}-\mathbf{a})\cdot(\mathbf{u}\times\mathbf{v})|/|\mathbf{u}\times\mathbf{v}|$ for the distance between skew lines comes from the same geometry. Strang, *Introduction to Linear Algebra*, covers projections and least squares beautifully.

WORKED SOLUTION

STEP I 2007, Question 8

Pure • Tangency of Cubic Curves

STEP I 2007, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q8

The key idea

Two curves touch at a point exactly when the equation for their intersections has a repeated root there. So look at the difference between the two cubics and expect a factor $(x-2)^2$. The coefficients $1, -6, 12, -8$ in the a-terms are the binomial coefficients of $(x-2)^3$, and spotting this makes both the factorisation and the sketches much easier.

Solution

Touching at $(2,8)$ and the other intersection

Grouping the terms that contain a,

$$ax^3 - 6ax^2 + 12ax - 8a = a(x-2)^3, \qquad \text{so} \qquad y = a(x-2)^3 + 12x - 16. \qquad (*)$$

The intersections with $y = x^3$ are the roots of

$$a(x-2)^3 + 12x - 16 - x^3 = 0.$$

Now $x^3 - 12x + 16$ vanishes at $x = 2$, and dividing out gives $x^3 - 12x + 16 = (x-2)^2(x+4)$. Hence

$$\begin{aligned} a(x-2)^3 + 12x - 16 - x^3 &= a(x-2)^3 - (x-2)^2(x+4) \\ &= (x-2)^2\big[(a-1)x - 2a - 4\big]. \end{aligned}$$

At $x = 2$ both curves pass through $(2,8)$, and the factor $(x-2)^2$ shows that $x = 2$ is a repeated root of the intersection equation. Equivalently, the gradients agree: from $(*)$, $y' = 3a(x-2)^2 + 12$, which is 12 at $x = 2$, and the gradient of $y = x^3$ there is $3 \cdot 2^2 = 12$. So the curves touch at $(2,8)$, for every value of a.

Any other intersection comes from $(a-1)x = 2a + 4$.

- If $a \neq 1$, then $x = \dfrac{2a+4}{a-1}$, which is never 2 (that would need $2a + 4 = 2a - 2$). So

there is exactly one other point,

$$\left(\frac{2a+4}{a-1}, \left(\frac{2a+4}{a-1}\right)^3\right).$$

- If $a = 1$, the bracket is the constant -6, so there is no other point of intersection.

Ask yourself Which curve is on top, and where?

The factorisation answers this for free. Since $(x-2)^2 \geq 0$, the sign of $(*)$ minus x^3 is the sign of the linear factor $(a-1)x - 2a - 4$. That tells us on which side of the other intersection the curve $(*)$ lies above $y = x^3$, which is most of what a sketch needs.

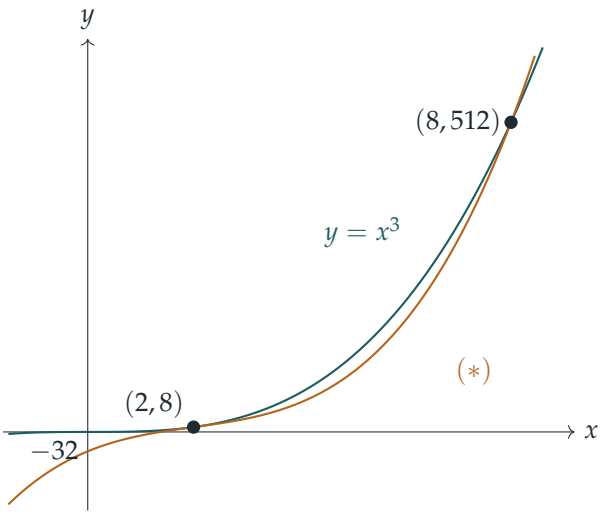
In all three sketches, $(*)$ has a point of inflection at $(2,8)$ with gradient 12, because $y'' = 6a(x-2)$ changes sign there when $a \neq 0$. So $(*)$ and $y = x^3$ share the tangent $y = 12x - 16$ at $(2,8)$; $y = x^3$ is concave up there while $(*)$ is not.

Part (i): $a = 2$

Here $y = 2(x-2)^3 + 12x - 16$ and

$$y - x^3 = (x-2)^2(x-8).$$

The other intersection is $(8, 512)$. The curve $(*)$ lies below $y = x^3$ for $x < 8$ (touching at $x = 2$) and above it for $x > 8$. Its gradient $6(x-2)^2 + 12$ is always positive, so it is increasing with no turning points. It meets the y-axis at $(0, -32)$ and crosses the x-axis once, near $x \approx 1.37$.



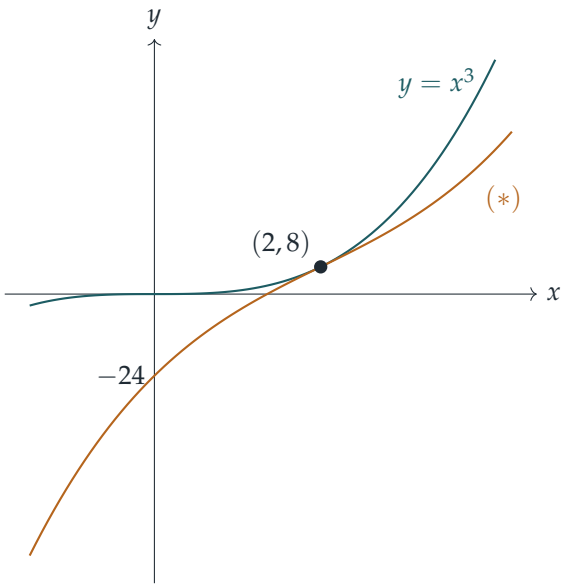
(The vertical scale is heavily compressed, so the touching at $(2,8)$ looks flat; the two curves share the tangent there.)

Part (ii): $a = 1$

Here $y = (x-2)^3 + 12x - 16 = x^3 - 6x^2 + 24x - 24$, and

$$y - x^3 = -6(x-2)^2 \leq 0.$$

So $(*)$ lies below $y = x^3$ everywhere except at $(2, 8)$, where it touches; there is no other intersection. The gap $6(x-2)^2$ grows as we move away from $x = 2$ in either direction. The gradient $3(x-2)^2 + 12$ is always positive, so $(*)$ is increasing. It meets the y-axis at $(0, -24)$ and crosses the x-axis once, near $x \approx 1.36$.



Part (iii): $a = -2$

Now the other intersection has $x = \dfrac{-4+4}{-3} = 0$, so it is the origin $(0, 0)$, and

$$y - x^3 = (x-2)^2(-3x) = -3x(x-2)^2.$$

So $(*)$ is above $y = x^3$ for $x < 0$ and below it for $x > 0$, touching at $(2, 8)$.

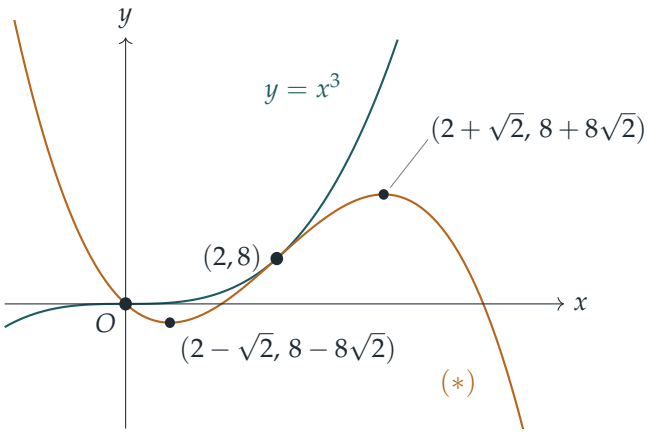
Expanding, $y = -2(x-2)^3 + 12x - 16 = -2x^3 + 12x^2 - 12x = -2x(x^2 - 6x + 6)$, which crosses the x-axis at $x = 0$ and $x = 3 \pm \sqrt{3}$ (about 1.27 and 4.73). The gradient is

$$y' = -6(x-2)^2 + 12,$$

which is zero when $(x-2)^2 = 2$, that is, $x = 2 \pm \sqrt{2}$. Substituting,

$$x = 2 - \sqrt{2}: \quad y = 8 - 8\sqrt{2} \approx -3.31 \quad \text{(minimum)},$$
$$x = 2 + \sqrt{2}: \quad y = 8 + 8\sqrt{2} \approx 19.3 \quad \text{(maximum)}.$$

Since the leading coefficient is negative, $y \to +\infty$ as $x \to -\infty$ and $y \to -\infty$ as $x \to +\infty$. At the origin $(*)$ has gradient -12, while $y = x^3$ has gradient 0, so the curves genuinely cross there.



Checks

- At $x = 2$, $(*)$ gives $a \cdot 0 + 24 - 16 = 8$ for every a, and $y' = 12$, matching $y = x^3$.
- With $a = 2$ and $x = 8$, the neat form gives $2 \cdot 6^3 + 96 - 16 = 512 = 8^3$, confirming the point $(8, 512)$.
- With $a = 0$, $(*)$ is the tangent line $y = 12x - 16$, and the formula gives the other intersection at $x = -4$; indeed $x^3 - 12x + 16 = (x-2)^2(x+4)$.
- We checked the factorisation exactly for thousands of random rational a and x, and the turning values $8 \pm 8\sqrt{2} \approx 19.31,\ -3.31$ in part (iii) numerically.

Common slips

- Forgetting the case $a = 1$, where the formula for the other intersection breaks down because there isn't one.
- Showing only that both curves pass through $(2, 8)$. Touching also needs equal gradients, or equivalently a repeated root.
- In part (iii), drawing the curves as touching at the origin. They cross there, with different gradients.

Beyond STEP

A repeated root of $f(x) - g(x)$ is the algebraic shadow of tangency: the curves agree to first order at that point. Pushing this further, the order of contact counts how many derivatives agree, and it is the idea behind Taylor polynomials, which are the

polynomials with the highest possible order of contact with a function at a point. Bartle and Sherbert, *Introduction to Real Analysis*, develops Taylor's theorem carefully. Siklos, *Advanced Problems in Mathematics*, has more curve sketching in the STEP style.

WORKED SOLUTION

STEP I 2007, Question 9

Mechanics • Limiting Friction on Slopes

STEP I 2007, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

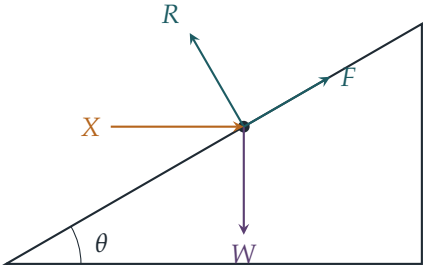
step.maths.org/questions/07-s1-q9

The key idea

This is two cases of limiting friction, which differ only in the direction of the friction force. In each case we resolve along and perpendicular to the plane and eliminate the normal reaction, which gives each horizontal force as W times a ratio of trigonometric expressions. Dividing one by the other removes W and leaves a relation between k, μ and θ alone. The inequality then follows from the fact that $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$ can never exceed $\frac{1}{2}$.

Solution

We model the body as a particle and assume friction is limiting, $F = \mu R$, in both cases. Write $s = \sin\theta$ and $c = \cos\theta$ to keep the algebra compact.



The horizontal force pushes the particle into the plane, so it has a component Xc up the slope and a component Xs into the plane. The weight has a component Ws down the slope and Wc into the plane.

About to slide down: force X

The particle is on the point of moving down, so friction μR acts *up* the plane (as drawn). Resolving perpendicular to the plane, then along it,

$$R = Wc + Xs, \qquad Xc + \mu R = Ws.$$

Substituting for R,

$$Xc + \mu Wc + \mu Xs = Ws \quad \Longrightarrow \quad X(c + \mu s) = W(s - \mu c). \tag{1}$$

About to slide up: force kX

Now the particle is on the point of moving up, so friction acts *down* the plane. Similarly,

$$R' = Wc + kXs, \qquad kXc = Ws + \mu R',$$

so

$$kX(c - \mu s) = W(s + \mu c). \tag{2}$$

Ask yourself Are we allowed to divide by everything in sight here?

Yes, given the physical set-up. A positive force X is *needed* to stop the particle sliding down, so the particle would slide without it, which means $\tan\theta > \mu$, i.e. $s - \mu c > 0$. Then (1) gives $X > 0$, and $c + \mu s > 0$ anyway. For a horizontal push to be able to drive the particle up the slope at all, (2) needs $c - \mu s > 0$, i.e. $\tan\theta < 1/\mu$. Together these force $\mu < \tan\theta < 1/\mu$, and in particular $\mu < 1$; we shall need this at the end.

The relation

Dividing (2) by (1) removes both X and W:

$$k\,\frac{c - \mu s}{c + \mu s} = \frac{s + \mu c}{s - \mu c} \quad \Longrightarrow \quad k(c - \mu s)(s - \mu c) = (s + \mu c)(c + \mu s).$$

Expanding each side and using $s^2 + c^2 = 1$:

$$(c - \mu s)(s - \mu c) = sc - \mu c^2 - \mu s^2 + \mu^2 sc = (1 + \mu^2)sc - \mu,$$
$$(s + \mu c)(c + \mu s) = sc + \mu s^2 + \mu c^2 + \mu^2 sc = (1 + \mu^2)sc + \mu.$$

So $k(1 + \mu^2)sc - k\mu = (1 + \mu^2)sc + \mu$, which rearranges to

$$(k - 1)(1 + \mu^2)\sin\theta\cos\theta = \mu(k + 1),$$

as required.

The inequality

The right-hand side is positive (since $\mu > 0$ and $k > 0$), and so is $(1 + \mu^2)\sin\theta\cos\theta$ for an acute θ; hence $k > 1$. (This is no surprise: pushing the particle up takes more than holding it.) Now

$$\sin\theta\cos\theta = \tfrac{1}{2}\sin 2\theta \leq \tfrac{1}{2},$$

and since $(k-1)(1+\mu^2) > 0$ we may multiply this inequality by it:

$$\mu(k+1) = (k-1)(1+\mu^2)\sin\theta\cos\theta \leq \tfrac{1}{2}(k-1)(1+\mu^2).$$

Doubling and collecting the terms in k:

$$2\mu k + 2\mu \leq k(1+\mu^2) - (1+\mu^2) \quad \Longrightarrow \quad k(1-2\mu+\mu^2) \geq 1 + 2\mu + \mu^2,$$

that is, $k(1-\mu)^2 \geq (1+\mu)^2$. We showed above that $\mu < 1$, so $(1-\mu)^2 > 0$ and we can divide by it without changing the direction:

$$k \geq \frac{(1+\mu)^2}{(1-\mu)^2}.$$

Equality holds exactly when $\sin 2\theta = 1$, i.e. $\theta = 45°$.

Ask yourself Why must we know $\mu < 1$ before the last step?

If $\mu = 1$ we would be dividing by zero (and the line before would read $0 \geq 4$, which is false); if $\mu > 1$ dividing would still be legal, but the conditions on θ would already have failed. Physically, with $\mu \geq 1$ the two requirements $\tan\theta > \mu \geq 1$ and $\tan\theta < 1/\mu \leq 1$ contradict each other: either the particle does not need holding up at all, or no horizontal push can ever start it up the slope. So the question implicitly has $\mu < 1$.

Checks

- Take $\mu = \frac{1}{2}$ and $\theta = 45°$, so $s = c = 1/\sqrt{2}$. From (1) and (2), $X = W \cdot \frac{1/2}{3/2} = \frac{1}{3}W$ and $kX = W \cdot \frac{3/2}{1/2} = 3W$, so $k = 9 = (1.5/0.5)^2$: the equality case. With $\theta = 60°$ instead, the same formulae give $k \approx 25.2 > 9$.
- If $\mu \to 0$ then $X = kX = W\tan\theta$ and $k \to 1$, matching both the identity (both sides 0) and the bound 1.
- A numerical check of the identity and the inequality over 10^5 random choices of $\mu \in (0,1)$ and admissible θ found no failures.

Common slips

- Taking $R = W\cos\theta$: the horizontal force also presses the particle into the plane and adds $X\sin\theta$ to the reaction.
- Keeping friction in the same direction in both cases. It always opposes the motion that is about to happen.
- Dividing by $(1-\mu)^2$ without saying why it is positive (and non-zero).

Beyond STEP

A neat way to see this is through the *angle of friction* λ, with $\tan\lambda = \mu$: the normal reaction and limiting friction combine into a single force at angle λ to the normal, and (1) and (2) become $X = W\tan(\theta - \lambda)$ and $kX = W\tan(\theta + \lambda)$. So $k = \tan(\theta + \lambda)/\tan(\theta - \lambda)$, and the bound is the statement that this ratio is smallest at $\theta = 45^\circ$. Friction-angle arguments of this kind, and the idea of a "friction cone", reappear in university statics and in the theory of contact. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and force analysis with the care first-year courses expect.

WORKED SOLUTION

STEP I 2007, Question 10

Mechanics • Displacement-Time Graphs

STEP I 2007, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q10

The key idea

On a displacement–time graph every rider and every army is a straight line (or two straight segments), and "passing each other" simply means two lines crossing. So we write down the equation of each line, measuring displacement from the Saxon army, and intersect them. The inequalities in the question are exactly the conditions that the crossings happen in the order we expect, and the final part is about what goes on when one of them fails.

Solution

Setting up the graph

We model the armies and horsemen as points moving along a straight line. Let s be the displacement from the (stationary) Saxon army, towards the Normans, and start the clock when the armies are d apart. Then:

- Saxon army: $s = 0$.
- Norman army: $s = d - ut$.
- Saxon horseman, outward: $s = xt$. He reaches the Norman army when $xt = d - ut$, so at $t_S = \dfrac{d}{x+u}$, at $s = \dfrac{xd}{x+u}$. Coming back at speed x,

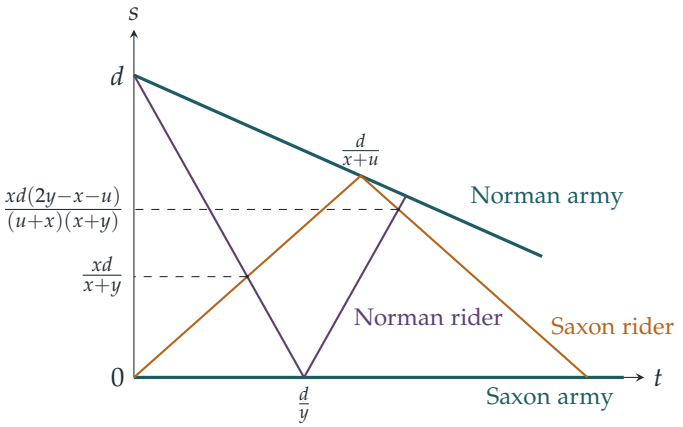
$$s = \frac{xd}{x+u} - x\left(t - \frac{d}{x+u}\right) = \frac{2xd}{x+u} - xt,$$

  and he is home at $t = \dfrac{2d}{x+u}$.
- Norman horseman, outward: $s = d - yt$. He reaches the Saxon army at $t_N = \dfrac{d}{y}$, and

coming back at speed y his line is

$$s = y\left(t - \frac{d}{y}\right) = yt - d.$$

Since $y > u$ he does catch his own army, when $yt - d = d - ut$, i.e. at $t = \dfrac{2d}{y+u}$.



(The picture is drawn with $u = 1$, $x = 2$, $y = 4$, $d = 1$.)

The first passing

Both riders are on their outward journeys, so we intersect $s = xt$ with $s = d - yt$: $(x+y)t = d$, giving

$$t = \frac{d}{x+y}, \qquad s = \frac{xd}{x+y}.$$

This is genuinely before either has turned round, because $\frac{d}{x+y}$ is less than both $\frac{d}{y}$ and $\frac{d}{x+u}$ (using $y > u$ for the second).

The second passing

Now both riders are on their way home: we intersect $s = \dfrac{2xd}{x+u} - xt$ with $s = yt - d$. Adding $xt + d$ to both sides,

$$(x+y)t = d + \frac{2xd}{x+u} = \frac{d(3x+u)}{x+u}, \qquad \text{so} \qquad t_2 = \frac{d(3x+u)}{(x+u)(x+y)}.$$

Then

$$s = \frac{2xd}{x+u} - \frac{xd(3x+u)}{(x+u)(x+y)} = \frac{xd\big(2(x+y) - (3x+u)\big)}{(x+u)(x+y)} = \frac{xd(2y-x-u)}{(u+x)(x+y)},$$

which is the second distance required.

Ask yourself How do we know both riders really are on their return legs at time t_2?

This is where the inequalities come in. The Norman rider has turned if $t_2 \geq \frac{d}{y}$:

$$y(3x+u) \geq (x+u)(x+y) \iff 3xy \geq x^2+xy+ux \iff u \leq 2y-x.$$

The Saxon rider has turned if $t_2 \geq \frac{d}{x+u}$:

$$3x+u \geq x+y \iff u \geq y-2x.$$

So the condition $y-2x < u < 2y-x$ is exactly what makes t_2 come after both turning points. The same condition $u < 2y-x$ also makes $s > 0$, so the Saxon rider is not yet home. (There is one more, unstated, requirement: the Norman rider must not have reached his own army first. Working through $t_2 \leq \frac{2d}{y+u}$ reduces to $(x-u)(2x-y+u) \geq 0$, and since $2x-y+u > 0$ this means $x \geq u$, i.e. the Saxon rider is at least as fast as the Norman army. That is surely intended: otherwise, the moment he turned round, the army would overtake him.)

Case (i): $u > 2y-x$

Now $t_2 < \frac{d}{y}$: the lines for the Saxon rider's return and the Norman rider's *outward* journey cross first. Since $y > u$, this case needs $x > y$: the Saxon rider is the faster one. He reaches the Norman army, turns, and catches up with the Norman rider from behind while both are heading towards the Saxon army. (Intersecting $s = \frac{2xd}{x+u} - xt$ with $s = d - yt$ puts this at $\frac{xd(x+u-2y)}{(x+u)(x-y)}$ from the Saxon army.) He is home at $\frac{2d}{x+u}$, before the Norman rider arrives at $\frac{d}{y}$, so the two do not meet head-on a second time: the Norman rider simply arrives at the Saxon army, where his opponent is already back.

Case (ii): $u < y-2x$

Now $t_2 < \frac{d}{x+u}$: the Norman rider is much faster. He reaches the Saxon army, turns, and overtakes the Saxon rider, who is still riding out towards the Normans. Intersecting $s = yt - d$ with $s = xt$, this happens at $t = \frac{d}{y-x}$, at $\frac{xd}{y-x}$ from the Saxon army. The Norman rider then gets home at $\frac{2d}{y+u}$, before the Saxon rider reaches the Norman army at $\frac{d}{x+u}$ (because $2(x+u) < y+u$), and there is no further passing on the open field.

Checks

- With $d=1$, $u=1$, $x=2$, $y=4$ (as in the picture), the formulae give passings at $\frac{1}{3}$ and $\frac{5}{9}$. Stepping the motion numerically in time steps of 10^{-5} gives crossings at 0.3333 and 0.5556.
- If $y = x+u$ the two distances are equal, both being $\frac{x}{x+y}d$. That is a nice special value to test.

- As $u \to 2y - x$ the second distance tends to 0: the second passing happens right at the Saxon army, which is the boundary with case (i).

Common slips

- Writing the return leg of a rider as $s = -xt$ or $s = yt$, forgetting to shift the time origin to the moment of turning.
- Using the Norman army's position at $t = 0$, rather than at the moment the Saxon rider arrives, for his turning point.
- In the final part, saying only that "they do not meet". In both cases they do pass a second time, but going the same way: one overtakes the other.

Beyond STEP

Graphs like this are *world lines*, the basic tool of special relativity. There, light signals sent back and forth between moving observers are analysed in exactly this way (the radar method), and the slopes of the lines encode the relative velocities. The algebra here, with u added to one speed and subtracted from another, is Galilean relative velocity. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both the Galilean and the relativistic pictures.

WORKED SOLUTION

STEP I 2007, Question 11

Mechanics • Projectile Motion

STEP I 2007, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

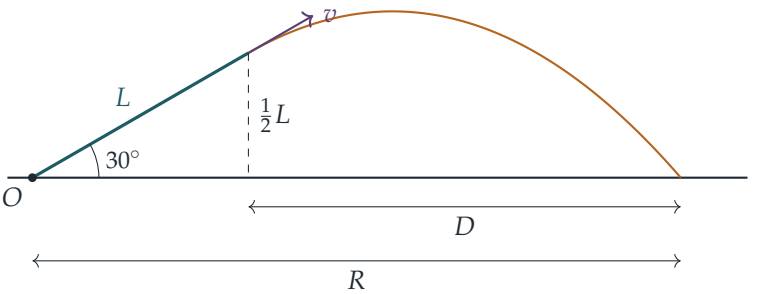
step.maths.org/questions/07-s1-q11

The key idea

Split the motion into two stages. Inside the smooth tube the only force along the tube is the component of gravity, so energy (or constant-acceleration formulae) gives the exit speed. After that it is an ordinary projectile, launched at 30° from a point a height $\frac{1}{2}L$ above the finish level. Eliminating the flight time gives the quadratic for D; differentiating it *implicitly* gives $\mathrm{d}D/\mathrm{d}L$ without ever solving for D.

Solution

We model the particle as a point mass, the tube as smooth (so the normal reaction does no work and has no component along the tube), and ignore air resistance.



Inside the tube

Resolving along the tube, the only force component is $mg\sin 30^\circ = \frac{1}{2}mg$ down the slope, so the deceleration is $\frac{1}{2}g$. Over distance L,

$$v^2 = u^2 - 2\cdot\tfrac{1}{2}g\cdot L = u^2 - gL,$$

where v is the exit speed. (We need $u^2 > gL$ for the particle to leave the tube at all.) The top of the tube is a height $L\sin 30^\circ = \frac{1}{2}L$ above the lower end and a horizontal distance

$L\cos 30^\circ = \frac{\sqrt{3}}{2}L$ from it.

The projectile stage

The particle leaves along the tube, so its velocity has horizontal component $\frac{\sqrt{3}}{2}v$ and vertical component $\frac{1}{2}v$ upwards. Measuring from the exit point, after time t

$$x = \tfrac{\sqrt{3}}{2}vt, \qquad y = \tfrac{1}{2}vt - \tfrac{1}{2}gt^2.$$

It returns to the level of the lower end when $y = -\frac{1}{2}L$, at which moment $x = D$. From the first equation $t = \dfrac{2D}{\sqrt{3}\,v}$, and substituting into the second,

$$\frac{D}{\sqrt{3}} - \frac{2gD^2}{3v^2} = -\frac{L}{2}.$$

Multiplying through by $-6v^2$,

$$4gD^2 - 2\sqrt{3}\,v^2 D - 3Lv^2 = 0,$$

and putting $v^2 = u^2 - gL$,

$$4gD^2 - 2\sqrt{3}\,(u^2 - gL)D - 3L(u^2 - gL) = 0,$$

which is the required equation. (Of its two roots, D is the positive one: the product of the roots is $-3Lv^2/(4g) < 0$.)

The derivative

For fixed u, D depends on L, so we differentiate the equation with respect to L, using the product rule on each term containing both:

$$8gD\frac{\mathrm{d}D}{\mathrm{d}L} - 2\sqrt{3}(u^2 - gL)\frac{\mathrm{d}D}{\mathrm{d}L} + 2\sqrt{3}\,gD - 3(u^2 - gL) + 3gL = 0.$$

The last two terms combine to $-3(u^2 - 2gL)$. Collecting the $\mathrm{d}D/\mathrm{d}L$ terms,

$$\left(8gD - 2\sqrt{3}(u^2 - gL)\right)\frac{\mathrm{d}D}{\mathrm{d}L} = -\left(2\sqrt{3}\,gD - 3(u^2 - 2gL)\right),$$

so

$$\frac{\mathrm{d}D}{\mathrm{d}L} = -\frac{2\sqrt{3}\,gD - 3(u^2 - 2gL)}{8gD - 2\sqrt{3}(u^2 - gL)},$$

as required.

Ask yourself Why does the $-3L(u^2 - gL)$ term produce two pieces?

Because L appears twice in it: differentiating $-3Lu^2 + 3gL^2$ gives $-3u^2 + 6gL$, which is $-3(u^2 - gL) + 3gL$. Forgetting the L inside the bracket is the easiest way to miss the $2gL$ in the answer.

Stationary R

The final horizontal displacement from the lower end is the horizontal run of the tube plus D:

$$R = \tfrac{\sqrt{3}}{2}L + D, \qquad \text{so} \qquad \frac{\mathrm{d}R}{\mathrm{d}L} = \frac{\sqrt{3}}{2} + \frac{\mathrm{d}D}{\mathrm{d}L}.$$

Hence $\mathrm{d}R/\mathrm{d}L = 0$ exactly when $\mathrm{d}D/\mathrm{d}L = -\frac{\sqrt{3}}{2}$, that is (multiplying up by the denominator, which we check below is non-zero)

$$2\sqrt{3}\,gD - 3(u^2 - 2gL) = \tfrac{\sqrt{3}}{2}\big(8gD - 2\sqrt{3}(u^2 - gL)\big) = 4\sqrt{3}\,gD - 3(u^2 - gL).$$

The $3u^2$ terms cancel, leaving $6gL - 2\sqrt{3}\,gD = 3gL$, so $2\sqrt{3}\,gD = 3gL$, i.e.

$$2D = L\sqrt{3}.$$

Every step is reversible, so $\mathrm{d}R/\mathrm{d}L = 0$ when $2D = L\sqrt{3}$, as required.

To find R, substitute $D = \frac{\sqrt{3}}{2}L$ into the quadratic:

$$4g \cdot \tfrac{3}{4}L^2 - 2\sqrt{3}(u^2 - gL) \cdot \tfrac{\sqrt{3}}{2}L - 3L(u^2 - gL) = 3gL^2 - 6L(u^2 - gL) = 0.$$

Since $L > 0$, this gives $gL = 2(u^2 - gL)$, so $L = \dfrac{2u^2}{3g}$. (Then $v^2 = u^2 - gL = \frac{1}{3}u^2 > 0$, so the particle does leave the tube, and the denominator $8gD - 2\sqrt{3}v^2 = \frac{8\sqrt{3}}{3}u^2 - \frac{2\sqrt{3}}{3}u^2 = 2\sqrt{3}\,u^2$ is non-zero.) Finally

$$R = \tfrac{\sqrt{3}}{2}L + D = \sqrt{3}\,L = \frac{2\sqrt{3}\,u^2}{3g} = \frac{2u^2}{\sqrt{3}\,g}.$$

Checks

- With $L = 0$ the quadratic gives $D = \sqrt{3}u^2/(2g)$, the familiar range $u^2 \sin 60^\circ / g$ for a launch at 30°.
- With $L = u^2/g$ the particle just reaches the top with $v = 0$, so $D = 0$ and $R = \frac{\sqrt{3}}{2}u^2/g$ again. The stationary value $2u^2/(\sqrt{3}g) \approx 1.155\,u^2/g$ lies between these and above both, so it is a maximum.
- Numerically, with $u = g = 1$, scanning L over $[0, 1]$ gives a largest R of 1.1547 at $L \approx 0.6667$, matching $2/\sqrt{3}$ and $L = \frac{2}{3}$; stepping the motion directly also reproduces the quadratic for D.

Common slips

- Taking the deceleration in the tube to be g rather than $g \sin 30^\circ = \frac{1}{2}g$.
- Setting $y = 0$ instead of $y = -\frac{1}{2}L$: the particle lands below its launch point.
- Writing $R = D$, forgetting the horizontal length $\frac{\sqrt{3}}{2}L$ of the tube itself.

Beyond STEP

Differentiating an equation $F(D, L) = 0$ without solving it is implicit differentiation, and in first-year analysis it becomes the implicit function theorem: $\mathrm{d}D/\mathrm{d}L = -F_L/F_D$, valid wherever $F_D \neq 0$, which is exactly the denominator we checked. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for the mechanics.

WORKED SOLUTION

STEP I 2007, Question 12

Probability and Statistics • Conditional Probability

STEP I 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q12

The key idea

For the second sweet we don't know what the bags look like, because that depends on what happened first. So we condition: split according to everything that happened before the second draw, work out the chance of red in each case, and add up using the law of total probability. In part (ii) there's a useful shortcut: the expected number of red sweets in a bag that has just received a random sweet is easy to write down, so we needn't split on the colour of the transferred sweet more than once.

Solution

Part (i)

Let R_1 be the event that the first sweet is red and R_2 the event that the second sweet is red. Every sweet is equally likely to be drawn first, so

$$\mathrm{P}(R_1) = \frac{a}{N}.$$

For R_2, condition on the colour of the first sweet. If the first is red, $N-1$ sweets remain, of which $a-1$ are red; if it is not red, $N-1$ remain, of which a are red. (Here $N \geq 2$ guarantees that a second sweet exists.) By the law of total probability,

$$\begin{aligned}\mathrm{P}(R_2) &= \mathrm{P}(R_1)\,\mathrm{P}(R_2 \mid R_1) + \mathrm{P}(R_1')\,\mathrm{P}(R_2 \mid R_1') \\ &= \frac{a}{N}\cdot\frac{a-1}{N-1} + \frac{N-a}{N}\cdot\frac{a}{N-1} \\ &= \frac{a\big[(a-1)+(N-a)\big]}{N(N-1)} = \frac{a(N-1)}{N(N-1)} = \frac{a}{N}.\end{aligned}$$

So $\mathrm{P}(R_2) = \mathrm{P}(R_1)$, as required. (The same is true by symmetry: every ordered pair of distinct sweets is equally likely, so the second sweet is just as likely to be any particular sweet as the first is.)

Part (ii)

Here the "first sweet" is the one transferred after the first toss, and the "second sweet" is the one chosen after the second toss. Let R_1, R_2 be the events that these are red. The two tosses are independent, and each is independent of which sweets are chosen.

The first sweet. Conditioning on the first toss,

$$\mathrm{P}(R_1) = p \cdot \frac{a}{N} + q \cdot \frac{b}{N} = \frac{pa + qb}{N}.$$

The second sweet. There are four cases for the two tosses.

Heads, Heads. A sweet is taken from bag 1, and then another sweet is taken from bag 1. That is exactly the situation of part (i) applied to bag 1 (with N sweets, a red), so the second sweet is red with probability a/N.

Heads, Tails. A sweet moves from bag 1 to bag 2, which now has $N+1$ sweets; then we draw from bag 2. Bag 2 has $b+1$ red sweets with probability a/N and b red sweets with probability $1 - a/N$, so

$$\mathrm{P}(R_2 \mid \mathrm{HT}) = \frac{a}{N} \cdot \frac{b+1}{N+1} + \left(1 - \frac{a}{N}\right) \frac{b}{N+1} = \frac{b + \frac{a}{N}}{N+1} = \frac{Nb + a}{N(N+1)}.$$

? **Ask yourself** Why does the second line collapse so neatly?

The numerator is the expected number of red sweets in bag 2 after the transfer: the b it started with, plus the chance a/N that the incoming sweet is red. Since bag 2 has a fixed size $N+1$ in this case, the chance of drawing red is that expected number divided by $N+1$.

Tails, Heads. By the same argument with the bags swapped, bag 1 now holds $N+1$ sweets with an expected $a + b/N$ red ones, so

$$\mathrm{P}(R_2 \mid \mathrm{TH}) = \frac{Na + b}{N(N+1)}.$$

Tails, Tails. Part (i) applied to bag 2 gives probability b/N.

Now combine the four cases, with probabilities p^2, pq, qp, q^2:

$$\begin{aligned}
\mathrm{P}(R_2) &= p^2 \frac{a}{N} + pq \frac{(Nb + a) + (Na + b)}{N(N+1)} + q^2 \frac{b}{N} \\
&= p^2 \frac{a}{N} + pq \frac{(N+1)(a+b)}{N(N+1)} + q^2 \frac{b}{N} \\
&= \frac{p^2 a + pqa + pqb + q^2 b}{N} \\
&= \frac{pa(p+q) + qb(p+q)}{N} = \frac{pa + qb}{N},
\end{aligned}$$

using $p + q = 1$ in the last step. This equals $\mathrm{P}(R_1)$, which is what we had to show.

Checks

- If $p = 1$, everything happens in bag 1, and the answer a/N for both sweets is exactly part (i). If $p = 0$ we get part (i) for bag 2.
- If $a = b = 0$ there are no red sweets and both probabilities are 0; if $a = b = N$ every sweet is red and both are 1.
- With $N = 5$, $a = 2$, $b = 4$, $p = 0.3$ the formula gives $(0.6 + 2.8)/5 = 0.68$. A simulation of 200 000 runs of the whole process gave 0.679 for the first sweet and 0.681 for the second, and exact enumeration over all small N, a, b agrees.

Common slips

- Forgetting that after a transfer the receiving bag has $N + 1$ sweets, not N, and the giving bag has $N - 1$.
- In the HH and TT cases, treating the second draw as independent of the first. It isn't, but part (i) tells us the answer is unchanged anyway, so use it rather than redoing the calculation.

Beyond STEP

Both parts are instances of a deep idea: if you draw from an urn without looking at what you took, your knowledge of the urn hasn't changed, so the next draw has the same distribution. Formally, the sequence of draws is *exchangeable*, and in part (ii) the quantity $p \times (\text{proportion red in bag } 1) + q \times (\text{proportion red in bag } 2)$, the chance that the next sweet drawn is red, is a *martingale*: a quantity whose expected future value equals its present value. Martingales are central to modern probability, from gambling to finance. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet conditional expectation and these ideas.

WORKED SOLUTION

STEP I 2007, Question 13

Probability and Statistics • Conditional Probability

STEP I 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.1, *Precise Conditioning*: https://beyondstep.keithmcnulty.org/ch12/conditioning.html

The key idea

Although the discs come out one at a time, every set of four discs is equally likely, so most of the work is counting combinations. The interest is in the conditions. Knowing *which* disc came first, or simply *that* a particular disc was taken, leaves the other three discs as a random choice from the ten that remain. Knowing only that *some* numbered disc was taken is a different kind of information altogether, and the last two parts are there to show the difference.

Solution

Let N be the number of numbered discs among the four taken. Since the bag is shaken, every ordered sequence of four distinct discs is equally likely, and so every unordered set of four discs is equally likely too. There are

$$\binom{11}{4} = 330$$

such sets. A set with exactly k numbered discs is made by choosing k of the 5 numbered discs and $4-k$ of the 6 blanks, so

$$\mathrm{P}(N=k) = \frac{\binom{5}{k}\binom{6}{4-k}}{330}.$$

Part (i)

All four numbered means $k=4$:

$$\mathrm{P}(N=4) = \frac{\binom{5}{4}\binom{6}{0}}{330} = \frac{5}{330} = \frac{1}{66}.$$

Equivalently, drawing in order, $\frac{5}{11} \cdot \frac{4}{10} \cdot \frac{3}{9} \cdot \frac{2}{8} = \frac{1}{66}$.

Part (ii)

Let T_1 be the event that disc "3" is taken first. Given T_1, the shaken bag now holds 10 discs, 4 numbered and 6 blank, and the next three discs are a random selection from these. All four are numbered exactly when these three are all numbered, so

$$\mathrm{P}(N = 4 \mid T_1) = \frac{\binom{4}{3}}{\binom{10}{3}} = \frac{4}{120} = \frac{1}{30}.$$

Part (iii)

Still given T_1, we need exactly one more numbered disc among the three drawn from the remaining ten:

$$\mathrm{P}(N = 2 \mid T_1) = \frac{\binom{4}{1}\binom{6}{2}}{\binom{10}{3}} = \frac{4 \times 15}{120} = \frac{1}{2}.$$

Part (iv)

Now let T be the event that disc "3" is taken at some point.

? **Ask yourself** Does it matter whether disc "3" came first or at some other draw?

No. Given T, the four discs taken are disc "3" together with three others, and by symmetry every set of three from the other ten discs is equally likely. That is exactly the situation of part (iii), where the order of the draws never mattered. So the answer is again $\frac{1}{2}$.

To confirm this directly: $\mathrm{P}(T) = \binom{10}{3}/\binom{11}{4} = \frac{120}{330} = \frac{4}{11}$, and the sets containing disc "3", one other numbered disc and two blanks number $\binom{4}{1}\binom{6}{2} = 60$, so

$$\mathrm{P}(N = 2 \mid T) = \frac{60/330}{120/330} = \frac{1}{2}.$$

Part (v)

Let F be the event that the first disc is numbered. Split F according to which number appears on the first disc: F is the disjoint union of the five events "disc j is taken first", $j = 1, \ldots, 5$, each of probability $\frac{1}{11}$. The argument of part (iii) did not use the fact that the number was 3, so for each j

$$\mathrm{P}(N = 2 \mid \text{disc } j \text{ first}) = \tfrac{1}{2}.$$

Hence, by the law of total probability,

$$\mathrm{P}(N = 2 \mid F) = \sum_{j=1}^{5} \frac{1/11}{5/11} \cdot \frac{1}{2} = \frac{1}{2}.$$

(Directly: given F, the other three discs are a random three from 4 numbered and 6 blank, which is part (iii) again.)

Part (vi)

Let A be the event that at least one numbered disc is taken, so A is the event $N \geq 1$. This is *not* the same sort of information as before: it does not single out a disc and leave the rest random. Instead we use the definition of conditional probability. Since $N = 2$ implies A,

$$\mathrm{P}(N = 2 \mid A) = \frac{\mathrm{P}(N = 2)}{\mathrm{P}(N \geq 1)}.$$

Ask yourself Why can't we just say "one numbered disc is known, so the other three are random from the remaining ten"?

Because we don't know *which* disc is the known one. A set with two numbered discs satisfies A in the same way as a set with four; there is no single disc we can set aside. So the condition reweights the sets according to N itself, and we must count.

We have

$$\mathrm{P}(N = 2) = \frac{\binom{5}{2}\binom{6}{2}}{330} = \frac{10 \times 15}{330} = \frac{150}{330}, \qquad \mathrm{P}(N = 0) = \frac{\binom{6}{4}}{330} = \frac{15}{330},$$

so $\mathrm{P}(N \geq 1) = \frac{315}{330}$ and

$$\mathrm{P}(N = 2 \mid A) = \frac{150}{315} = \frac{10}{21}.$$

Checks

The distribution of N should add up: the counts $\binom{5}{k}\binom{6}{4-k}$ for $k = 0, \ldots, 4$ are $15, 100, 150, 60, 5$, which total 330. In parts (ii) and (iii) the conditional distribution of the number of further numbered discs has counts $20, 60, 36, 4$ out of 120, again summing correctly. We also checked all six answers ($\frac{1}{66}$, $\frac{1}{30}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{10}{21}$) by listing all $11 \times 10 \times 9 \times 8 = 7920$ ordered draws with exact fractions. Note that $\frac{10}{21} \approx 0.476$ is a little below $\frac{1}{2}$, as it should be: conditioning on $N \geq 1$ only removes the $N = 0$ case, so it leaves more weight on $N = 1$ than knowing a particular disc does.

Common slips

- Treating parts (iv) and (vi) as the same kind of condition. "Disc 3 is taken" names a disc; "a numbered disc is taken" does not, and the answers differ.
- In part (vi), forgetting that the condition is $N \geq 1$ and dividing by $\mathrm{P}(N = 1)$ or by $\frac{5}{11}$ instead.
- Counting ordered draws in the numerator and unordered sets in the denominator. Choose one and stick to it.

Beyond STEP

The contrast between parts (v) and (vi) is the same trap as the famous "boy or girl" paradox: being told that a *specific* child is a boy gives a different answer from being told that *at least one* is. At university this is made precise by treating conditioning on an event as restricting and renormalising the probability measure, and the counts here are the hypergeometric distribution. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both discuss these paradoxes carefully.

WORKED SOLUTION

STEP I 2007, Question 14

Probability and Statistics • Poisson Parameter Uniqueness

STEP I 2007, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q14

The key idea

Each piece of information gives an equation of the form $h(\lambda) = \text{constant}$, where h is some function of the mean. Whether that equation pins down λ uniquely depends entirely on the *shape* of h on $\lambda > 0$: a strictly monotonic h hits each value at most once, whereas an h that rises and then falls hits most values twice and only its maximum value once. So the whole question is about finding where each function is monotonic and where it peaks.

Solution

Throughout, $\lambda > 0$ (a Poisson mean is positive) and $\mathrm{P}(X = k) = \mathrm{e}^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$

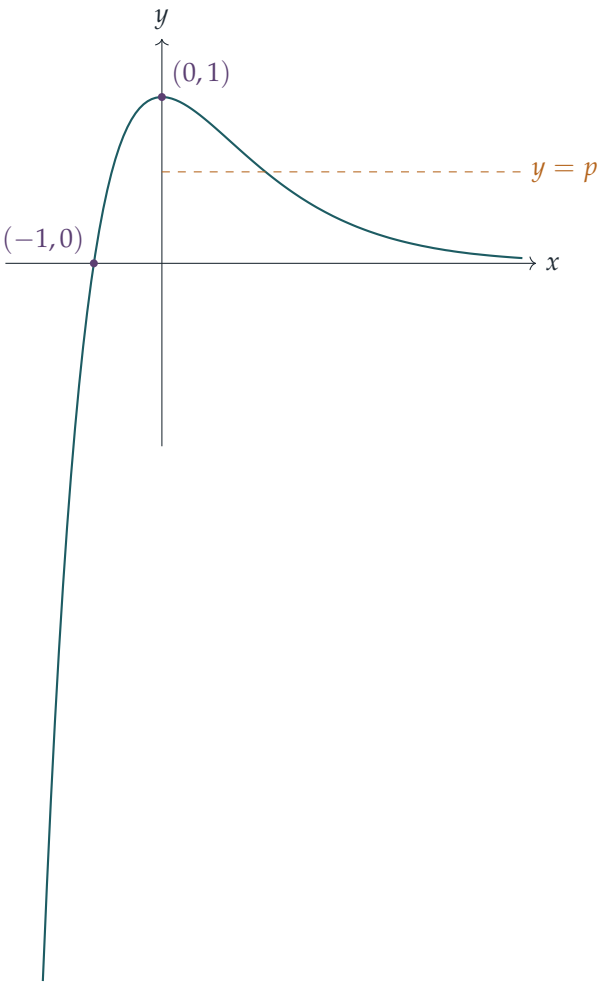
Part (i)

Let $f(x) = (x+1)\mathrm{e}^{-x}$. By the product rule,

$$f'(x) = \mathrm{e}^{-x} - (x+1)\mathrm{e}^{-x} = -x\,\mathrm{e}^{-x}.$$

Since $\mathrm{e}^{-x} > 0$, $f'(x)$ has the opposite sign to x: f is increasing for $x < 0$ and decreasing for $x > 0$. So the only turning point is a maximum at $(0, 1)$, which is also where the curve crosses the y-axis. The curve meets the x-axis where $x + 1 = 0$, at $(-1, 0)$.

For the ends: as $x \to \infty$, e^{-x} beats the linear factor and $f(x) \to 0$ from above (the x-axis is an asymptote); as $x \to -\infty$, both factors grow in size and $f(x) \to -\infty$.



Now the probability. We have

$$\mathrm{P}(X \geq 2) = 1 - \mathrm{P}(X = 0) - \mathrm{P}(X = 1) = 1 - \mathrm{e}^{-\lambda} - \lambda\mathrm{e}^{-\lambda} = 1 - f(\lambda),$$

so the information says exactly that

$$f(\lambda) = (1 + \lambda)\mathrm{e}^{-\lambda} = p.$$

On $\lambda > 0$ the function f is continuous and strictly decreasing (since $f'(\lambda) = -\lambda\mathrm{e}^{-\lambda} < 0$), falling from $f(0) = 1$ towards 0 as $\lambda \to \infty$. So it takes every value in $(0, 1)$ exactly once: by the intermediate value theorem there is at least one solution, and strict monotonicity rules out a second. In the sketch, the horizontal line $y = p$ with $0 < p < 1$ meets the right-hand branch exactly once. Hence λ is uniquely determined.

Part (ii)

Here $\mathrm{P}(X = 1) = \lambda\mathrm{e}^{-\lambda} = q$. Write $g(\lambda) = \lambda\mathrm{e}^{-\lambda}$. Then

$$g'(\lambda) = (1 - \lambda)\mathrm{e}^{-\lambda},$$

so g is strictly increasing on $(0,1]$ and strictly decreasing on $[1,\infty)$, with maximum value $g(1)=\mathrm{e}^{-1}$. Also $g(\lambda)\to 0$ as $\lambda\to 0^+$ and as $\lambda\to\infty$. (In fact $g(\lambda)=\mathrm{e}^{-1}f(\lambda-1)$, so the graph of g is the curve from part (i) shifted one unit right and scaled vertically by e^{-1}.)

Ask yourself How many times can a horizontal line meet a curve that rises to a peak and then falls?

Each monotonic piece is hit at most once, so:

- if $0<q<\mathrm{e}^{-1}$, there is one solution in $(0,1)$ and another in $(1,\infty)$, so two possible values of λ;
- if $q>\mathrm{e}^{-1}$, there are no solutions;
- if $q=\mathrm{e}^{-1}$, the only solution is the peak itself, $\lambda=1$.

So the information determines a unique λ for exactly one value of q, namely

$$q=\mathrm{e}^{-1},\qquad \lambda=1.$$

Part (iii)

By the definition of conditional probability, and since the event $\{X=1\}$ lies inside $\{X\le 2\}$,

$$\mathrm{P}(X=1\mid X\le 2)=\frac{\mathrm{P}(X=1)}{\mathrm{P}(X\le 2)}=\frac{\lambda\mathrm{e}^{-\lambda}}{\mathrm{e}^{-\lambda}\left(1+\lambda+\frac{1}{2}\lambda^2\right)}=\frac{2\lambda}{2+2\lambda+\lambda^2}.$$

Call this $k(\lambda)$. Dividing top and bottom by λ (allowed, as $\lambda>0$),

$$k(\lambda)=\frac{2}{\lambda+\dfrac{2}{\lambda}+2}.$$

Ask yourself What is the smallest value of $\lambda+2/\lambda$ for $\lambda>0$?

By the AM–GM inequality, $\lambda+\dfrac{2}{\lambda}\ge 2\sqrt{\lambda\cdot\dfrac{2}{\lambda}}=2\sqrt{2}$, with equality exactly when $\lambda=2/\lambda$, that is $\lambda=\sqrt{2}$. Moreover $\lambda+2/\lambda$ is strictly decreasing on $(0,\sqrt{2}]$ and strictly increasing on $[\sqrt{2},\infty)$ (its derivative $1-2/\lambda^2$ changes sign only at $\sqrt{2}$), and it tends to ∞ at both ends. So k has the same rise-then-fall shape as g in part (ii), with maximum

$$k(\sqrt{2})=\frac{2}{2\sqrt{2}+2}=\frac{1}{\sqrt{2}+1}=\sqrt{2}-1.$$

Exactly as before: for $0<r<\sqrt{2}-1$ there are two values of λ, for $r>\sqrt{2}-1$ there are none, and for $r=\sqrt{2}-1$ there is exactly one. So λ is uniquely determined only when

$$r=\sqrt{2}-1,\qquad \lambda=\sqrt{2}.$$

Alternatively, clear denominators in $k(\lambda) = r$ to get the quadratic $r\lambda^2 - 2(1-r)\lambda + 2r = 0$. Its roots have product $2 > 0$ and sum $2(1-r)/r > 0$, so any real roots are both positive; there is exactly one root precisely when the discriminant $4\left[(1-r)^2 - 2r^2\right] = 4(1 - 2r - r^2)$ vanishes, giving $r = \sqrt{2} - 1$ (taking the positive root) and the double root $\lambda = (1-r)/r = \sqrt{2}$.

Checks

- Part (ii): $\lambda = 1$ gives $\mathrm{P}(X = 1) = \mathrm{e}^{-1} \approx 0.3679$; a numerical scan of $\lambda\mathrm{e}^{-\lambda}$ over $0 < \lambda < 40$ never exceeds this, and $q = 0.3$ gives two crossings.
- Part (iii): with $\lambda = \sqrt{2}$, $\dfrac{2\sqrt{2}}{2 + 2\sqrt{2} + 2} = \dfrac{\sqrt{2}}{2 + \sqrt{2}} = \sqrt{2} - 1 \approx 0.4142$, and a scan confirms this is the maximum; $r = 0.35$ gives two values of λ, $r = 0.45$ none.
- Sanity: as $\lambda \to 0$ almost all of the mass is at 0, and as $\lambda \to \infty$ the mass given $X \leq 2$ piles up at 2, so in both limits $\mathrm{P}(X = 1 \mid X \leq 2) \to 0$, as the formula says.

Common slips

- Forgetting that $\lambda > 0$: in part (i) the curve also exists for $x < 0$, but only the branch $x > 0$ is relevant to the probability.
- Claiming uniqueness in part (i) without saying why: you need both that a solution exists (continuity, values running from 1 down to 0) and that there is no second one (strict monotonicity).
- In part (iii), using $\mathrm{P}(X = 1 \text{ and } X \leq 2) = \mathrm{P}(X = 1)\,\mathrm{P}(X \leq 2)$; the events are not independent, but $\{X = 1\}$ is contained in $\{X \leq 2\}$.

Beyond STEP

This is a first taste of *identifiability*: in statistics we ask whether an observed quantity determines the parameter of a model, and the answer is yes exactly when the map from parameter to observable is one-to-one. Here $\mathrm{P}(X \geq 2)$ identifies λ, but $\mathrm{P}(X = 1)$ and $\mathrm{P}(X = 1 \mid X \leq 2)$ generally do not. The same rise-and-fall shape reappears when you maximise a likelihood: $\lambda\mathrm{e}^{-\lambda}$ is the likelihood of λ after seeing a single observation $X = 1$, and its peak at $\lambda = 1$ is the maximum likelihood estimate. Blitzstein and Hwang, *Introduction to Probability*, is a lively place to read more about the Poisson distribution and conditioning.

WORKED SOLUTION

STEP II 2007, Question 1

Pure • Binomial Approximations

STEP II 2007, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q1

The key idea

A binomial series converges quickly only when the small quantity really is small, so the whole question is about rewriting a surd as a simple rational multiple of $\sqrt{1+\text{something tiny}}$. To do that we look for a perfect square (or cube) that, multiplied by 3 or 6, lands just next to a power of 10. Then k is small compared with 100 (or 1000), and a handful of terms gives the accuracy we need.

Solution

Part (i)

With $x = \frac{k}{100}$, the binomial series for $(1+x)^{1/2}$ has coefficients

$$\binom{\frac{1}{2}}{1} = \frac{1}{2}, \qquad \binom{\frac{1}{2}}{2} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})}{2!} = -\frac{1}{8}, \qquad \binom{\frac{1}{2}}{3} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})}{3!} = \frac{1}{16}.$$

Substituting $x = k/100$,

$$\left(1+\frac{k}{100}\right)^{1/2} \approx 1 + \frac{k}{200} - \frac{k^2}{80\,000} + \frac{k^3}{16\,000\,000}.$$

(a) With $k = 8$ the left-hand side is $\sqrt{1.08}$, and the point is that $108 = 3\times 36$:

$$\sqrt{1.08} = \frac{\sqrt{108}}{10} = \frac{6\sqrt{3}}{10} = \frac{3\sqrt{3}}{5}, \qquad \text{so} \qquad \sqrt{3} = \frac{5}{3}\sqrt{1.08}.$$

The series gives

$$\sqrt{1.08} \approx 1 + 0.04 - 0.0008 + 0.000032 = 1.039232,$$

and therefore

$$\sqrt{3} \approx \frac{5}{3}\times 1.039232 = 1.7320533\ldots \approx 1.73205 \text{ to five decimal places.}$$

(b)

Ask yourself Which value of k makes $1+\frac{k}{100}$ equal to 6 times a nice square?

We want $100+k=6m^2$ for some integer m with $6m^2$ close to 100, so that k is small. Since $6\times 16=96$, take $m=4$ and $k=-4$:

$$\sqrt{0.96}=\frac{\sqrt{96}}{10}=\frac{4\sqrt{6}}{10}=\frac{2\sqrt{6}}{5},\qquad\text{so}\qquad\sqrt{6}=\frac{5}{2}\sqrt{0.96}.$$

With $k=-4$ every term after the first is negative:

$$\sqrt{0.96}\approx 1-0.02-0.0002-0.000004=0.979796,$$

and so

$$\sqrt{6}\approx\frac{5}{2}\times 0.979796=2.44949 \text{ to five decimal places.}$$

This k is half the size of the one in (a), so the neglected terms are even smaller here. (Something like $k=50$, from $1.5=6/4$, also gives $\sqrt{6}$ exactly, but with $x=\frac{1}{2}$ the series converges far too slowly for five decimal places.)

Part (ii)

The first two terms give

$$\left(1+\frac{k}{1000}\right)^{1/3}\approx 1+\frac{k}{3000}.$$

Ask yourself Which cube, multiplied by 3, is just above 1000?

We need $1000+k=3m^3$ with k small. The cubes near $1000/3\approx 333$ are $6^3=216$ and $7^3=343$, and $3\times 343=1029$ is very close to 1000. So take $k=29$:

$$\sqrt[3]{1.029}=\frac{\sqrt[3]{3\times 7^3}}{10}=\frac{7\sqrt[3]{3}}{10},\qquad\text{so}\qquad\sqrt[3]{3}=\frac{10}{7}\sqrt[3]{1.029}.$$

Using the two-term approximation,

$$\sqrt[3]{3}\approx\frac{10}{7}\left(1+\frac{29}{3000}\right)=\frac{10}{7}\cdot\frac{3029}{3000}=\frac{3029}{2100},$$

which is the required approximation.

Checks

A calculator gives $\sqrt{3}=1.7320508\ldots$ and $\sqrt{6}=2.4494897\ldots$, which agree with both answers to five decimal places. For part (ii), $3029/2100=1.44238\ldots$ while $\sqrt[3]{3}=1.44225\ldots$. The error of about 1.3×10^{-4} matches the size of the first neglected term, $\frac{10}{7}\cdot\frac{1}{9}\left(\frac{29}{1000}\right)^2\approx 1.3\times 10^{-4}$, so the approximation is a slight overestimate.

Common slips

- Getting the third coefficient wrong: $\frac{1}{2} \cdot (-\frac{1}{2}) \cdot (-\frac{3}{2})/6 = +\frac{1}{16}$, not $-\frac{1}{16}$.
- Forgetting to multiply back by $\frac{5}{3}$, $\frac{5}{2}$ or $\frac{10}{7}$ after evaluating the series.
- Choosing a large k (such as $k = 50$ in (b)), for which three or four terms are nowhere near enough.

Beyond STEP

How many terms you need comes down to bounding the remainder of a Taylor series. Lagrange's form of the remainder lets you prove, rather than just believe, that these approximations are correct to five places. The same trick of rescaling into the region where a series converges quickly is how calculators and computer libraries actually compute roots and logarithms. For the theory of Taylor series and their remainders, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2007, Question 2

Pure • Cubic Curves and Area

STEP II 2007, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q2

The key idea

Everything about a cubic's turning points is packed into its derivative. Because the turning points are at $x = p$ and $x = q$, the derivative must be a multiple of $(x-p)(x-q)$, and comparing coefficients gives b and c at once. After that, the cleanest way to get $m - n$ is to integrate the derivative between the turning points, and the area becomes easy once we notice that $y - n$ has a *double* root at $x = q$.

Solution

Part (i)

Differentiating,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 2bx + c.$$

This quadratic vanishes at the turning points $x = p$ and $x = q$, and its leading coefficient is 6, so

$$6x^2 - 2bx + c = 6(x-p)(x-q) = 6x^2 - 6(p+q)x + 6pq.$$

Comparing coefficients,

$$b = 3(p+q), \qquad c = 6pq.$$

Two facts we shall use later. First, since the coefficient of x^3 is positive, $\mathrm{d}y/\mathrm{d}x = 6(x-p)(x-q)$ is positive outside the roots and negative between them, so the curve rises, falls, then rises: the maximum comes first, and $p < q$.

Second, the condition $n > 0$ restricts q. Substituting $x = q$,

$$n = 2q^3 - 3(p+q)q^2 + 6pq^2 = q^2(3p - q),$$

so $n > 0$ means $q < 3p$. Altogether $0 < p < q < 3p$.

Part (ii)

The curve passes through the origin, and

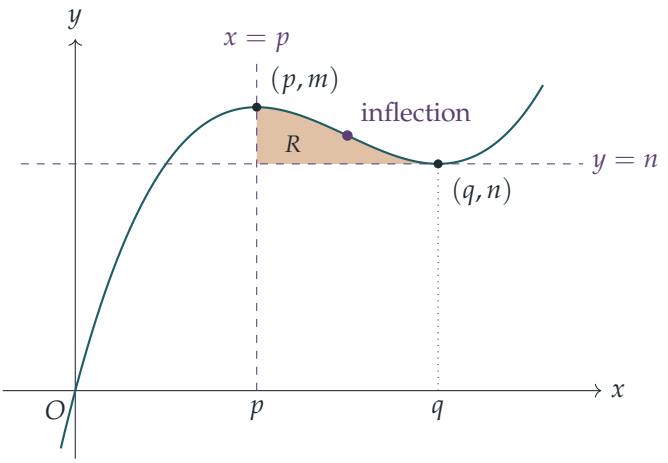
$$y = x\big(2x^2 - bx + c\big),$$

where the quadratic factor has discriminant

$$b^2 - 8c = 9(p+q)^2 - 48pq = 3(3p-q)(p-3q).$$

Since $q < 3p$ and $p < q < 3q$, the first bracket is positive and the second negative, so the discriminant is negative and the origin is the only point where the curve meets the x-axis. So the curve comes up from $-\infty$, crosses the axis at O, rises to the maximum (p, m), falls to the minimum (q, n), still above the axis, and then rises for ever.

The point of inflection is where $\mathrm{d}^2y/\mathrm{d}x^2 = 12x - 2b = 0$, that is, at $x = \frac{b}{6} = \frac{1}{2}(p+q)$, halfway between the turning points.



The region R (shaded) lies between $x = p$ and $x = q$, below the curve and above the line $y = n$; the curve touches $y = n$ at the minimum.

Symmetry. Let $s = \frac{1}{2}(p+q)$, $h = q - p$, and measure from the inflection point by putting $x = s + t$. Then $x - p = t + \frac{1}{2}h$ and $x - q = t - \frac{1}{2}h$, so

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 6\big(t^2 - \tfrac{1}{4}h^2\big),$$

which is an even function of t. Its integral from 0, namely $y - y(s) = 2t^3 - \frac{3}{2}h^2t$, is therefore an odd function of t. So $y(s+t) - y(s) = -\big(y(s-t) - y(s)\big)$: the curve has *rotational symmetry of order* 2 *about its point of inflection* (a half-turn about that point maps the curve to itself). In particular the half-turn swaps the two turning points, so the inflection point is the midpoint of (p, m) and (q, n), namely $\big(\frac{1}{2}(p+q), \frac{1}{2}(m+n)\big)$.

Part (iii)

The drop from the maximum to the minimum is the integral of the gradient:

$$m - n = y(p) - y(q) = -\int_p^q 6(x-p)(x-q)\,\mathrm{d}x.$$

Ask yourself Is there a quick way to evaluate $\int_p^q (x-p)(x-q)\,\mathrm{d}x$ without expanding?

Substitute $x = p + u$, so $x - q = u - h$ with $h = q - p$, and u runs from 0 to h:

$$\int_0^h u(u-h)\,\mathrm{d}u = \frac{h^3}{3} - \frac{h^3}{2} = -\frac{h^3}{6}.$$

Hence

$$m - n = -6\Big(-\frac{(q-p)^3}{6}\Big) = (q-p)^3,$$

as required.

Part (iv)

On $p \le x \le q$ the curve is decreasing from m down to n, so it lies on or above $y = n$, and

$$\text{area of } R = \int_p^q (y-n)\,\mathrm{d}x.$$

The cubic $y - n = 2x^3 - bx^2 + cx - n$ vanishes at $x = q$ and so does its derivative, so $x = q$ is a double root and

$$y - n = 2(x-q)^2(x-r)$$

for some r. Comparing the x^2 coefficients, $-2(2q + r) = -b = -3(p+q)$, so $r = \frac{1}{2}(3p - q)$. With $h = q - p$ again and $u = x - q$ (running from $-h$ to 0),

$$x - r = u + q - \tfrac{1}{2}(3p - q) = u + \tfrac{3}{2}h.$$

Therefore

$$\begin{aligned}\text{area of } R &= \int_{-h}^0 2u^2\big(u + \tfrac{3}{2}h\big)\,\mathrm{d}u = \Big[\tfrac{1}{2}u^4 + hu^3\Big]_{-h}^0 \\ &= -\big(\tfrac{1}{2}h^4 - h^4\big) = \tfrac{1}{2}h^4 = \tfrac{1}{2}(q-p)^4,\end{aligned}$$

which is the required result.

Checks

Take $p = 1$, $q = 2$ (allowed, since $q < 3p$). Then $b = 9$, $c = 12$, and $y = 2x^3 - 9x^2 + 12x$ has $y' = 6(x-1)(x-2)$, with $m = y(1) = 5$ and $n = y(2) = 4 > 0$. So $m - n = 1 = (q-p)^3$. The inflection point is $(1.5, 4.5)$, the midpoint of the turning points, and a numerical integration of $y - 4$ over $[1, 2]$ gives $0.5000 = \frac{1}{2}(q-p)^4$. We also checked all four results for 2000 random pairs with $0 < p < q < 3p$.

Common slips

- Taking $q < p$: for a positive cubic coefficient the maximum is always to the left of the minimum.
- Integrating y instead of $y - n$, which adds the rectangle of height n under the line $y = n$.
- Describing the symmetry as reflection in the line $x = \frac{1}{2}(p + q)$; a cubic has half-turn symmetry about a point, not mirror symmetry.

Beyond STEP

Every cubic is a half-turn symmetric curve about its inflection point, because shifting x to remove the x^2 term (the "depressed cubic" substitution used in Cardano's formula) leaves an odd function plus a constant. The double-root trick in part (iv) is the start of the theory of tangency and multiplicity of roots, which becomes central in algebraic geometry. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP II 2007, Question 3

Pure • Integration by Substitution

STEP II 2007, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q3

The key idea

Everything rests on the standard integral of $1/(a^2+x^2)$, so the whole question is about recognising that shape after a substitution. In (i)(a) the numerator $\cos x$ is the derivative of $\sin x$, which is a strong hint. In (i)(b) the half-angle substitution turns $1+\sin^2 x$ into a quartic in t, and the point is to see that the same thing can be run backwards: in (ii) we spot which trigonometric integral produces the new quartic.

Solution

The standard integral

Let $a \neq 0$ and put $x = a\tan\theta$ with $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. As θ runs over this interval, x runs over every real number exactly once (whatever the sign of a), so the substitution is valid, and $\tan\theta = x/a$ gives $\theta = \arctan(x/a)$. Now $\mathrm{d}x = a\sec^2\theta\,\mathrm{d}\theta$ and

$$a^2 + x^2 = a^2(1+\tan^2\theta) = a^2\sec^2\theta,$$

so

$$\int \frac{1}{a^2+x^2}\,\mathrm{d}x = \int \frac{a\sec^2\theta}{a^2\sec^2\theta}\,\mathrm{d}\theta = \int \frac{1}{a}\,\mathrm{d}\theta = \frac{\theta}{a} + \text{constant} = \frac{1}{a}\arctan\frac{x}{a} + \text{constant},$$

as required.

Part (i)(a)

Substitute $u = \sin x$, so $\mathrm{d}u = \cos x\,\mathrm{d}x$, and u goes from 0 to 1 as x goes from 0 to $\frac{1}{2}\pi$. Using the standard integral with $a = 1$,

$$I = \int_0^1 \frac{1}{1+u^2}\,\mathrm{d}u = \Big[\arctan u\Big]_0^1 = \frac{\pi}{4}.$$

Part (i)(b)

With $t = \tan\frac{1}{2}x$ we have the standard half-angle results

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2}{1+t^2}\,\mathrm{d}t,$$

the last because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2\frac{1}{2}x = \frac{1}{2}(1+t^2)$. The limits $x = 0$ and $x = \frac{1}{2}\pi$ become $t = 0$ and $t = \tan\frac{1}{4}\pi = 1$. Next,

$$1 + \sin^2 x = \frac{(1+t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{1+6t^2+t^4}{(1+t^2)^2}.$$

Putting the pieces together, the factors of $1+t^2$ cancel completely:

$$I = \int_0^1 \frac{1-t^2}{1+t^2}\cdot\frac{(1+t^2)^2}{1+6t^2+t^4}\cdot\frac{2}{1+t^2}\,\mathrm{d}t = 2\int_0^1 \frac{1-t^2}{1+6t^2+t^4}\,\mathrm{d}t.$$

Dividing by 2 gives $\displaystyle\int_0^1 \frac{1-t^2}{1+6t^2+t^4}\,\mathrm{d}t = \tfrac{1}{2}I$, as required. (By (i)(a), its value is $\pi/8$.)

Part (ii)

? **Ask yourself** Where could the quartic $1+14t^2+t^4$ have come from?

In (i)(b) the $6t^2$ arose as $2t^2$ from squaring $1+t^2$, plus $4t^2 = (2t)^2$ from $\sin^2 x$. To get $14t^2$ we need an extra $12t^2 = 3(2t)^2$, that is, $3\sin^2 x$ in place of $\sin^2 x$. Indeed, with $t = \tan\frac{1}{2}x$,

$$1 + 3\sin^2 x = \frac{(1+t^2)^2 + 12t^2}{(1+t^2)^2} = \frac{1+14t^2+t^4}{(1+t^2)^2}.$$

So exactly the calculation of (i)(b), with $1+3\sin^2 x$ in place of $1+\sin^2 x$, gives

$$\int_0^{\frac{1}{2}\pi} \frac{\cos x}{1+3\sin^2 x}\,\mathrm{d}x = 2\int_0^1 \frac{1-t^2}{1+14t^2+t^4}\,\mathrm{d}t.$$

Evaluate the left-hand side as in (i)(a), with $u = \sin x$:

$$\int_0^1 \frac{1}{1+3u^2}\,\mathrm{d}u = \frac{1}{3}\int_0^1 \frac{1}{\frac{1}{3}+u^2}\,\mathrm{d}u = \frac{1}{3}\cdot\sqrt{3}\Big[\arctan\left(\sqrt{3}\,u\right)\Big]_0^1 = \frac{1}{\sqrt{3}}\cdot\frac{\pi}{3},$$

using the standard integral with $a = 1/\sqrt{3}$. Halving,

$$\int_0^1 \frac{1-t^2}{1+14t^2+t^4}\,\mathrm{d}t = \frac{\pi}{6\sqrt{3}} = \frac{\sqrt{3}\,\pi}{18}.$$

Checks

Simpson's rule gives $I \approx 0.785398$ (which is $\pi/4$), the integral in (i)(b) ≈ 0.392699 (which is $\pi/8$), and the integral in (ii) ≈ 0.302300, matching $\sqrt{3}\,\pi/18 \approx 0.302300$. A quick sanity check: both quartic integrands equal 1 at $t = 0$ and 0 at $t = 1$, and the one with $14t^2$ in the denominator is smaller throughout, so its integral should be smaller than $\pi/8 \approx 0.393$, as it is.

Common slips

- Forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}x$, which loses the factor of 2 in (i)(b).
- Writing $\int \frac{\mathrm{d}u}{1+3u^2} = \arctan(\sqrt{3}u)$ without the factor $\frac{1}{\sqrt{3}}$; take out the 3 first, or substitute $v = \sqrt{3}u$.
- Leaving the limits in x after substituting: $x = \frac{1}{2}\pi$ corresponds to $t = 1$.

Beyond STEP

There is a direct route too: dividing top and bottom by t^2 and putting $v = t + 1/t$ turns $\frac{1-t^2}{1+kt^2+t^4}$ into $-\frac{1}{v^2+k-2}$, another arctan integral. The general lesson, that rational functions of $\sin x$ and $\cos x$ always become rational functions of t under $t = \tan\frac{1}{2}x$ and so can always be integrated in principle, is the start of the systematic theory of integration in elementary terms. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2007, Question 4

Pure • Trigonometric Equations

STEP II 2007, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q4

The key idea

Expand everything with the compound-angle formulae and divide by $\cos A \cos B$: the equation becomes a relation in $t = \tan A$ and $u = \tan B$ with a tu term, a t term, a u term and a constant. Such an expression factorises into two linear brackets exactly when "completing the rectangle" leaves nothing over, and the leftover turns out to be a multiple of $\beta^2 + \gamma^2 - \alpha^2$. The three equations then all have $\alpha = 2$, $\beta = \sqrt{3}$, $\gamma = 1$, which satisfy the condition; the only work is choosing A and B correctly.

Solution

The factorisation

By the compound-angle formulae,

$$\begin{aligned}&\alpha\sin(A-B) + \beta\cos(A+B) - \gamma\sin(A+B)\\ &\quad= \alpha(\sin A\cos B - \cos A\sin B) + \beta(\cos A\cos B - \sin A\sin B)\\ &\qquad- \gamma(\sin A\cos B + \cos A\sin B).\end{aligned}$$

Since $\cos A\cos B \neq 0$ we may divide by it. Writing $t = \tan A$, $u = \tan B$, the equation is equivalent to $P(t,u) = 0$, where

$$P(t,u) = \alpha(t-u) + \beta(1-tu) - \gamma(t+u) = -\beta tu + (\alpha-\gamma)t - (\alpha+\gamma)u + \beta.$$

Since $\beta \neq 0$, put

$$m = -\frac{\alpha+\gamma}{\beta}, \qquad n = \frac{\alpha-\gamma}{\beta}.$$

Then $-\beta(t-m)(u-n) = -\beta tu + \beta n\,t + \beta m\,u - \beta mn = -\beta tu + (\alpha-\gamma)t - (\alpha+\gamma)u + \frac{\alpha^2-\gamma^2}{\beta}$, which matches P except in the constant term. Hence, for all t and u,

$$P(t,u) = -\beta(t-m)(u-n) + \frac{\beta^2+\gamma^2-\alpha^2}{\beta}. \qquad (*)$$

If. When $\alpha^2 = \beta^2 + \gamma^2$ the constant in $(*)$ vanishes, so (as $\beta \neq 0$) the equation is $(\tan A - m)(\tan B - n) = 0$, with m and n depending only on α, β, γ.

Only if. Suppose the equation reduces to $(t - m')(u - n') = 0$ for constants m', n', so that $P(t, u) = 0$ holds exactly when $t = m'$ or $u = n'$. Write $c = (\beta^2 + \gamma^2 - \alpha^2)/\beta$ and suppose $c \neq 0$. Every pair (m', u) is then a solution, so $P(m', u) = -\beta(m' - m)(u - n) + c$ is zero for every u. If $m' \neq m$ this is a non-constant linear function of u, which is not always zero; if $m' = m$ it equals $c \neq 0$, which is never zero. Either way we have a contradiction, so $c = 0$, that is, $\alpha^2 = \beta^2 + \gamma^2$. (If "reduces to" is read as "P is a constant multiple of $(t - m)(u - n)$", comparing the tu coefficients forces the multiple to be $-\beta$, and then $(*)$ again forces $c = 0$.)

Ask yourself What happens in the parts below if $\cos x = 0$, where we are not allowed to divide?

Multiply $(*)$ back through by $\cos A \cos B$. When $\alpha^2 = \beta^2 + \gamma^2$ we get

$$\alpha \sin(A - B) + \beta \cos(A + B) - \gamma \sin(A + B) = -\beta(\sin A - m \cos A)(\sin B - n \cos B), \qquad (**)$$

and both sides are now just combinations of sines and cosines, so this identity holds for *all* A and B (both sides are continuous and agree whenever $\cos A \cos B \neq 0$). Using $(**)$ we never need to treat $\cos A = 0$ separately.

In all three parts $\alpha = 2$, $\beta = \sqrt{3}$, $\gamma = 1$, and $\alpha^2 = 4 = \beta^2 + \gamma^2$. So

$$m = -\frac{3}{\sqrt{3}} = -\sqrt{3}, \qquad n = \frac{1}{\sqrt{3}},$$

and the equation is $(\sin A + \sqrt{3}\cos A)(\sin B - \frac{1}{\sqrt{3}}\cos B) = 0$. Note that $\sin A + \sqrt{3}\cos A = 0$ forces $\cos A \neq 0$ (otherwise $\sin A = 0$ too), so it is the same as $\tan A = -\sqrt{3}$; likewise the second bracket vanishes exactly when $\tan B = \frac{1}{\sqrt{3}}$.

Part (i)

Take $A = x$ and $B = \frac{1}{4}\pi$, so that $A - B = x - \frac{1}{4}\pi$ and $A + B = x + \frac{1}{4}\pi$. Since $\tan \frac{1}{4}\pi = 1 \neq \frac{1}{\sqrt{3}}$, the second bracket is non-zero, so we need $\tan x = -\sqrt{3}$:

$$x = \tfrac{2}{3}\pi \quad \text{or} \quad x = \tfrac{5}{3}\pi.$$

Part (ii)

Now $A = x$ and $B = \frac{1}{6}\pi$. But $\tan \frac{1}{6}\pi = \frac{1}{\sqrt{3}} = n$, so the second bracket is $\sin \frac{1}{6}\pi - \frac{1}{\sqrt{3}}\cos \frac{1}{6}\pi = \frac{1}{2} - \frac{1}{2} = 0$. By $(**)$ the left-hand side minus the right-hand side is identically zero: the equation is an identity, and *every* x with $0 \leq x < 2\pi$ satisfies it.

Part (iii)

We need $A - B = x + \frac{1}{3}\pi$ and $A + B = 3x$. Adding and subtracting,

$$A = 2x + \tfrac{1}{6}\pi, \qquad B = x - \tfrac{1}{6}\pi.$$

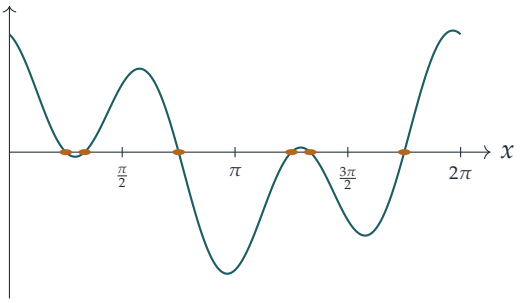
So the solutions are given by $\tan(2x + \frac{1}{6}\pi) = -\sqrt{3}$ or $\tan(x - \frac{1}{6}\pi) = \frac{1}{\sqrt{3}}$.

For the first, $2x + \frac{1}{6}\pi = \frac{2}{3}\pi + k\pi$, so $x = \frac{1}{4}\pi + \frac{1}{2}k\pi$ for integer k. In $[0, 2\pi)$ this gives $x = \frac{1}{4}\pi,\ \frac{3}{4}\pi,\ \frac{5}{4}\pi,\ \frac{7}{4}\pi$.

For the second, $x - \frac{1}{6}\pi = \frac{1}{6}\pi + k\pi$, so $x = \frac{1}{3}\pi + k\pi$, giving $x = \frac{1}{3}\pi,\ \frac{4}{3}\pi$.

These six values are distinct, so the solutions are

$$x = \tfrac{1}{4}\pi,\ \tfrac{1}{3}\pi,\ \tfrac{3}{4}\pi,\ \tfrac{5}{4}\pi,\ \tfrac{4}{3}\pi,\ \tfrac{7}{4}\pi.$$



The graph of the left-hand side minus the right-hand side of (iii) crosses zero exactly at the six marked points.

Checks

- In (i), at $x = \frac{2}{3}\pi$: $2\sin\frac{5}{12}\pi \approx 1.932$, $\sqrt{3}\cos\frac{11}{12}\pi \approx -1.673$, and $\sin\frac{11}{12}\pi \approx 0.259$; indeed $1.932 - 1.673 = 0.259$.
- In (ii), try $x = 0$: $2\sin(-\frac{1}{6}\pi) + \sqrt{3}\cos\frac{1}{6}\pi = -1 + \frac{3}{2} = \frac{1}{2} = \sin\frac{1}{6}\pi$. Any other x works too.
- In (iii), at $x = \frac{1}{4}\pi$: $2\sin\frac{7}{12}\pi \approx 1.932$ and $\sin\frac{3}{4}\pi - \sqrt{3}\cos\frac{3}{4}\pi \approx 0.707 + 1.225 = 1.932$. A numerical root search on $[0, 2\pi)$ finds exactly the six roots above.

Common slips

- Taking $A = x + \frac{1}{3}\pi$ and B from $3x$ in (iii): you must solve $A - B = x + \frac{1}{3}\pi$, $A + B = 3x$ simultaneously.
- Losing solutions in (iii) by stopping at $k = 0, 1$: the period of $\tan(2x + \frac{1}{6}\pi)$ is $\frac{1}{2}\pi$, so there are four solutions from that bracket.
- Missing that (ii) is an identity and hunting for particular solutions.

Beyond STEP

The expression $-\beta tu + (\alpha - \gamma)t - (\alpha + \gamma)u + \beta$ is a degenerate case of a conic: its zero set is a rectangular hyperbola unless a determinant-like quantity vanishes, when it splits into two lines. Deciding when a quadratic form factorises is a first step in classifying conics with matrices and eigenvalues, a topic in first-year linear algebra; see Strang, *Introduction to Linear Algebra*. Siklos, *Advanced Problems in Mathematics*, has more trigonometric equations in this spirit.

WORKED SOLUTION

STEP II 2007, Question 5

Pure • Iterated Functions

STEP II 2007, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q5

The key idea

The shape $\dfrac{x+\sqrt{3}}{1-\sqrt{3}\,x}$ is the compound-angle formula for $\tan(A+B)$ with $\tan B = \sqrt{3}$, that is $B = \frac{\pi}{3}$. So f simply adds $\frac{\pi}{3}$ to an angle, and since tan has period π, three steps bring us back where we started. In part (iii) the same trick works with sin and $\frac{\pi}{6}$, but there is a catch: $\sqrt{1-t^2}$ is a cosine only when the angle lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and that changes the answer.

Solution

Part (i)

Put $y = f(x) = \dfrac{x+\sqrt{3}}{1-\sqrt{3}\,x}$. Then $f^2(x) = \dfrac{y+\sqrt{3}}{1-\sqrt{3}\,y}$, and multiplying top and bottom by $1-\sqrt{3}\,x$,

$$f^2(x) = \frac{(x+\sqrt{3})+\sqrt{3}(1-\sqrt{3}\,x)}{(1-\sqrt{3}\,x)-\sqrt{3}(x+\sqrt{3})} = \frac{2\sqrt{3}-2x}{-2-2\sqrt{3}\,x} = \frac{x-\sqrt{3}}{1+\sqrt{3}\,x}.$$

Now apply f once more, this time with $y = f^2(x)$, multiplying top and bottom by $1+\sqrt{3}\,x$:

$$f^3(x) = \frac{(x-\sqrt{3})+\sqrt{3}(1+\sqrt{3}\,x)}{(1+\sqrt{3}\,x)-\sqrt{3}(x-\sqrt{3})} = \frac{4x}{4} = x.$$

So f^3 is the identity. (The two excluded points are exactly what keeps every step defined: $f(x)$ needs $x \neq 1/\sqrt{3}$, $f^2(x)$ needs $x \neq -1/\sqrt{3}$, and f maps $\mathbb{R} \setminus \{\pm 1/\sqrt{3}\}$ onto itself, so we can keep going.)

Since $2007 = 3 \times 669$,

$$f^{2007}(x) = \left(f^3\right)^{669}(x) = x.$$

Part (ii)

Let $x = \tan\theta$. For any angle α with $\tan\alpha$ defined and $1 - \sqrt{3}\tan\alpha \neq 0$, the compound-angle formula with $\tan\frac{\pi}{3} = \sqrt{3}$ gives

$$f(\tan\alpha) = \frac{\tan\alpha + \tan\frac{\pi}{3}}{1 - \tan\alpha\tan\frac{\pi}{3}} = \tan\big(\alpha + \tfrac{\pi}{3}\big).$$

With $\alpha = \theta$ this is the case $n = 1$. If $f^k(x) = \tan(\theta + \frac{1}{3}k\pi)$, then applying the formula with $\alpha = \theta + \frac{1}{3}k\pi$,

$$f^{k+1}(x) = f\big(\tan(\theta + \tfrac{1}{3}k\pi)\big) = \tan\big(\theta + \tfrac{1}{3}(k+1)\pi\big),$$

so by induction $f^n(x) = \tan(\theta + \frac{1}{3}n\pi)$ for every positive integer n.

Nothing breaks along the way: the formula fails only if some angle $\theta + \frac{1}{3}k\pi$ is congruent to $\frac{\pi}{2}$ or $\frac{\pi}{6}$ modulo π, and that would force $\theta = \pm\frac{\pi}{6}$ or $\pm\frac{\pi}{2}$, that is, $x = \pm 1/\sqrt{3}$ or x infinite, which are excluded. This also explains part (i): adding $\frac{\pi}{3}$ three times adds π, the period of tan.

Part (iii)

Since $|t| \leq 1$ we can write $t = \sin\varphi$ with $\varphi = \arcsin t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

? **Ask yourself** Is $\sqrt{1 - \sin^2\alpha}$ always equal to $\cos\alpha$?

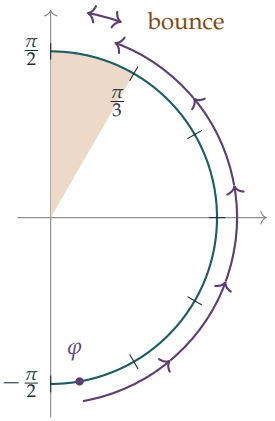
Only when $\cos\alpha \geq 0$. For an angle $\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ it is fine, so

$$g(\sin\alpha) = \cos\tfrac{\pi}{6}\sin\alpha + \sin\tfrac{\pi}{6}\cos\alpha = \sin\big(\alpha + \tfrac{\pi}{6}\big). \qquad (*)$$

To apply $(*)$ again we must rewrite the answer as sin of an angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. There are two cases.

- If $\alpha \leq \frac{\pi}{3}$, then $\alpha + \frac{\pi}{6} \leq \frac{\pi}{2}$ is already in range.
- If $\frac{\pi}{3} \leq \alpha \leq \frac{\pi}{2}$, then $\alpha + \frac{\pi}{6}$ may overshoot $\frac{\pi}{2}$, but $\sin(\alpha + \frac{\pi}{6}) = \sin\big(\pi - \alpha - \frac{\pi}{6}\big) = \sin\big(\frac{5\pi}{6} - \alpha\big)$, and $\frac{5\pi}{6} - \alpha \in [\frac{\pi}{3}, \frac{\pi}{2}]$.

(At $\alpha = \frac{\pi}{3}$ both give $\frac{\pi}{2}$.) So the in-range angle *climbs* by $\frac{\pi}{6}$ until it reaches $[\frac{\pi}{3}, \frac{\pi}{2}]$, and from then on it is *reflected* by $\alpha \mapsto \frac{5\pi}{6} - \alpha$, which is undone by doing it again: the angle bounces between two values.



Each application of g moves the angle $\frac{\pi}{6}$ round the right half of the unit circle; once in the shaded zone $[\frac{\pi}{3}, \frac{\pi}{2}]$ it flips back and forth.

Now let m be the number of climbs needed: the least integer $m \geq 0$ with $\varphi + \frac{m\pi}{6} \geq \frac{\pi}{3}$. Since $\varphi \geq -\frac{\pi}{2}$, $m \leq 5$, and in terms of t:

t	$[\frac{\sqrt{3}}{2}, 1]$	$[\frac{1}{2}, \frac{\sqrt{3}}{2})$	$[0, \frac{1}{2})$	$[-\frac{1}{2}, 0)$	$[-\frac{\sqrt{3}}{2}, -\frac{1}{2})$	$[-1, -\frac{\sqrt{3}}{2})$
m	0	1	2	3	4	5

Write $g^n(t) = \sin\varphi_n$ with $\varphi_n \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\varphi_0 = \varphi$. For $n \leq m$ every earlier angle was below $\frac{\pi}{3}$, so $\varphi_n = \varphi + \frac{n\pi}{6}$. Then $\varphi_m \in [\frac{\pi}{3}, \frac{\pi}{2}]$, so $\varphi_{m+1} = \frac{5\pi}{6} - \varphi_m$, $\varphi_{m+2} = \varphi_m$, and so on, alternating. Since $\sin\varphi_{m+1} = \sin(\varphi_m + \frac{\pi}{6}) = \sin\big(\varphi + \frac{(m+1)\pi}{6}\big)$, we get, with $\varphi = \arcsin t$ and m from the table,

$$g^n(t) = \begin{cases} \sin\big(\varphi + \frac{n\pi}{6}\big), & n \leq m+1, \\ \sin\big(\varphi + \frac{m\pi}{6}\big), & n > m+1,\ n-m \text{ even}, \\ \sin\big(\varphi + \frac{(m+1)\pi}{6}\big), & n > m+1,\ n-m \text{ odd}. \end{cases}$$

In particular, for every t and every $n \geq 5$, $g^{n+2}(t) = g^n(t)$.

For example, $t = 0$ has $\varphi = 0$, $m = 2$, and the orbit is $0 \mapsto \frac{1}{2} \mapsto \frac{\sqrt{3}}{2} \mapsto 1 \mapsto \frac{\sqrt{3}}{2} \mapsto 1 \mapsto \cdots$.

Checks

- At $x = 1$: $f(1) = \frac{1+\sqrt{3}}{1-\sqrt{3}} = -2 - \sqrt{3}$, $f^2(1) = \frac{1-\sqrt{3}}{1+\sqrt{3}} = \sqrt{3} - 2$, and $f(\sqrt{3} - 2) = 1$, matching $\tan\frac{\pi}{4}$, $\tan\frac{7\pi}{12}$, $\tan\frac{11\pi}{12}$.
- Directly, $g(\frac{\sqrt{3}}{2}) = \frac{3}{4} + \frac{1}{4} = 1$ and $g(1) = \frac{\sqrt{3}}{2}$: the bounce is real. The naive guess $\sin\frac{5\pi}{6} = \frac{1}{2}$ for $g^5(0)$ is wrong; iterating gives 1.
- Numerically, iterating g nineteen times from 5000 random $t \in [-1, 1]$ agreed with the boxed formula to within 2×10^{-12}.

Common slips

- Answering (iii) with $\sin(\varphi + \frac{n\pi}{6})$ for all n. That forgets that $\sqrt{1-t^2} \geq 0$, so the angle can never pass $\frac{\pi}{2}$.
- Arithmetic slips in f^2: multiply numerator and denominator through by $1 - \sqrt{3}\,x$ before simplifying.

Beyond STEP

A map $x \mapsto \frac{ax+b}{cx+d}$ is a Möbius transformation, and composing two of them corresponds to multiplying their matrices $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Here f has matrix $2\left(\begin{smallmatrix} \cos\frac{\pi}{3} & \sin\frac{\pi}{3} \\ -\sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{smallmatrix}\right)$, a scaled rotation by $\frac{\pi}{3}$, whose cube is a multiple of the identity, which is why $f^3(x) = x$. Part (iii) is a small taste of dynamical systems: points that are *eventually periodic* rather than periodic. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of Möbius transformations.

WORKED SOLUTION

STEP II 2007, Question 6

Pure • First-Order Differential Equations

STEP II 2007, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q6

The key idea

Part (i) hands us two functions whose derivatives both have $\sqrt{3+x^2}$ in the denominator; a suitable combination of them has derivative a multiple of $\sqrt{3+x^2}$ itself, so the integral comes for free. In part (ii) the differential equation is a quadratic in $\frac{dy}{dx}$. Solving that quadratic gives two ordinary first-order equations, and each one needs exactly the integral from part (i).

Solution

Part (i)

Write $s = \sqrt{3+x^2}$, so that $s > 0$ for every real x and $\dfrac{ds}{dx} = \dfrac{x}{s}$. Note also that $x + s > 0$, since $s > |x|$, so the logarithm is defined for all x.

By the chain rule,

$$\frac{d}{dx}\ln(x+s) = \frac{1+\dfrac{x}{s}}{x+s} = \frac{\dfrac{s+x}{s}}{x+s} = \frac{1}{\sqrt{3+x^2}}.$$

By the product rule,

$$\frac{d}{dx}(xs) = s + x\cdot\frac{x}{s} = \frac{s^2+x^2}{s} = \frac{3+2x^2}{\sqrt{3+x^2}}.$$

? **Ask yourself** How do these two derivatives give $\int \sqrt{3+x^2}\,dx$?

We want a numerator of $3+x^2$ over the denominator s, since $\dfrac{3+x^2}{s} = s$. The second

derivative has numerator $3+2x^2$; adding 3 times the first derivative gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(xs+3\ln(x+s)\Big)=\frac{3+2x^2+3}{s}=\frac{2(3+x^2)}{s}=2\sqrt{3+x^2}.$$

Therefore

$$\int\sqrt{3+x^2}\,\mathrm{d}x=\tfrac{1}{2}x\sqrt{3+x^2}+\tfrac{3}{2}\ln\big(x+\sqrt{3+x^2}\,\big)+C.$$

Part (ii)

Write $p=\dfrac{\mathrm{d}y}{\mathrm{d}x}$. The equation is the quadratic $3p^2+2xp-1=0$ in p, so

$$p=\frac{-2x\pm\sqrt{4x^2+12}}{6}=\frac{-x\pm\sqrt{3+x^2}}{3}.$$

The discriminant $4x^2+12$ is always positive, so there are two different real roots at every x. Their difference, $\frac{2}{3}\sqrt{3+x^2}$, is never zero, so a solution can't switch from one root to the other part-way along. (A derivative has the intermediate value property, so it can't jump between two values that are always at least $\frac{2}{3}\sqrt{3}$ apart.) So every solution satisfies one of

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{x}{3}+\frac{1}{3}\sqrt{3+x^2}\qquad\text{or}\qquad\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{x}{3}-\frac{1}{3}\sqrt{3+x^2}$$

for all x. Integrating with part (i),

$$y=-\frac{x^2}{6}\pm\Big(\tfrac{1}{6}x\sqrt{3+x^2}+\tfrac{1}{2}\ln\big(x+\sqrt{3+x^2}\,\big)\Big)+C.$$

At $x=1$ we have $\sqrt{3+x^2}=2$ and $\ln(x+\sqrt{3+x^2})=\ln 3$.

Taking the + sign: $y(1)=0$ gives $0=-\frac{1}{6}+\frac{1}{3}+\frac{1}{2}\ln 3+C$, so $C=-\frac{1}{6}-\frac{1}{2}\ln 3$. Combining the logarithms,

$$y=\frac{x\sqrt{3+x^2}-x^2-1}{6}+\frac{1}{2}\ln\left(\frac{x+\sqrt{3+x^2}}{3}\right).$$

Taking the − sign: $y(1)=0$ gives $0=-\frac{1}{6}-\frac{1}{3}-\frac{1}{2}\ln 3+C$, so $C=\frac{1}{2}+\frac{1}{2}\ln 3$, and

$$y=\frac{3-x^2-x\sqrt{3+x^2}}{6}-\frac{1}{2}\ln\left(\frac{x+\sqrt{3+x^2}}{3}\right).$$

These are the two solutions through $(1,0)$, and both are valid for all real x.

Checks

- Put $x=1$ in each answer: the bracket is $\frac{2-1-1}{6}=0$ (or $\frac{3-1-2}{6}=0$), and $\ln\frac{1+2}{3}=0$. So $y(1)=0$ in both.

- The two roots for p add up to $-\frac{2x}{3}$, so the two solutions should add up to $\frac{1-x^2}{3}$. The square roots and logarithms do cancel, leaving exactly $\frac{2-2x^2}{6} = \frac{1-x^2}{3}$.
- Numerically, Simpson's rule gives $\int_{-1}^{2} \sqrt{3+x^2}\,\mathrm{d}x \approx 5.949681$, matching the formula in part (i). Stepping each branch of the equation from $(1, 0)$ to $x = 4$ by Runge–Kutta gives $y \approx 0.584957$ and $y \approx -5.584957$, which match the two formulae. (Their sum is $-5 = \frac{1-16}{3}$.)

Common slips

- Treating $3p^2 + 2xp = 1$ as linear, or trying to separate variables, instead of solving it as a quadratic in p.
- Using one constant C for both branches. Each sign needs its own constant, fixed by $y(1) = 0$.
- Forgetting the factor $\frac{1}{3}$ in front of $\sqrt{3+x^2}$, so the part (i) integral is used with the wrong multiple.

Beyond STEP

An equation like $F\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$ that is not linear in the derivative is *implicit*. Through a point there can be several solutions, one for each root of the equation in p. Where two roots meet, *singular solutions* can appear (the envelope of a family of curves, as in Clairaut's equation). Here the roots never meet, so there are exactly two solutions. The $\ln\left(x + \sqrt{3+x^2}\right)$ in part (i) is $\operatorname{arsinh}(x/\sqrt{3})$ plus a constant, and the integral is the standard $x = \sqrt{3}\sinh u$ substitution in disguise. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and existence and uniqueness of their solutions.

WORKED SOLUTION

STEP II 2007, Question 7

Pure • Jensen's Inequality

STEP II 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q7

This question has a full guided treatment in *Beyond STEP*, Section 3.3, *The Power of Convexity*: https://beyondstep.keithmcnulty.org/ch03/convexity.html

The key idea

Jensen's inequality says that for a concave function the average of the values is at most the value at the average. Each part comes down to spotting which concave function to use and which numbers to average. For the triangle, the angles have a fixed sum, so their average is known. For the AM–GM inequality, the logarithm turns a product into a sum. Parts (a) and (b) then need the right list of numbers to feed into AM–GM, including some constants.

Solution

Concavity

If $f(x) = \sin x$ then $f''(x) = -\sin x$. For $0 < x < \pi$ we have $\sin x > 0$, so $f''(x) < 0$ and $\sin x$ is concave there.

If $f(x) = \ln x$ then $f'(x) = 1/x$ and $f''(x) = -1/x^2$. This is negative for every $x > 0$, so $\ln x$ is concave for $x > 0$.

Part (i)

The angles of a triangle satisfy $A, B, C > 0$ and $A + B + C = \pi$, so each lies in $(0, \pi)$, where sin is concave. Jensen's inequality with $n = 3$ gives

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin\left(\frac{A + B + C}{3}\right) = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Multiplying by 3,

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2},$$

as required. Equality holds exactly when $A = B = C = \pi/3$, the equilateral triangle.

Part (ii)

Take $f(x) = \ln x$, which is concave on $x > 0$, and put $x_k = t_k > 0$. Jensen's inequality gives

$$\frac{1}{n}\sum_{k=1}^{n} \ln t_k \leq \ln\left(\frac{t_1 + t_2 + \cdots + t_n}{n}\right).$$

The left-hand side is $\frac{1}{n}\ln(t_1 t_2 \cdots t_n) = \ln \sqrt[n]{t_1 t_2 \cdots t_n}$. Since e^x is increasing, we may exponentiate both sides without changing the direction of the inequality:

$$\sqrt[n]{t_1 t_2 \cdots t_n} \leq \frac{t_1 + t_2 + \cdots + t_n}{n},$$

with equality if and only if $t_1 = t_2 = \cdots = t_n$. (For $n = 1$ this is trivially an equality, which is consistent.)

(a)

? **Ask yourself** Which four numbers should we average?

The left-hand side has four terms, so use $n = 4$. We need the geometric mean to produce xyz, and x^4, y^4, z^4 give $\sqrt[4]{x^4y^4z^4} = xyz$. The constant 16 is the fourth term, and $\sqrt[4]{16} = 2$. So take $t_1 = x^4$, $t_2 = y^4$, $t_3 = z^4$, $t_4 = 16$, all positive:

$$\frac{x^4 + y^4 + z^4 + 16}{4} \geq \sqrt[4]{16\,x^4y^4z^4} = 2xyz,$$

using $x, y, z > 0$ to take the fourth root. Multiplying by 4 gives

$$x^4 + y^4 + z^4 + 16 \geq 8xyz,$$

as required. Equality needs $x^4 = y^4 = z^4 = 16$, that is $x = y = z = 2$.

(b) Now the powers are fifth powers, so we want $n = 5$ with x^5, y^5, z^5 among the numbers. To get exactly xyz from the fifth root, the other two numbers must multiply to 1. The natural choice is $t_4 = t_5 = 1$:

$$\frac{x^5 + y^5 + z^5 + 1 + 1}{5} \geq \sqrt[5]{x^5y^5z^5 \cdot 1 \cdot 1} = xyz.$$

Rearranging,

$$x^5 + y^5 + z^5 - 5xyz \geq -2$$

for all positive x, y, z.

Ask yourself Does this show that the minimum is -2?

Not on its own: a lower bound is the minimum only if it is attained. Equality in AM–GM holds when all five numbers are equal, $x^5 = y^5 = z^5 = 1$, so $x = y = z = 1$. There the expression is $1 + 1 + 1 - 5 = -2$. Hence the minimum value is

$$\boxed{-2}, \quad \text{attained at } x = y = z = 1.$$

Checks

- In (i), a right-angled isosceles triangle gives $1 + 2 \cdot \frac{\sqrt{2}}{2} \approx 2.414$, below $3\sqrt{3}/2 \approx 2.598$. Random sampling of 200 000 triangles never exceeded 2.598.
- In (a), at $x = y = z = 2$ both sides equal 64.
- In (b), a numerical minimisation from random starting points always converged to $(1, 1, 1)$ with value -2. Also, $x = y = z = 1.1$ gives $3(1.61051) - 5(1.331) \approx -1.82 > -2$.

Common slips

- Forgetting that exponentiating preserves the inequality only because e^x is increasing; it deserves a word.
- In (b), stopping at "≥ -2" without exhibiting values of x, y, z where -2 is reached.
- Using AM–GM with $n = 3$ on x^5, y^5, z^5, which gives $x^5 + y^5 + z^5 \geq 3(xyz)^{5/3}$ and doesn't compare cleanly with $5xyz$.

Beyond STEP

Jensen's inequality is a statement about convex (or concave) functions, and its natural home is analysis and probability: in the language of expectation it reads $\mathrm{E}[f(X)] \leq f(\mathrm{E}[X])$ for concave f. The trick in (a) and (b), padding with constants so that the geometric mean comes out right, is the starting point of the theory of weighted AM–GM and Hölder's inequality. For more on inequalities and convexity, try Bartle and Sherbert, *Introduction to Real Analysis*; for the probabilistic version, Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 2007, Question 8

Pure • Ceva's Theorem with Vectors

STEP II 2007, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q8

The key idea

Every point of the plane can be written as $\beta\mathbf{b} + \gamma\mathbf{c}$, and part (i) tells us exactly when such a point lies on BC: the coefficients add up to 1. So to find where a line meets BC (or CA, or AB) we write a general point of the line in terms of $\mathbf{b}$ and $\mathbf{c}$ and impose the right condition on the coefficients. Each ratio then comes out in terms of β and γ, and the product collapses.

Solution

Throughout, a ratio such as $BD : DC$ means a *signed* ratio: $\overrightarrow{BD} = k\overrightarrow{DC}$ gives $BD/DC = k$, which is negative when D lies outside the segment BC. Because A, B, C are not collinear, $\mathbf{b}$ and $\mathbf{c}$ are non-parallel and non-zero, so we may compare coefficients: $p\mathbf{b} + q\mathbf{c} = p'\mathbf{b} + q'\mathbf{c}$ forces $p = p'$ and $q = q'$.

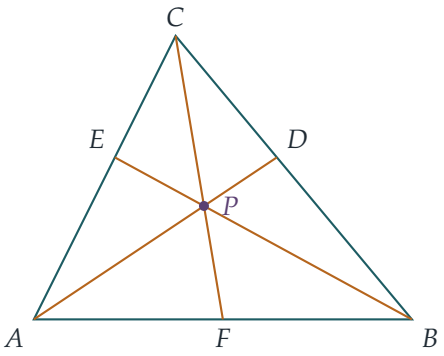
Part (i)

If $s + t = 1$ then

$$s\mathbf{b} + t\mathbf{c} = (1 - t)\mathbf{b} + t\mathbf{c} = \mathbf{b} + t(\mathbf{c} - \mathbf{b}),$$

which is the vector equation of the line through B in the direction $\overrightarrow{BC}$, with t as parameter. As t runs over all real numbers, X covers the whole of it. So the locus of X is the (infinite) straight line through B and C; the segment BC is the part with $0 \leq t \leq 1$.

Conversely, a point $\beta\mathbf{b} + \gamma\mathbf{c}$ lies on line BC if and only if $\beta + \gamma = 1$, by comparing coefficients. This is the test we use from now on. In particular $\beta + \gamma \neq 1$ says that P is not on BC, and $\beta, \gamma \neq 0$ say that P is not on CA or AB.



Part (ii)

A general point of line AP is $\lambda(\beta\mathbf{b}+\gamma\mathbf{c}) = \lambda\beta\,\mathbf{b}+\lambda\gamma\,\mathbf{c}$. By part (i) it lies on BC exactly when $\lambda\beta+\lambda\gamma = 1$, that is,

$$\lambda = \frac{1}{\beta+\gamma}.$$

(This needs $\beta+\gamma \neq 0$. If $\beta+\gamma = 0$ then $\overrightarrow{AP} = \beta(\mathbf{b}-\mathbf{c})$ is parallel to BC and D does not exist, so the question implicitly assumes $\beta+\gamma \neq 0$.) Hence

$$\mathbf{d} = \frac{\beta\mathbf{b}+\gamma\mathbf{c}}{\beta+\gamma}.$$

Now

$$\overrightarrow{BD} = \mathbf{d}-\mathbf{b} = \frac{\beta\mathbf{b}+\gamma\mathbf{c}-(\beta+\gamma)\mathbf{b}}{\beta+\gamma} = \frac{\gamma}{\beta+\gamma}(\mathbf{c}-\mathbf{b}),$$
$$\overrightarrow{DC} = \mathbf{c}-\mathbf{d} = \frac{(\beta+\gamma)\mathbf{c}-\beta\mathbf{b}-\gamma\mathbf{c}}{\beta+\gamma} = \frac{\beta}{\beta+\gamma}(\mathbf{c}-\mathbf{b}).$$

Both are multiples of the same vector $\mathbf{c}-\mathbf{b}$, so $\beta\overrightarrow{BD} = \gamma\overrightarrow{DC}$, and since $\beta \neq 0$,

$$BD : DC = \gamma : \beta,$$

as required.

Part (iii)

We find E and F in the same way, but now the target lines are CA (multiples of $\mathbf{c}$) and AB (multiples of $\mathbf{b}$).

? **Ask yourself** How do we tell when a point $p\mathbf{b}+q\mathbf{c}$ is on line CA?

Line CA passes through the origin A in the direction $\mathbf{c}$, so its points are exactly the multiples of $\mathbf{c}$: the coefficient of $\mathbf{b}$ must be zero. Similarly, a point is on AB exactly when its $\mathbf{c}$-coefficient is zero.

Finding E. A general point of line BP is

$$\mathbf{b} + \mu(\beta\mathbf{b} + \gamma\mathbf{c} - \mathbf{b}) = \big(1 - \mu(1-\beta)\big)\mathbf{b} + \mu\gamma\,\mathbf{c}.$$

Setting the $\mathbf{b}$-coefficient to zero gives $\mu = 1/(1-\beta)$ (we need $\beta \neq 1$; if $\beta = 1$ then $\overrightarrow{BP} = \gamma\mathbf{c}$ is parallel to CA and E does not exist). So

$$\mathbf{e} = e\,\mathbf{c}, \qquad e = \frac{\gamma}{1-\beta}.$$

Then $\overrightarrow{CE} = (e-1)\mathbf{c}$ and $\overrightarrow{EA} = -e\,\mathbf{c}$, and $e \neq 0$ because $\gamma \neq 0$, so

$$\frac{CE}{EA} = \frac{e-1}{-e} = \frac{1-e}{e} = \frac{(1-\beta-\gamma)/(1-\beta)}{\gamma/(1-\beta)} = \frac{1-\beta-\gamma}{\gamma}.$$

Finding F. Swapping the roles of B and C (that is, of $\mathbf{b} \leftrightarrow \mathbf{c}$ and $\beta \leftrightarrow \gamma$), line CP meets AB at $\mathbf{f} = f\,\mathbf{b}$ with $f = \beta/(1-\gamma)$, assuming $\gamma \neq 1$. Then $\overrightarrow{AF} = f\,\mathbf{b}$ and $\overrightarrow{FB} = (1-f)\mathbf{b}$, where $1 - f = (1-\beta-\gamma)/(1-\gamma) \neq 0$ because $\beta + \gamma \neq 1$. So

$$\frac{AF}{FB} = \frac{f}{1-f} = \frac{\beta}{1-\beta-\gamma}.$$

The product. Using part (ii) for the middle factor,

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = \frac{\beta}{1-\beta-\gamma} \times \frac{\gamma}{\beta} \times \frac{1-\beta-\gamma}{\gamma} = 1,$$

where every cancelled factor is non-zero because $\beta \neq 0$, $\gamma \neq 0$ and $\beta + \gamma \neq 1$. This is the required result. (It holds for signed ratios, and so also for the lengths themselves.)

Checks

- Centroid: $\beta = \gamma = \frac{1}{3}$ makes P the centroid. Then $BD : DC = 1 : 1$, $CE/EA = \frac{1/3}{1/3} = 1$ and $AF/FB = 1$, so all three cevians are medians, as they should be.
- With $A = (0,0)$, $B = (4,0)$, $C = (1.5,3)$, $\beta = 0.3$, $\gamma = 0.4$ (the diagram), we get $D = (2.571, 1.714)$, $E = \frac{4}{7}\mathbf{c}$ and $F = (2,0)$, giving $\frac{AF}{FB} = 1$, $\frac{BD}{DC} = \frac{4}{3}$, $\frac{CE}{EA} = \frac{3}{4}$, with product 1.
- We checked the formulas for D, E, F, all three signed ratios and the product exactly (with fractions) on over 1700 random triangles and random β, γ, including points P outside the triangle.

Common slips

- Describing the locus in (i) as the segment BC: s and t are not restricted to be positive, so it is the whole line.

- Getting the ratio upside down: $\overrightarrow{BD}$ is proportional to γ, the coefficient of $\mathbf{c}$, so D is pulled towards C when γ is large.
- Forgetting that the parameters along BP and CP are measured from B and C, not from the origin.

Beyond STEP

Part (iii) is Ceva's theorem, and $(1-\beta-\gamma, \beta, \gamma)$ are the *barycentric coordinates* of P with respect to triangle ABC: they are proportional to the areas of triangles PBC, PCA, PAB, and the three ratios are just quotients of them. Barycentric coordinates are the natural setting for affine geometry, where lines and ratios along lines make sense but lengths and angles do not; they reappear in computer graphics and in the finite-element method. Strang's *Introduction to Linear Algebra* is a good place to see coordinates relative to a basis treated systematically.

WORKED SOLUTION

STEP II 2007, Question 9

Mechanics • Oblique Impact with Restitution

STEP II 2007, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q9

The key idea

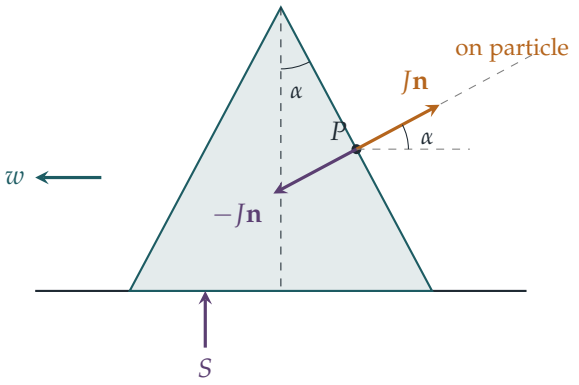
Everything is smooth, so the only impulse between the particle and the cone acts along the common normal at the point of impact, and the table can only push vertically. That tells us two things at once: the particle's velocity *along* the surface is unchanged, and horizontal momentum of the particle and cone together is conserved. Combine that with Newton's law of restitution along the normal and the cone's speed drops out. Part (ii) is part (i) applied to the normal component of the falling particle's velocity.

Solution

Model the particle as a point and the cone as a rigid body that (as we are told) neither rotates nor moves vertically, so just after impact it slides horizontally with some speed w. Work in the vertical plane through the axis of the cone and the point of impact P. Since the generator through P makes angle α with the vertical axis, the outward normal at P makes angle α with the *horizontal*. Take unit vectors

$$\mathbf{n} = (\cos\alpha,\ \sin\alpha) \text{ (outward normal)}, \qquad \mathbf{t} = (\sin\alpha,\ -\cos\alpha) \text{ (down the slope)},$$

with the first coordinate horizontal (away from the axis) and the second vertically upwards.



Let the impulse on the particle be $J\mathbf{n}$ ($J > 0$), so the cone receives $-J\mathbf{n}$, whose horizontal part is $J\cos\alpha$, directed away from P. The table supplies a vertical impulse $S = J\sin\alpha$ upwards, which is exactly what is needed to cancel the downward part of $-J\mathbf{n}$; that is why the cone has no vertical velocity. So

$$Mw = J\cos\alpha, \tag{1}$$

with the cone moving horizontally away from P.

Part (i)

The particle arrives with velocity $-u\mathbf{n}$, which has no component along $\mathbf{t}$. The surfaces are smooth, so the impulse on the particle is purely along $\mathbf{n}$ and cannot change its component along $\mathbf{t}$. That component is therefore still zero after the impact, and the particle's velocity is a multiple of $\mathbf{n}$: it moves along the normal line at P (outwards, or still inwards but more slowly if the cone runs away from it fast enough).

After impact the particle's velocity is $(-u + J/m)\,\mathbf{n}$. The cone moves with speed w horizontally away from P (to the left in the diagram), so its velocity is $(-w, 0)$, whose component along $\mathbf{n}$ is $-w\cos\alpha$.

> **Ask yourself** Newton's law of restitution compares relative velocities along which direction, and what is the cone's contribution?

Restitution acts along the common normal $\mathbf{n}$. Before impact the speed of approach along $\mathbf{n}$ is u (the cone is at rest). After impact the speed of separation is the particle's $\mathbf{n}$-component minus the cone's:

$$\left(-u + \frac{J}{m}\right) - (-w\cos\alpha) = eu \quad\Longrightarrow\quad J = m\big((1+e)u - w\cos\alpha\big).$$

Substituting into (1),

$$Mw = m\cos\alpha\,\big((1+e)u - w\cos\alpha\big) \quad\Longrightarrow\quad \boxed{w = \frac{m(1+e)u\cos\alpha}{M + m\cos^2\alpha}}.$$

Equivalently, horizontal momentum is conserved (the table's impulse is vertical): $m(-u\cos\alpha) = m\big(-u + \frac{J}{m}\big)\cos\alpha - Mw$, which is (1) again.

Part (ii)

Now the particle arrives with velocity $(0, -u)$. Resolve it:

$$(0,-u)\cdot\mathbf{n} = -u\sin\alpha, \qquad (0,-u)\cdot\mathbf{t} = u\cos\alpha.$$

The tangential part $u\cos\alpha$ is unaffected by the impact (smooth surfaces again) and plays no further role: the impulse $J\mathbf{n}$, equation (1) and the restitution equation involve only normal components. So the working of part (i) applies unchanged with the normal approach speed u replaced by $u\sin\alpha$:

$$w = \frac{m(1+e)(u\sin\alpha)\cos\alpha}{M+m\cos^2\alpha} = \frac{mu(1+e)\sin\alpha\cos\alpha}{M+m\cos^2\alpha},$$

as required.

Maximising over α. With M, m, e, u fixed we need to maximise $f(\alpha) = \dfrac{\sin\alpha\cos\alpha}{M+m\cos^2\alpha}$ for $0 < \alpha < \frac{\pi}{2}$.

> **Ask yourself** Differentiating this quotient is messy. Is there a substitution that makes it tidy?

Divide the top and bottom by $\cos^2\alpha$ and put $T = \tan\alpha > 0$, using $\sec^2\alpha = 1 + T^2$:

$$f = \frac{\tan\alpha}{M\sec^2\alpha + m} = \frac{T}{MT^2 + (M+m)} = \frac{1}{MT + \dfrac{M+m}{T}}.$$

By the AM–GM inequality (both terms are positive),

$$MT + \frac{M+m}{T} \geq 2\sqrt{M(M+m)},$$

with equality exactly when $MT = (M+m)/T$, i.e. $T^2 = (M+m)/M$. So f, and hence w, is greatest there. Since $T \mapsto \alpha$ is a bijection from $(0,\infty)$ to $(0, \frac{\pi}{2})$ this is the unique maximum, and

$$\cos^2\alpha = \frac{1}{1+\tan^2\alpha} = \frac{1}{1+\frac{M+m}{M}} = \frac{M}{2M+m}, \qquad \text{so} \qquad \cos\alpha = \sqrt{\frac{M}{2M+m}}$$

(taking the positive root as α is acute). The greatest speed is $w_{\max} = \dfrac{mu(1+e)}{2\sqrt{M(M+m)}}$.

Checks

- If $\alpha \to \frac{\pi}{2}$ the cone is a flat disc: a vertical fall gives no horizontal push and $w \to 0$. In part (i) with $\alpha \to \frac{\pi}{2}$ the normal is vertical, and again $w \to 0$. Good.
- In part (i) with $\alpha \to 0$ (a vertical wall) we get $w = \dfrac{m(1+e)u}{M+m}$, the familiar answer for a direct collision with a free block of mass M.

- If $m \ll M$ the best angle has $\cos^2\alpha \to \frac{1}{2}$, i.e. $\alpha = 45°$, which is where $\sin\alpha\cos\alpha$ is greatest.
- Numerically, with $M = 3$, $m = 1$, $e = \frac{1}{2}$, $u = 4$, $\alpha = \frac{\pi}{6}$, part (ii) gives $w = \frac{2\sqrt{3}}{5} \approx 0.693$. A direct impulse calculation (solving for J from restitution) agrees with both formulas for random parameters, and a grid search over α confirms the maximising angle.

Common slips

- Forgetting that the cone moves, and writing the restitution equation as if the separation speed were just the particle's speed eu. The cone's velocity has a component $w\cos\alpha$ along the normal.
- Conserving momentum along the normal. The table's impulse has a component along $\mathbf{n}$, so only *horizontal* momentum is conserved.
- Taking the normal to make angle α with the vertical. It is the surface that makes angle α with the vertical, so the normal makes angle α with the horizontal.

Beyond STEP

The table here acts as a *constraint*: it removes one degree of freedom from the cone, and its impulse is whatever it has to be to enforce that. At university this becomes the theory of impulsive constraints, which is neatly handled with generalised coordinates in Lagrangian mechanics, and the projection onto the normal is the same idea behind the effective mass of a constrained body. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of collisions and impulses in two dimensions.

WORKED SOLUTION

STEP II 2007, Question 10

Mechanics • Statics with Friction

STEP II 2007, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q10

The key idea

This is a statics problem with only one awkward ingredient: the position of the centre of mass. Once we know where G sits on the axis, taking moments about the contact point gets rid of both the normal reaction and the friction, and gives P directly. Horizontal resolution then says friction equals P in size, and the friction law $|F| \leq \mu R$ does the rest. The modulus appears because, depending on α, P may have to push or pull.

Solution

The model

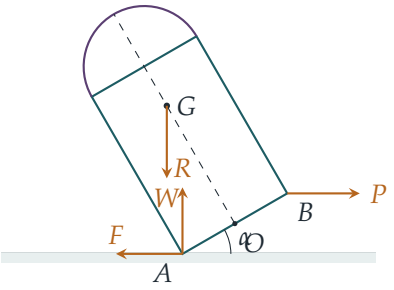
We treat the figure as a rigid body touching the plane at a single point A, the lowest point of the rim of the flat base (the free end of the cylinder). Let B be the highest point of that base, so AB is a diameter of length $2r$ making angle α with the horizontal. The forces are: the weight W at the centre of mass G; the normal reaction R and the friction F at A; and the horizontal force P at B.

The centre of mass

Measure distances along the axis from the centre O of the base. The cylinder has volume $3\pi r^3$, so mass $3\pi r^3\rho$, with centre of mass at $\frac{3}{2}r$. The hemisphere has volume $\frac{2}{3}\pi r^3$ and density 3ρ, so mass $2\pi r^3\rho$; its centre of mass is $\frac{3}{8}r$ beyond its flat face, which is at $3r$, so at $\frac{27}{8}r$. The total mass is $5\pi r^3\rho$, and

$$OG = \frac{3\cdot\frac{3}{2}r + 2\cdot\frac{27}{8}r}{5} = \frac{\frac{9}{2}r + \frac{27}{4}r}{5} = \frac{9}{4}r.$$

So G lies on the axis, $\frac{9}{4}r$ from the base, and $W = 5\pi r^3\rho g$.



Equilibrium

Take A as origin, with x horizontal (in the direction from A towards the foot of B) and y vertically up. Let P have horizontal component X in the x-direction, so X may be positive or negative; similarly let the friction be F in the x-direction.

The base runs from A in the direction $(\cos\alpha, \sin\alpha)$, so $B = (2r\cos\alpha, 2r\sin\alpha)$ and $O = (r\cos\alpha, r\sin\alpha)$. The axis is perpendicular to the base, pointing into the solid, in the direction $(-\sin\alpha, \cos\alpha)$. Hence the horizontal coordinate of G is

$$x_G = r\cos\alpha - \tfrac{9}{4}r\sin\alpha.$$

Resolving vertically, $R = W$. Resolving horizontally, $F + X = 0$, so $|F| = |X|$.

Ask yourself Which point should we take moments about?

About A: then R and F have no moment, leaving only W and P. The weight acts vertically through G, so its moment about A is Wx_G (clockwise when $x_G > 0$). The horizontal force at B acts at height $2r\sin\alpha$, so its moment is $X \cdot 2r\sin\alpha$ (anticlockwise when $X < 0$, clockwise when $X > 0$). Taking clockwise as positive,

$$Wx_G + 2r\sin\alpha\, X = 0 \quad\Longrightarrow\quad X = -\frac{W\left(r\cos\alpha - \frac{9}{4}r\sin\alpha\right)}{2r\sin\alpha} = W\left(\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right),$$

dividing by $\sin\alpha$, which is positive since $0 < \alpha < \frac{1}{2}\pi$.

The friction condition

Since the plane is rough and the solid does not slip at A, we need $|F| \le \mu R$. Using $|F| = |X|$ and $R = W > 0$,

$$W\left|\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right| \le \mu W \quad\Longrightarrow\quad \mu \ge \left|\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right|,$$

which is the required result.

The sign of X tells us which way P acts. If $\tan\alpha > \frac{4}{9}$, then G lies beyond A on the side away from B, the figure would topple backwards on its own, and P must pull B away from the solid; if $\tan\alpha < \frac{4}{9}$, P must push. In either case friction opposes P, and the modulus covers both.

Checks

- If $\tan\alpha = \frac{4}{9}$, then $x_G = 0$: G is directly above A, the figure balances on its own, $P = 0$, and no friction is needed. The bound gives $\mu \geq |\frac{9}{8} - \frac{9}{8}| = 0$, as it should.
- At $\alpha = 45^\circ$ the bound is $\mu \geq \frac{5}{8}$.
- Numerically, slicing the solid into thin discs gives total mass $5\pi\rho r^3$ and $OG = 2.2500r$, and solving the three equilibrium equations directly for 10 000 random values of α gives $|F|/R$ equal to $|\frac{9}{8} - \frac{1}{2}\cot\alpha|$ every time.

Common slips

- Forgetting that the hemisphere has three times the density, and so taking its mass as $\frac{2}{3}\pi r^3\rho$; this moves G and changes the constant $\frac{9}{8}$.
- Measuring the hemisphere's centre of mass $\frac{3}{8}r$ from the base of the cylinder rather than from its own flat face at height $3r$.
- Assuming the direction of P (or of friction) and so losing the modulus: the result must hold whichever way P acts.

Beyond STEP

Finding centres of mass of composite bodies is the discrete shadow of a volume integral: $\bar{\mathbf{r}} = \int \rho\,\mathbf{r}\,\mathrm{d}V \big/ \int \rho\,\mathrm{d}V$, which is how the $\frac{3}{8}r$ for a hemisphere is derived in the first place. In first-year mechanics this grows into the theory of rigid bodies, where the same integrals with r^2 in place of $\mathbf{r}$ give moments of inertia. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP II 2007, Question 11

Mechanics • Projectiles in Three Dimensions

STEP II 2007, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q11

The key idea

Write the motion as a single position vector: the horizontal velocity is constant, and only the vertical component feels gravity. Once $\mathbf{r}(t)$ is down on paper, the distance from O is just $|\mathbf{r}|$, and the shortest distance comes from minimising a quadratic. The last part works because the bullet is so fast that the particle barely moves while it is in flight.

Solution

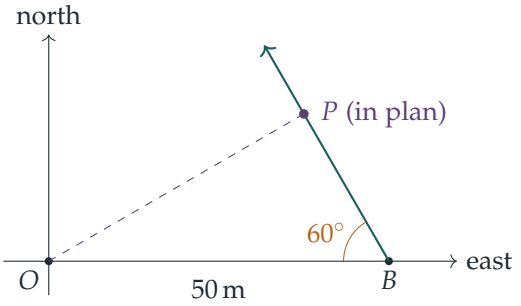
We model the particle as a point moving under gravity alone ($g = 10\,\mathrm{m\,s^{-2}}$, no air resistance), and take O as the origin, so B has position vector $50\mathbf{i}$.

Part (i)

Components of the initial velocity. If $\tan\alpha = \frac{1}{2}$ then $\cos\alpha = \frac{2}{\sqrt{5}}$ and $\sin\alpha = \frac{1}{\sqrt{5}}$ (a 1, 2, $\sqrt{5}$ triangle). So the initial velocity has

$$\text{horizontal speed } 25 \cdot \tfrac{2}{\sqrt{5}} = 10\sqrt{5}, \qquad \text{vertical speed } 25 \cdot \tfrac{1}{\sqrt{5}} = 5\sqrt{5}.$$

Which horizontal direction? The direction makes 60° with the line OB. If the particle headed east-ish, away from O, it could never pass O at all; to pass to the north of O it must head back towards the west and northwards. So the horizontal direction is at 60° to $\overrightarrow{BO}$ (west), on the north side, giving the unit vector $-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$.



The horizontal velocity is therefore

$$10\sqrt{5}\left(-\tfrac{1}{2}\mathbf{i}+\tfrac{\sqrt{3}}{2}\mathbf{j}\right)=-5\sqrt{5}\,\mathbf{i}+5\sqrt{15}\,\mathbf{j},$$

and it stays constant, since gravity acts vertically. Vertically, $z=5\sqrt{5}\,t-\frac{1}{2}(10)t^2$. Adding the starting point $50\mathbf{i}$,

$$\mathbf{r}(t)=\left(50-5\sqrt{5}\,t\right)\mathbf{i}+5\sqrt{15}\,t\,\mathbf{j}+\left(5\sqrt{5}\,t-5t^2\right)\mathbf{k}.$$

The distance from O. Squaring and adding the components,

$$\begin{aligned}|\mathbf{r}|^2&=\left(2500-500\sqrt{5}\,t+125t^2\right)+375t^2+\left(125t^2-50\sqrt{5}\,t^3+25t^4\right)\\&=25t^4-50\sqrt{5}\,t^3+625t^2-500\sqrt{5}\,t+2500.\end{aligned}$$

Compare this with

$$\begin{aligned}25\left(t^2-\sqrt{5}\,t+10\right)^2&=25\left(t^4+5t^2+100-2\sqrt{5}\,t^3+20t^2-20\sqrt{5}\,t\right)\\&=25t^4-50\sqrt{5}\,t^3+625t^2-500\sqrt{5}\,t+2500,\end{aligned}$$

which is the same. So $|\mathbf{r}|=5\left|t^2-\sqrt{5}\,t+10\right|$. The quadratic $t^2-\sqrt{5}\,t+10$ has discriminant $5-40<0$, so it is always positive and we can drop the modulus:

$$|\mathbf{r}|=5\left(t^2-\sqrt{5}\,t+10\right)\text{ m},$$

as required.

The point P. Completing the square,

$$t^2-\sqrt{5}\,t+10=\left(t-\tfrac{\sqrt{5}}{2}\right)^2+\tfrac{35}{4},$$

so the distance is least when $t=\frac{\sqrt{5}}{2}$, and the shortest distance is $5\cdot\frac{35}{4}=\frac{175}{4}=43.75\,\text{m}$. Substituting $t=\frac{\sqrt{5}}{2}$, so that $\sqrt{5}\,t=\frac{5}{2}$ and $t^2=\frac{5}{4}$:

$$\overrightarrow{OP}=\tfrac{75}{2}\,\mathbf{i}+\tfrac{25\sqrt{3}}{2}\,\mathbf{j}+\tfrac{25}{4}\,\mathbf{k}\quad\text{(in metres)}.$$

The horizontal part of $\overrightarrow{OP}$ has east component $\frac{75}{2}$ and north component $\frac{25\sqrt{3}}{2}$. A bearing θ is measured clockwise from north, so

$$\tan\theta=\frac{\text{east}}{\text{north}}=\frac{75/2}{25\sqrt{3}/2}=\sqrt{3},$$

and since both components are positive, $\theta=60^\circ$: the bearing of P from O is 060°.

Part (ii)

The vertical component of velocity is

$$\dot{z} = 5\sqrt{5} - 10t,$$

which is zero exactly when $t = \frac{\sqrt{5}}{2}$. This is the same time at which the particle is at P, so the particle is at its highest point (height $\frac{25}{4}$ m) at P.

Part (iii)

The bullet travels in a straight line from O to P at $350\,\mathrm{m\,s^{-1}}$ (we ignore gravity on it), covering $OP = \frac{175}{4}$ m. It takes

$$T = \frac{175/4}{350} = \frac{1}{8}\ \mathrm{s}.$$

? **Ask yourself** Where is the particle after this extra $\frac{1}{8}$ second?

By part (ii), at P the particle's vertical velocity is zero, so its velocity there is purely horizontal, of magnitude $10\sqrt{5}\,\mathrm{m\,s^{-1}}$. In the next T seconds it moves horizontally $10\sqrt{5}\,T$ and drops $\frac{1}{2}gT^2$. With $T = \frac{1}{8}$,

$$\text{horizontal shift} = \frac{10\sqrt{5}}{8} \approx 2.80\,\mathrm{m}, \qquad \text{drop} = 5 \cdot \frac{1}{64} \approx 0.08\,\mathrm{m}.$$

These are at right angles, so the distance from P is

$$\sqrt{\frac{500}{64} + \frac{25}{4096}} = \frac{5\sqrt{1281}}{64} \approx 2.80\,\mathrm{m},$$

which is approximately 3 m, as required. (The drop hardly matters: the answer is essentially $10\sqrt{5}/8$.) The marksman misses.

Checks

- At $t = 0$ the formula gives $5 \cdot 10 = 50$ m, the distance OB. Good.
- $|\overrightarrow{OP}|^2 = 37.5^2 + 21.65^2 + 6.25^2 = 1406.25 + 468.75 + 39.06 = 1914.06 = 43.75^2$, so the position vector of P agrees with the minimum distance.
- A numerical scan of $|\mathbf{r}(t)|$ gives a minimum of 43.75 m at $t \approx 1.118 = \frac{\sqrt{5}}{2}$, and stepping the particle on from P for 0.125 s gives a separation of 2.796 m.

Common slips

- Taking the horizontal direction as $\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$ (away from O). Then the particle never passes north of O, and the distance does not come out as given.

- Giving the bearing as 30°: that is the angle from east, but bearings are measured clockwise from north.

Beyond STEP

Part (iii) is a baby version of the interception problem: to hit a moving target you must aim where it *will* be, which leads to solving $|\mathbf{r}_{\text{target}}(t) - \mathbf{r}_{\text{bullet}}(t)| = 0$ for both the firing direction and the time. Minimising $|\mathbf{r}(t)|$ is also the idea behind "distance of closest approach" in orbital and collision problems, where squaring the distance to get a polynomial is the standard first move. Kleppner and Kolenkow's *An Introduction to Mechanics* develops vector kinematics of this kind properly.

WORKED SOLUTION

STEP II 2007, Question 12

Probability and Statistics • Geometric Distribution Maximum

STEP II 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q12

The key idea

Forget the rather fussy description of the procedure and look at each die on its own. Each die is thrown until it first shows a six, and the experiment ends when the *slower* of the two dice has got there. So the number of throws is the maximum of two independent geometric random variables, and the maximum is easiest to handle through the cumulative probability $\mathrm{P}(\max \le r)$.

Solution

Part (i)

Let X be the number of the throw on which the first die first shows a six, and define Y in the same way for the second die. Every throw of a die is independent of every other throw, so X and Y are independent, and each is geometric:

$$\mathrm{P}(X = k) = q^{k-1}p, \qquad \mathrm{P}(X \le r) = 1 - \mathrm{P}(\text{first } r \text{ throws all non-six}) = 1 - q^r.$$

How does the procedure fit this? While neither die has shown a six we throw both; once one has, we keep throwing only the other one. Either way, each die is thrown exactly until its first six (and never after it), and we stop as soon as both have shown a six. If they first show a six on the same throw, that is the case where both show a six together. So the number of throws is

$$N = \max(X, Y).$$

? **Ask yourself** Why work with $\mathrm{P}(N \le r)$ rather than $\mathrm{P}(N = r)$ directly?

The event $N \le r$ is simply "$X \le r$ *and* $Y \le r$", and independence turns that into a product. The event $N = r$ splits into awkward cases (one die finishes at r and the other earlier, or both at r). So

$$\mathrm{P}(N \le r) = \mathrm{P}(X \le r)\,\mathrm{P}(Y \le r) = (1 - q^r)^2,$$

and, for $r \geq 1$,

$$\begin{aligned}
\mathrm{P}(N = r) &= \mathrm{P}(N \leq r) - \mathrm{P}(N \leq r-1) \\
&= (1-q^r)^2 - (1-q^{r-1})^2 \\
&= \left[(1-q^r) - (1-q^{r-1})\right]\left[(1-q^r) + (1-q^{r-1})\right] \\
&= (q^{r-1} - q^r)(2 - q^{r-1} - q^r) \\
&= pq^{r-1}\,(2 - q^{r-1} - q^r),
\end{aligned}$$

using the difference of two squares and then $q^{r-1} - q^r = q^{r-1}(1-q) = pq^{r-1}$. This is the required result.

For the expectation, split the probability into its two pieces:

$$\mathrm{P}(N = r) = 2pq^{r-1} - pq^{2r-2}(1+q).$$

Differentiating the given result (or dividing it by x) gives $\sum_{r\geq 1} rx^{r-1} = (1-x)^{-2}$ for $|x| < 1$. Using it with $x = q$ and with $x = q^2$ (both less than 1 for $0 < p \leq 1$):

$$\begin{aligned}
\mathrm{E}(N) &= 2p\sum_{r=1}^{\infty} rq^{r-1} - p(1+q)\sum_{r=1}^{\infty} r(q^2)^{r-1} \\
&= \frac{2p}{(1-q)^2} - \frac{p(1+q)}{(1-q^2)^2} \\
&= \frac{2}{p} - \frac{p(1+q)}{p^2(1+q)^2} \\
&= \frac{2}{p} - \frac{1}{p(1+q)} \\
&= \frac{1+2q}{p(1+q)},
\end{aligned}$$

where we used $1 - q = p$ and $1 - q^2 = (1-q)(1+q) = p(1+q)$. In terms of p alone (put $q = 1-p$),

$$\mathrm{E}(N) = \frac{3-2p}{p(2-p)}.$$

Part (ii)

Over many experiments the mean number of throws should be close to $\mathrm{E}(N)$, so we estimate p by solving $\mathrm{E}(N) = m$:

$$m\,p(2-p) = 3 - 2p \quad \Longleftrightarrow \quad mp^2 - 2(m+1)p + 3 = 0,$$

so

$$p = \frac{(m+1) \pm \sqrt{(m+1)^2 - 3m}}{m} = \frac{m+1 \pm \sqrt{m^2 - m + 1}}{m}.$$

Ask yourself Which root do we want?

Every experiment takes at least one throw, so $m \geq 1$. Then $\sqrt{m^2 - m + 1} \geq 1$ (because $m^2 - m = m(m-1) \geq 0$), so the root with the plus sign is at least $(m+2)/m > 1$, which is impossible for a probability. The root with the minus sign is at most $m/m = 1$, by the same inequality, and it is positive because $(m+1)^2 > m^2 - m + 1$. So the estimate is

$$\hat{p} = \frac{m + 1 - \sqrt{m^2 - m + 1}}{m}.$$

Checks

- If $p = 1$ every experiment takes one throw: $\mathrm{E}(N) = \frac{1}{1 \cdot 1} = 1$, and $m = 1$ gives $\hat{p} = (2-1)/1 = 1$.
- The probabilities sum to 1: $\sum_r \mathrm{P}(N = r)$ telescopes to $\lim_{r \to \infty}(1 - q^r)^2 = 1$.
- For a fair die, $p = \frac{1}{6}$ gives $\mathrm{E}(N) = \frac{8/3}{11/36} = \frac{96}{11} \approx 8.73$, a bit more than the 6 throws one die needs on average, as it should be. A simulation of 200,000 experiments gave a mean of 8.71, and putting $m = 96/11$ into the formula for $\hat{p}$ returns $\frac{1}{6}$.

Common slips

- Counting each die's throw separately: a simultaneous throw of both dice counts as *one* throw, which is exactly why $N = \max(X, Y)$.
- Forgetting that the term $pq^{2r-2}(1+q)$ needs the sum with $x = q^2$, not $x = q$.
- Keeping both roots in part (ii) without checking which one can be a probability.

Beyond STEP

The trick $\mathrm{P}(\max \leq r) = \prod \mathrm{P}(X_i \leq r)$ is the standard way to handle maxima (and, with $\geq$, minima) of independent variables, the start of *order statistics*. Part (ii) is the *method of moments*: match the sample mean to the theoretical mean and solve for the parameter. First-year statistics compares it with maximum likelihood and asks how accurate such estimators are. For the probability side, Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

WORKED SOLUTION

STEP II 2007, Question 13

Probability and Statistics • Birthday Problem

STEP II 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 9.9, *The Birthday Problem*: https://beyondstep.keithmcnulty.org/ch09/birthday.html

The key idea

The probability that *no* two guests share a birthday is a product of factors of the form $\frac{365-r}{365}$, and products are awkward. The first part turns each factor into an exponential, and a product of exponentials is the exponential of a sum, which we can add up exactly. For the host, the guests' events are independent, so the complement is a single power, and the same approximation applies.

Solution

The approximation

For any real x,

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots .$$

Put $x = r/n$. Since r is much smaller than n, x is small and the terms in x^2 and beyond are negligible compared with x. So

$$e^{-r/n} \approx 1 - \frac{r}{n} = \frac{n-r}{n},$$

as required.

At least two guests sharing a birthday

Let A be the event that at least two of the k guests share a birthday. It is easier to find $P(A')$, the probability that all k birthdays are different. Take the guests one at a time. The first can have any birthday. Given that the first r guests have r different birthdays,

the $(r+1)$th guest must avoid those r days, which happens with probability $\frac{365-r}{365}$, because the birthdays are independent and uniform over the 365 days. Multiplying,

$$\mathrm{P}(A') = \prod_{r=0}^{k-1} \frac{365-r}{365}.$$

For a party of modest size, every r here is much smaller than 365, so by the first part, with $n = 365$,

$$\begin{aligned}\mathrm{P}(A') &\approx \prod_{r=0}^{k-1} \mathrm{e}^{-r/365} = \exp\Big(-\frac{1}{365}\sum_{r=0}^{k-1} r\Big) \\ &= \exp\Big(-\frac{1}{365}\cdot\frac{k(k-1)}{2}\Big) = \mathrm{e}^{-k(k-1)/730}.\end{aligned}$$

(The $r = 0$ factor is exactly $1 = \mathrm{e}^0$, so including it costs nothing.) Hence

$$\mathrm{P}(A) = 1 - \mathrm{P}(A') \approx 1 - \mathrm{e}^{-k(k-1)/730},$$

which is the required result.

Ask yourself Isn't it worrying that we approximate many factors at once? The errors might pile up.

Each factor's error is of order $(r/365)^2$, while the exponent itself is of order $k^2/730$. For the values of k we need (around 20), the largest $r/365$ is about 0.06, and the total relative error in the exponent is only a few per cent. The approximation gives 0.500 at $k = 23$ against a true value of 0.507.

Smallest party for an even chance

We need

$$1 - \mathrm{e}^{-k(k-1)/730} \geq \tfrac{1}{2} \iff \mathrm{e}^{-k(k-1)/730} \leq \tfrac{1}{2} \iff \frac{k(k-1)}{730} \geq \ln 2,$$

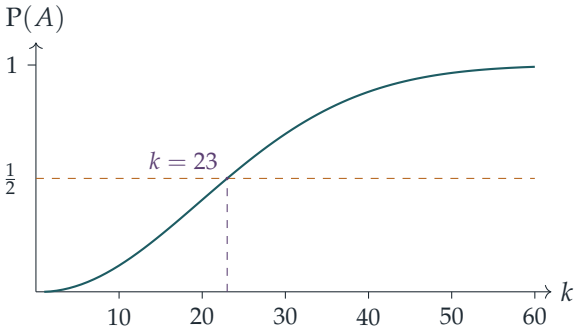
taking logarithms (the logarithm is increasing, so the inequality keeps its direction) and multiplying by -1 (which reverses it). Using $\ln 2 \approx \frac{253}{365}$, the condition becomes

$$k(k-1) \geq 730 \times \frac{253}{365} = 2 \times 253 = 506.$$

Since $k(k-1)$ increases with k for $k \geq 1$, and

$$22 \times 21 = 462 < 506, \qquad 23 \times 22 = 506,$$

the smallest value is $k = 23$.



Sharing the host's birthday

Now let m be the number of guests (the host is not counted among them). Let B be the event that at least one guest has the same birthday as the host. Whatever the host's birthday is, each guest independently misses it with probability $\frac{364}{365}$, so

$$\mathrm{P}(B') = \left(\frac{364}{365}\right)^m \approx \left(\mathrm{e}^{-1/365}\right)^m = \mathrm{e}^{-m/365},$$

using the first part with $n = 365$ and $r = 1$. This time $r/n = 1/365$ really is small, whatever the size of m.

We need $\mathrm{P}(B) \geq \frac{1}{2}$, that is $\mathrm{e}^{-m/365} \leq \frac{1}{2}$. Exactly as before this is

$$\frac{m}{365} \geq \ln 2 \approx \frac{253}{365} \iff m \geq 253.$$

So there must be at least 253 guests.

Ask yourself Why is the answer so much larger than 23?

In the first problem any pair of guests can match, and 23 guests make $\binom{23}{2} = 253$ pairs. In the second, only the m guest–host pairs count. To get the same number of chances of a match we need $m = 253$, and this is exactly why 253 appears in both answers.

Checks

- Exact products confirm both answers. The true probability of a shared birthday is 0.4757 for 22 guests and 0.5073 for 23. The true value of $1 - (364/365)^m$ is below $\frac{1}{2}$ for $m = 252$ and is 0.5005 for $m = 253$.
- $\frac{253}{365} = 0.69315\ldots$ and $\ln 2 = 0.69315\ldots$, agreeing to five decimal places, so the given approximation does not affect either answer.
- A simulation of 200 000 parties of 23 gave a shared birthday 50.7% of the time.

Common slips

- Computing the probability that two *particular* guests share a birthday, or adding $\frac{1}{365}$ once for each pair. Overlapping events can't simply be added; working with the complement avoids this.
- Using $\frac{364}{365}$ for every factor in the first product. The $(r+1)$th guest must avoid r days, not just one.
- Including the host in the count of guests for the last part. The host is a separate person, so 253 guests are needed.

Beyond STEP

The heuristic here is that a match among k objects in N boxes becomes likely once k is about $\sqrt{2N\ln 2} \approx 1.18\sqrt{N}$. It underlies "birthday attacks" in cryptography, which find collisions in an N-bit hash after about $2^{N/2}$ tries rather than 2^N. Behind the approximation is the Poisson limit: the number of matching pairs is roughly Poisson with mean $\binom{k}{2}/365$, and $\mathrm{P}(\text{no matches}) \approx \mathrm{e}^{-k(k-1)/730}$. Blitzstein and Hwang, *Introduction to Probability*, discusses the birthday problem and its Poisson approximation in detail.

WORKED SOLUTION

STEP II 2007, Question 14

Probability and Statistics • Piecewise Probability Density

STEP II 2007, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q14

The key idea

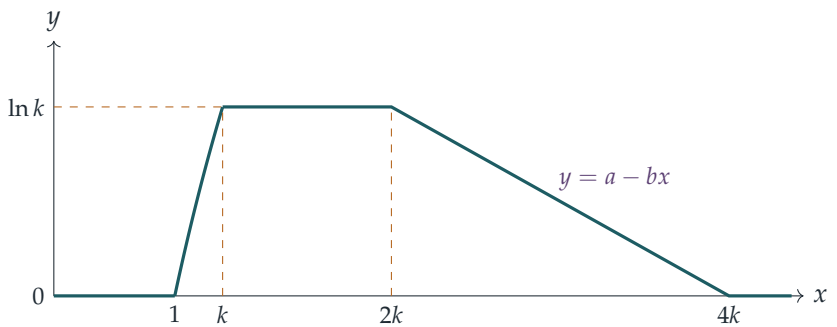
The word *continuous* is doing a lot of work. A density defined in pieces has to join up at every break point, and that gives us two equations for a and b straight away. The one remaining condition, that the total area under the graph is 1, then pins down k. For the median we add up areas piece by piece until we pass one half.

Solution

Throughout, $k > 1$: otherwise the interval $1 \leq x \leq k$ would be empty, and in any case we need $\ln k > 0$ for the density to be non-negative and not identically zero beyond $x = 1$.

Part (i)

On $[1, k]$ the graph is the usual logarithm curve, starting at 0 when $x = 1$ (so it joins the zero piece continuously), increasing and concave. It reaches height $\ln k$ at $x = k$, stays level at $\ln k$ until $x = 2k$, then falls in a straight line to 0 at $x = 4k$ (we show below that continuity forces exactly this), and is zero afterwards. The sketch uses the values we find in part (ii): $k = \mathrm{e}^{1/3} \approx 1.40$, so the plateau is at height $\frac{1}{3}$.



Part (ii)

Continuity. At $x = k$ both formulas give $\ln k$, so nothing new there. At $x = 2k$ and $x = 4k$ continuity requires

$$a - 2bk = \ln k, \qquad a - 4bk = 0.$$

Subtracting, $2bk = \ln k$, so

$$b = \frac{\ln k}{2k}, \qquad a = 4bk = 2\ln k.$$

Total probability. We split the area into three pieces. Integrating by parts (with $u = \ln x$, $\mathrm{d}v = \mathrm{d}x$),

$$\int_1^k \ln x \,\mathrm{d}x = \Big[x\ln x - x\Big]_1^k = k\ln k - k + 1.$$

The plateau is a rectangle of width k and height $\ln k$, with area $k\ln k$. The last piece is a triangle with base $4k - 2k = 2k$ and height $\ln k$, with area $\frac{1}{2}\cdot 2k \cdot \ln k = k\ln k$. So

$$1 = (k\ln k - k + 1) + k\ln k + k\ln k = 3k\ln k - k + 1,$$

which simplifies to

$$k(3\ln k - 1) = 0.$$

Since $k > 1$, we cannot have $k = 0$, so $\ln k = \frac{1}{3}$ and

$$k = \mathrm{e}^{1/3} \approx 1.396, \qquad a = 2\ln k = \tfrac{2}{3}, \qquad b = \frac{\ln k}{2k} = \frac{1}{6\,\mathrm{e}^{1/3}} \approx 0.119.$$

Note $b > 0$, so the last piece really does slope downwards, as drawn.

Part (iii)

We find which piece contains the median m by accumulating probability. Using $\ln k = \frac{1}{3}$,

$$\mathrm{P}(X \le k) = k\ln k - k + 1 = 1 - \tfrac{2k}{3} \approx 0.070,$$

and the plateau adds $k\ln k = \frac{k}{3}$, so

$$\mathrm{P}(X \le 2k) = 1 - \tfrac{2k}{3} + \tfrac{k}{3} = 1 - \tfrac{k}{3} \approx 0.535.$$

? **Ask yourself** Which piece of the graph does the median lie on?

Since $0.070 < \frac{1}{2} < 0.535$, the median lies on the plateau, $k \le m \le 2k$. There the density is constant at $\frac{1}{3}$, so

$$\mathrm{P}(X \le m) = 1 - \tfrac{2k}{3} + \tfrac{1}{3}(m - k) = \tfrac{1}{2}.$$

Multiplying by 3: $3 - 2k + m - k = \frac{3}{2}$, so

$$m = 3k - \tfrac{3}{2} = 3\mathrm{e}^{1/3} - \tfrac{3}{2} \approx 2.69.$$

This does lie between $k \approx 1.40$ and $2k \approx 2.79$, as it must.

Checks

- Numerically integrating the density with $k = \mathrm{e}^{1/3}$, $a = \frac{2}{3}$, $b = 1/(6\mathrm{e}^{1/3})$ (midpoint rule) gives total area 1.0000000, and $\int_1^{2.68684} f(x)\,\mathrm{d}x = 0.5000000$.
- Sanity check on the median: just over half the probability lies to the left of $2k$ (about 0.535), so the median should sit a little to the left of $2k \approx 2.79$; indeed $m \approx 2.69$.

Common slips

- Forgetting the lower limit in $\int_1^k \ln x\,\mathrm{d}x$: the $+1$ from $-[x\ln x - x]$ at $x = 1$ is what makes the equation for k factorise so neatly.
- Assuming the median is on the sloping piece because it is "in the middle" of the support $[1, 4k]$; always accumulate the areas first.

Beyond STEP

Here the density was built from pieces and the constants were fixed by continuity and normalisation. At university you will meet the *cumulative distribution function* $F(x) = \mathrm{P}(X \le x)$ as the basic object: it exists for every random variable, discrete or continuous, and quantities like the median are simply solutions of $F(m) = \frac{1}{2}$. Grimmett and Welsh, *Probability: An Introduction*, develops this viewpoint clearly.

WORKED SOLUTION

STEP III 2007, Question 1

Pure • Tangent Sums and Vieta

STEP III 2007, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q1

The key idea

The tangent of a sum of angles is a ratio of *symmetric* functions of the individual tangents, and symmetric functions of the roots of a polynomial are exactly what Vieta's formulae hand us in terms of the coefficients. For the final part, the trick is to turn the trigonometric equation into a quartic in $t = \tan\theta$; the symmetry of its coefficients then forces the tangent of the sum to vanish.

Solution

The tangent of a sum of four angles

Split the sum as $A + B$ with $A = \theta_1 + \theta_2$ and $B = \theta_3 + \theta_4$, so that

$$\tan A = \frac{t_1 + t_2}{1 - t_1 t_2}, \qquad \tan B = \frac{t_3 + t_4}{1 - t_3 t_4}.$$

Then, multiplying the top and bottom of $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$ by $(1 - t_1 t_2)(1 - t_3 t_4)$,

$$\tan(A+B) = \frac{(t_1 + t_2)(1 - t_3 t_4) + (t_3 + t_4)(1 - t_1 t_2)}{(1 - t_1 t_2)(1 - t_3 t_4) - (t_1 + t_2)(t_3 + t_4)}.$$

The numerator is $t_1 + t_2 + t_3 + t_4 - (t_1 t_3 t_4 + t_2 t_3 t_4 + t_1 t_2 t_3 + t_1 t_2 t_4)$, and the denominator is $1 - t_1 t_2 - t_3 t_4 + t_1 t_2 t_3 t_4 - (t_1 t_3 + t_1 t_4 + t_2 t_3 + t_2 t_4)$. Writing

$$\sigma_1 = \sum t_i, \quad \sigma_2 = \sum_{i<j} t_i t_j, \quad \sigma_3 = \sum_{i<j<k} t_i t_j t_k, \quad \sigma_4 = t_1 t_2 t_3 t_4$$

for the elementary symmetric functions, this is

$$\boxed{\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{\sigma_1 - \sigma_3}{1 - \sigma_2 + \sigma_4}.}$$

(The pattern is the same as for two or three angles: odd symmetric functions with alternating signs on top, even ones on the bottom.)

In terms of the coefficients

If $t_1, \ldots, t_4$ are the roots of $at^4 + bt^3 + ct^2 + dt + e = 0$, then $at^4 + bt^3 + ct^2 + dt + e = a(t - t_1)(t - t_2)(t - t_3)(t - t_4)$, and comparing coefficients (Vieta) gives

$$\sigma_1 = -\frac{b}{a}, \qquad \sigma_2 = \frac{c}{a}, \qquad \sigma_3 = -\frac{d}{a}, \qquad \sigma_4 = \frac{e}{a}.$$

Substituting and multiplying top and bottom by $a \neq 0$,

$$\boxed{\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{d - b}{a - c + e}.}$$

The final part

Since $1 + \cos 2\theta = 2\cos^2\theta$, the equation is

$$2p\cos^2\theta + \cos\theta\cos\alpha + \sin\theta\sin\alpha = 0.$$

Dividing by $\cos^2\theta$ (we ignore the special case $\cos\theta = 0$) gives

$$2p = -(\cos\alpha + t\sin\alpha)\sec\theta, \qquad t = \tan\theta.$$

The $\sec\theta$ is not a rational function of t, so we square, using $\sec^2\theta = 1 + t^2$:

$$4p^2 = (\cos\alpha + t\sin\alpha)^2(1 + t^2).$$

Expanding,

$$\sin^2\alpha\, t^4 + 2\sin\alpha\cos\alpha\, t^3 + t^2 + 2\sin\alpha\cos\alpha\, t + (\cos^2\alpha - 4p^2) = 0, \qquad (*)$$

using $\sin^2\alpha + \cos^2\alpha = 1$ for the t^2 coefficient.

Ask yourself Squaring can introduce extra solutions. Are the $\tan\theta_i$ really the four roots of $(*)$?

Squaring means $(*)$ is also satisfied by the solutions of $p\cos 2\theta - \cos(\theta - \alpha) + p = 0$. But replacing θ by $\theta + \pi$ leaves $t = \tan\theta$ and $\cos^2\theta$ unchanged and changes the sign of $\cos(\theta - \alpha)$. So for each real root t of $(*)$, of the two angles in $[0, 2\pi)$ with tangent t, exactly one satisfies the original equation (not both, since that would need $\cos(\theta - \alpha) = 0$ and hence $p\cos^2\theta = 0$, a special case). Thus the solutions in $[0, 2\pi)$ correspond one-to-one with the real roots of $(*)$, and the four given angles have as their tangents the four roots of $(*)$.

Now apply the previous result with $a = \sin^2\alpha$, $b = d = 2\sin\alpha\cos\alpha$, $c = 1$, $e = \cos^2\alpha - 4p^2$ (we need $a \neq 0$, i.e. $\sin\alpha \neq 0$; otherwise $(*)$ is not a quartic, another special case). The numerator is $d - b = 0$, and the denominator is

$$a - c + e = \sin^2\alpha - 1 + \cos^2\alpha - 4p^2 = -4p^2,$$

which is non-zero (if $p = 0$ the equation is just $\cos(\theta - \alpha) = 0$, with only two solutions). Hence

$$\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0,$$

so $\theta_1 + \theta_2 + \theta_3 + \theta_4 = n\pi$ for some integer n, as required.

Checks

- Put $t_3 = t_4 = 0$ in the four-angle formula: it collapses to $(t_1 + t_2)/(1 - t_1 t_2)$, the familiar two-angle formula.
- With $p = \frac{1}{2}$ and $\alpha = 3$, solving the equation numerically on $[0, 2\pi)$ gives four roots whose sum, divided by π, is an integer to within 10^{-9}. (Not every p and α give four solutions: $p = \frac{1}{2}$, $\alpha = 1$ gives only two.) The same held for every random (p, α) we tried that gave four solutions, and we also checked both formulae on thousands of random angles and quartics.

Common slips

- Sign errors in Vieta: the alternate coefficients b and d come with minus signs.
- Trying to write $\cos(\theta - \alpha)$ directly in terms of $\tan\theta$ without squaring, or squaring without checking that no unwanted angles slip in.

Beyond STEP

The expression of tan of a sum through elementary symmetric functions is one instance of a big theorem: every symmetric polynomial in the roots is a polynomial in the coefficients. It is also neatly explained by complex numbers, since $\prod(1 + \mathrm{i}t_k) = \prod \sec\theta_k \, \mathrm{e}^{\mathrm{i}\sum\theta_k}$, and its real and imaginary parts are $1 - \sigma_2 + \sigma_4$ and $\sigma_1 - \sigma_3$. Needham's *Visual Complex Analysis* develops this geometric way of thinking, and Siklos's *Advanced Problems in Mathematics* has more STEP problems in the same spirit.

WORKED SOLUTION

STEP III 2007, Question 2

Pure • Binomial Power Series

STEP III 2007, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q2

The key idea

Everything flows from one binomial series. Once we know the coefficients of $(1-4x)^{-1/2}$, differentiating the series pulls down a factor n (turning $(n!)^2$ into $n!\,(n-1)!$), and integrating it divides by $n+1$ (turning $(n!)^2$ into $n!\,(n+1)!$). So each later sum is the derivative or the integral of the first series, evaluated at a well-chosen point inside $|x| < \frac{1}{4}$.

Solution

Part (i)

Multiply and divide the product of the odd numbers by the missing even numbers $2\cdot 4\cdot 6\cdots(2n)$:

$$1\cdot 3\cdot 5\cdots(2n-1) = \frac{1\cdot 2\cdot 3\cdot 4\cdots(2n-1)(2n)}{2\cdot 4\cdots(2n)} = \frac{(2n)!}{2^n\,(1\cdot 2\cdots n)} = \frac{(2n)!}{2^n\,n!},$$

since each of the n even numbers $2k$ contributes a factor 2 and a factor k.

For $|4x| < 1$, that is $|x| < \frac{1}{4}$, the binomial series gives

$$(1-4x)^{-1/2} = 1 + \sum_{n=1}^{\infty} \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{n!}\,(-4x)^n.$$

The numerator has n factors, each negative, so it equals $(-1)^n\,\dfrac{1\cdot 3\cdots(2n-1)}{2^n}$. The $(-1)^n$ cancels the sign in $(-4)^n$, and $4^n/2^n = 2^n$, so the coefficient of x^n is

$$\frac{2^n\cdot 1\cdot 3\cdots(2n-1)}{n!} = \frac{2^n}{n!}\cdot\frac{(2n)!}{2^n\,n!} = \frac{(2n)!}{(n!)^2},$$

using the first result. Hence, for $|x| < \frac{1}{4}$,

$$\frac{1}{\sqrt{1-4x}} = 1 + \sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}\,x^n,$$

as required.

Part (ii)

A power series may be differentiated term by term inside its interval of convergence, here $|x| < \frac{1}{4}$. Differentiating both sides,

$$\frac{\mathrm{d}}{\mathrm{d}x}(1-4x)^{-1/2} = -\tfrac{1}{2}(1-4x)^{-3/2}\cdot(-4) = 2(1-4x)^{-3/2},$$

while on the right the constant disappears and x^n becomes nx^{n-1}. Since $n/(n!)^2 = 1/\big(n!\,(n-1)!\big)$,

$$2(1-4x)^{-3/2} = \sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\,x^{n-1}.$$

Multiplying by x gives the powers x^n that we want:

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\,x^n = \frac{2x}{(1-4x)^{3/2}} \qquad (|x| < \tfrac{1}{4}).$$

Now put $x = \frac{6}{25}$, which is allowed because $\frac{6}{25} < \frac{6}{24} = \frac{1}{4}$. Then $1-4x = \frac{1}{25}$, so $(1-4x)^{3/2} = \frac{1}{125}$, and

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\left(\frac{6}{25}\right)^n = 2\cdot\frac{6}{25}\cdot 125 = 60.$$

Part (iii)

> **Ask yourself** The factorials are now $(n+1)!\,n!$ rather than $(n!)^2$. What operation on the series produces that?

We need to divide the nth coefficient by $n+1$, since $(n+1)! = (n+1)\,n!$. Integrating x^n does exactly this: $\int_0^x t^n\,\mathrm{d}t = \dfrac{x^{n+1}}{n+1}$. Term-by-term integration of a power series is also valid inside its interval of convergence, so for $|x| < \frac{1}{4}$ we integrate the series of part (i) from 0 to x:

$$\int_0^x (1-4t)^{-1/2}\,\mathrm{d}t = \Big[-\tfrac{1}{2}(1-4t)^{1/2}\Big]_0^x = \frac{1-\sqrt{1-4x}}{2},$$

and so

$$\frac{1-\sqrt{1-4x}}{2} = x + \sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n+1)!}\,x^{n+1}.$$

Next we match the powers in the target sum. Since $3^{2n} = 9^n = 9^{n+1}/9$,

$$\frac{2^{n+1}}{3^{2n}} = 9\left(\frac{2}{9}\right)^{n+1},$$

so the target sum is 9 times our series at $x = \frac{2}{9}$. This is a legitimate value, since $\frac{2}{9} < \frac{2}{8} = \frac{1}{4}$. There, $1 - 4x = \frac{1}{9}$ and $\sqrt{1-4x} = \frac{1}{3}$, giving

$$\sum_{n=1}^{\infty} \frac{2^{n+1}(2n)!}{3^{2n}(n+1)!\,n!} = 9\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n+1)!}\left(\frac{2}{9}\right)^{n+1}$$
$$= 9\left(\frac{1-\frac{1}{3}}{2} - \frac{2}{9}\right) = 9\left(\frac{1}{3} - \frac{2}{9}\right) = 1,$$

as required. Notice that the leading term x (the integral of the constant 1) has to be subtracted, because the sum starts at $n = 1$.

Checks

- In (i), $n = 2$ gives $1 \cdot 3 = 3$ and $\frac{4!}{2^2\,2!} = 3$; the series begins $1 + 2x + 6x^2 + 20x^3 + \cdots$, which matches squaring: $(1 + 2x + 6x^2)^2(1 - 4x) = 1 + O(x^3)$.
- Summing the first 3000 terms of (ii) with exact fractions gives $60.000\ldots$, and the first 400 terms of (iii) give $1.000\ldots$ to machine precision.
- In (iii) the first term ($n = 1$) is $\frac{4\cdot 2}{9\cdot 2} = \frac{4}{9}$, already a good share of the total 1; the terms then decrease roughly like $(8/9)^n$.

Common slips

- Losing the signs in the binomial coefficient: there are n negative factors in the numerator and n in $(-4)^n$, so they cancel exactly.
- Forgetting to check that $\frac{6}{25}$ and $\frac{2}{9}$ lie inside $|x| < \frac{1}{4}$; outside it the series diverge.
- In (iii), forgetting the x term from integrating the constant 1, which would give the answer 3 instead of 1.

Beyond STEP

The numbers $\frac{(2n)!}{n!\,(n+1)!} = \frac{1}{n+1}\binom{2n}{n}$ that appear in part (iii) are the Catalan numbers $1, 1, 2, 5, 14, \ldots$, which count bracketings, binary trees and many other structures, and $\frac{1-\sqrt{1-4x}}{2x}$ is their generating function. The facts we used without proof, that a power series can be differentiated and integrated term by term strictly inside its radius of convergence, are standard theorems of first-year analysis; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2007, Question 3

Pure • Fibonacci Identities

STEP III 2007, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q3

The key idea

Every identity here is proved the same way: take the term with the largest index and split it using the recurrence, $F_m = F_{m-1} + F_{m-2}$. Doing that once turns the expression in (ii) into one with all indices shifted down by one; doing it twice brings you back to the same shape with the indices shifted down by two, which is exactly what an induction needs. For the divisibility in (iv), notice that $F_{2n-1} + F_{2n} = F_{2n+1}$, so F_{2n-1}^2 and F_{2n}^2 leave the same remainder on division by F_{2n+1}.

Solution

Part (i)

Adding consecutive terms from $F_1 = F_2 = 1$:

$$F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21.$$

Part (ii)

Let $k \geq 1$, so that every term is defined. Replace the largest-index term F_{2k+3} by $F_{2k+2} + F_{2k+1}$:

$$\begin{aligned} F_{2k+3}F_{2k+1} - F_{2k+2}^2 &= (F_{2k+2} + F_{2k+1})F_{2k+1} - F_{2k+2}^2 \\ &= F_{2k+1}^2 + F_{2k+2}\left(F_{2k+1} - F_{2k+2}\right) \\ &= F_{2k+1}^2 - F_{2k+2}F_{2k}, \end{aligned}$$

since $F_{2k+2} - F_{2k+1} = F_{2k}$ by the recurrence. This is $-F_{2k+2}F_{2k} + F_{2k+1}^2$, as required.

Part (iii)

Write $P(n)$ for the statement $F_{2n+1}F_{2n-1} - F_{2n}^2 = 1$.

Base case. For $n = 1$: $F_3F_1 - F_2^2 = 2 \cdot 1 - 1^2 = 1$, so $P(1)$ holds.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$. By part (ii),

$$F_{2k+3}F_{2k+1} - F_{2k+2}^2 = F_{2k+1}^2 - F_{2k+2}F_{2k}.$$

Ask yourself Part (ii) only shifted the indices down by one. How do we get all the way back to $P(k)$?

Apply the same trick once more, now splitting the largest-index term $F_{2k+2} = F_{2k+1} + F_{2k}$:

$$\begin{aligned} F_{2k+1}^2 - F_{2k+2}F_{2k} &= F_{2k+1}^2 - (F_{2k+1} + F_{2k})F_{2k} \\ &= F_{2k+1}(F_{2k+1} - F_{2k}) - F_{2k}^2 \\ &= F_{2k+1}F_{2k-1} - F_{2k}^2, \end{aligned}$$

using $F_{2k+1} - F_{2k} = F_{2k-1}$ (valid since $2k+1 \geq 3$). By the hypothesis $P(k)$ this equals 1. Chaining the two displays,

$$F_{2k+3}F_{2k+1} - F_{2k+2}^2 = 1,$$

which is $P(k+1)$. By induction, $P(n)$ holds for all $n \geq 1$.

Deduction. Rearranging $P(n)$,

$$F_{2n}^2 + 1 = F_{2n+1}F_{2n-1},$$

and F_{2n-1} is an integer (every F_m is, by the recurrence), so $F_{2n}^2 + 1$ is divisible by F_{2n+1}.

Part (iv)

By the recurrence, $F_{2n-1} = F_{2n+1} - F_{2n}$. Squaring,

$$F_{2n-1}^2 = F_{2n+1}^2 - 2F_{2n+1}F_{2n} + F_{2n}^2,$$

so

$$F_{2n-1}^2 + 1 = F_{2n+1}(F_{2n+1} - 2F_{2n}) + (F_{2n}^2 + 1).$$

The first bracket is a multiple of F_{2n+1} (its cofactor is an integer), and $F_{2n}^2 + 1$ is a multiple of F_{2n+1} by part (iii). Hence $F_{2n-1}^2 + 1$ is divisible by F_{2n+1}.

In fact we can find the quotient. Using $F_{2n}^2 + 1 = F_{2n+1}F_{2n-1}$,

$$F_{2n-1}^2 + 1 = F_{2n+1}(F_{2n+1} - 2F_{2n} + F_{2n-1}) = F_{2n+1}(2F_{2n-1} - F_{2n}),$$

and for $n \geq 2$ the bracket is $F_{2n-1} - F_{2n-2} = F_{2n-3}$. So $F_{2n-1}^2 + 1 = F_{2n+1}F_{2n-3}$, a close cousin of the identity in (iii).

Checks

With $n = 3$: $F_6^2 + 1 = 65 = 13 \times 5 = F_7F_5$, and $F_5^2 + 1 = 26 = 13 \times 2 = F_7F_3$. With $n = 4$: $F_7^2 + 1 = 170 = 34 \times 5 = F_9F_5$. A computer confirmed the identities in (ii) and (iii), both divisibility statements and the quotient F_{2n-3} for every n up to 37.

Common slips

- Trying the induction using (ii) alone: it shifts the indices by only one, and the resulting expression $F_{2k+1}^2 - F_{2k+2}F_{2k}$ is not yet the hypothesis. One more use of the recurrence is needed.
- Forgetting the base case, or checking it with the wrong indices ($n = 1$ involves F_1, F_2, F_3).

Beyond STEP

Part (iii) is half of Cassini's identity, $F_{m+1}F_{m-1} - F_m^2 = (-1)^m$, which drops out instantly from determinants: the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^m$ has entries F_{m+1}, F_m, F_m, F_{m-1}, and its determinant is $(-1)^m$. Divisibility facts such as (iv) are the start of the arithmetic of linear recurrences, where congruences do most of the work. Davenport's *The Higher Arithmetic* is a good next step for the number theory, and Strang's *Introduction to Linear Algebra* for the matrix viewpoint.

WORKED SOLUTION

STEP III 2007, Question 4

Pure • Parametric Curves and Curvature

STEP III 2007, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q4

The key idea

Everything rests on a clean formula for $\dot{x}$. The awkward term $\ln\tan\frac{1}{2}t$ differentiates to $1/\sin t$, and then $\dot{x}$ collapses to $a\cos^2 t/\sin t$. Once we have $\dot{x}$ and $\dot{y}$, the gradient, the tangent, the curvature and the normal all follow by straightforward coordinate geometry. Writing the unit tangent as $(\cos t, \sin t)$ keeps the final part very short.

Solution

The gradient

Since $\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\tan\frac{1}{2}t = \dfrac{\frac{1}{2}\sec^2\frac{1}{2}t}{\tan\frac{1}{2}t} = \dfrac{1}{2\sin\frac{1}{2}t\cos\frac{1}{2}t} = \dfrac{1}{\sin t}$, we get

$$\dot{x} = a\left(-\sin t + \frac{1}{\sin t}\right) = a\,\frac{1-\sin^2 t}{\sin t} = \frac{a\cos^2 t}{\sin t},$$
$$\dot{y} = a\cos t.$$

Both are positive for $0 < t < \frac{1}{2}\pi$, so

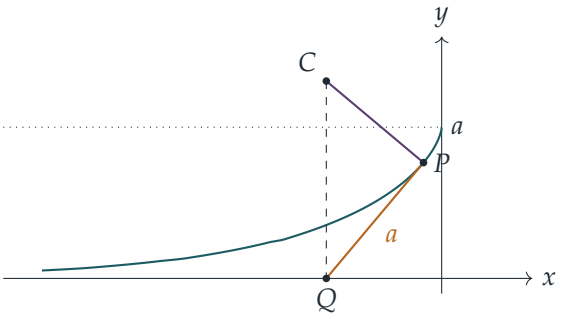
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dot{y}}{\dot{x}} = a\cos t\cdot\frac{\sin t}{a\cos^2 t} = \tan t,$$

as required.

The sketch

- As $t \to 0^+$, $\tan\frac{1}{2}t \to 0^+$, so $x \to -\infty$ while $y \to 0$. The x-axis is an asymptote.
- As $t \to \frac{1}{2}\pi^-$, $x \to a(0+\ln 1) = 0$ and $y \to a$. The curve ends at $(0, a)$, where the gradient $\tan t \to \infty$, so the tangent there is vertical.
- Since $\dot{x} > 0$, x increases with t, and so does the gradient $\tan t$. So the curve rises steadily and is concave upwards, with y between 0 and a.

This curve is the *tractrix*. The figure also shows the points P, Q and C from the later parts.



The tangent: $PQ = a$

The tangent at $P = (x, y)$ has gradient $\tan t$. It meets $y = 0$ after a horizontal run of $y/\tan t$ back from P, so

$$x_Q = x - \frac{y}{\tan t} = x - \frac{a \sin t \cos t}{\sin t} = x - a\cos t.$$

Hence $\overrightarrow{QP} = (a\cos t,\ a\sin t)$, and

$$PQ = \sqrt{a^2\cos^2 t + a^2 \sin^2 t} = a.$$

So the length of the tangent from the curve to the x-axis is always a. (For later use, $x_Q = a \ln \tan \frac{1}{2}t$.)

The radius of curvature

Write $\dot{x} = a\left(\dfrac{1}{\sin t} - \sin t\right)$ to differentiate it easily:

$$\ddot{x} = a\Big(-\frac{\cos t}{\sin^2 t} - \cos t\Big), \qquad \ddot{y} = -a\sin t.$$

Then

$$\begin{aligned}\dot{x}\ddot{y} - \dot{y}\ddot{x} &= \frac{a\cos^2 t}{\sin t}(-a\sin t) + a\cos t \cdot a\Big(\frac{\cos t}{\sin^2 t} + \cos t\Big) \\ &= a^2\Big(-\cos^2 t + \frac{\cos^2 t}{\sin^2 t} + \cos^2 t\Big) = a^2\cot^2 t,\end{aligned}$$

and

$$\dot{x}^2 + \dot{y}^2 = a^2\cos^2 t\Big(\frac{\cos^2 t}{\sin^2 t} + 1\Big) = \frac{a^2\cos^2 t}{\sin^2 t} = a^2\cot^2 t.$$

Since $\cot t > 0$ on our range, $(\dot{x}^2 + \dot{y}^2)^{3/2} = a^3\cot^3 t$, and therefore

$$\rho = \frac{a^3\cot^3 t}{a^2\cot^2 t} = a\cot t,$$

as required.

The centre of curvature: CQ is vertical

The tangent at P points along the unit vector $(\cos t, \sin t)$, so the normal points along $(-\sin t, \cos t)$ or its negative.

? **Ask yourself** Which of the two normal directions is "above the curve"?

The vector $(-\sin t, \cos t)$ has positive y-component, since $\cos t > 0$. Moving a short way along it from P takes us above the curve, because the curve is concave upwards (and in any case the other direction takes us downwards). So

$$\overrightarrow{OC} = \overrightarrow{OP} + \rho(-\sin t, \cos t) = \big(x - a\cot t \sin t,\ a\sin t + a\cot t\cos t\big).$$

The x-coordinate is

$$x_C = x - a\cos t = x_Q$$

by the tangent part. So C and Q have the same x-coordinate, and CQ is parallel to the y-axis, as required. (The y-coordinate simplifies to $a\sin t + \dfrac{a\cos^2 t}{\sin t} = \dfrac{a}{\sin t}$, so $C = \big(a\ln\tan\tfrac{1}{2}t,\ a\operatorname{cosec} t\big)$.)

Checks

- With $a = 1$ and $t = \frac{5\pi}{18}$ $(50°)$, $P \approx (-0.1202, 0.7660)$ and $Q \approx (-0.7630, 0)$, giving $PQ = 1.0000$. The formula gives $C \approx (-0.7630, 1.3054)$, directly above Q.
- A numerical check over 2000 random values of a and t, with $\dot{x}$, $\ddot{x}$ found by finite differences, agreed with $\mathrm{d}y/\mathrm{d}x = \tan t$, $PQ = a$, $\rho = a\cot t$ and $x_C = x_Q$ to within rounding error.
- As $t \to \frac{1}{2}\pi$, $\rho \to 0$: the curve bends ever more sharply as it approaches $(0, a)$ (its mirror image in the y-axis meets it there in a cusp). As $t \to 0$, $\rho \to \infty$, matching the curve flattening onto its asymptote.

Common slips

- Differentiating $\ln\tan\frac{1}{2}t$ and forgetting the factor $\frac{1}{2}$ from the chain rule. The answer must come out as $1/\sin t$.
- Leaving $\dot{x}$ as $a(-\sin t + \operatorname{cosec} t)$ without simplifying. The form $a\cos^2 t/\sin t$ is what makes every later part tidy.
- Using the normal direction $(\sin t, -\cos t)$, which points below the curve and puts C in the wrong place.

Beyond STEP

The tractrix is the path traced by a weight dragged on a string of length a by someone walking along the x-axis, and $PQ = a$ is exactly that property. The point C traces the *evolute* of the curve, the locus of its centres of curvature. Here $C = (a \ln \tan \frac{1}{2}t,\ a \operatorname{cosec} t)$ turns out to lie on a catenary. Rotating the tractrix about its asymptote gives the pseudosphere, a surface of constant negative curvature and a model of hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful geometric account of the pseudosphere.

WORKED SOLUTION

STEP III 2007, Question 5

Pure • Higher Derivatives and Induction

STEP III 2007, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q5

The key idea

Everything becomes simple once we write r and $\mathrm{d}\theta/\mathrm{d}x$ in terms of θ alone. From $\coth\theta = x$ we get $r = 1/\sinh\theta$, and then $\mathrm{d}r/\mathrm{d}x = \cosh\theta$ and $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$. Differentiating $\cosh n\theta / r^n$ then produces exactly the compound-angle formula for $\cosh(n+1)\theta$, which sets up an induction.

Solution

Setting up

Since $x > 1$, $\coth\theta = x$ has a unique solution with $\theta > 0$, so $\sinh\theta > 0$. Using $\coth^2\theta - 1 = \operatorname{cosech}^2\theta$,

$$r^2 = x^2 - 1 = \coth^2\theta - 1 = \frac{1}{\sinh^2\theta}, \qquad \text{so} \qquad r = \frac{1}{\sinh\theta},$$

taking the positive root because both r and $\sinh\theta$ are positive.

Two derivatives will do all the work. Differentiating $\coth\theta = x$ with respect to x,

$$-\operatorname{cosech}^2\theta \frac{\mathrm{d}\theta}{\mathrm{d}x} = 1 \quad \Longrightarrow \quad \frac{\mathrm{d}\theta}{\mathrm{d}x} = -\sinh^2\theta = -\frac{1}{r^2}.$$

Differentiating $r^2 = x^2 - 1$ gives $2r\dfrac{\mathrm{d}r}{\mathrm{d}x} = 2x$, so

$$\frac{\mathrm{d}r}{\mathrm{d}x} = \frac{x}{r} = \coth\theta\,\sinh\theta = \cosh\theta.$$

The first two derivatives

Since $y = \ln r^2 = 2\ln r$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}x} = \frac{2\cosh\theta}{r},$$

as required. For the second derivative it is cleanest to write $1/r = \sinh\theta$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2\cosh\theta\sinh\theta = \sinh 2\theta$. By the chain rule,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2\cosh 2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}x} = -\frac{2\cosh 2\theta}{r^2},$$

as required.

The third derivative

Now write $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -2\cosh 2\theta\sinh^2\theta$ and use the product rule, with $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$:

$$\begin{aligned}\frac{\mathrm{d}^3y}{\mathrm{d}x^3} &= -2\big(2\sinh 2\theta\sinh^2\theta + 2\cosh 2\theta\sinh\theta\cosh\theta\big)\frac{\mathrm{d}\theta}{\mathrm{d}x}\\ &= 4\sinh^3\theta\big(\cosh 2\theta\cosh\theta + \sinh 2\theta\sinh\theta\big)\\ &= 4\sinh^3\theta\cosh 3\theta,\end{aligned}$$

by the identity $\cosh(A+B) = \cosh A\cosh B + \sinh A\sinh B$. So

$$\frac{\mathrm{d}^3y}{\mathrm{d}x^3} = \frac{4\cosh 3\theta}{r^3}.$$

The general formula

The pattern $2\cosh\theta/r$, $-2\cosh 2\theta/r^2$, $4\cosh 3\theta/r^3$ suggests the coefficients $2\cdot 0!$, $-2\cdot 1!$, $2\cdot 2!$, so we conjecture

$$\frac{\mathrm{d}^ny}{\mathrm{d}x^n} = (-1)^{n-1}\,\frac{2\,(n-1)!\,\cosh n\theta}{r^n} \qquad (n \geq 1).$$

Ask yourself What is the general version of the calculation we just did for the third derivative?

It is the derivative of $\cosh n\theta/r^n = \cosh n\theta\,\sinh^n\theta$. By the product and chain rules, with $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\big(\cosh n\theta\,\sinh^n\theta\big) &= \big(n\sinh n\theta\,\sinh^n\theta + n\cosh n\theta\,\sinh^{n-1}\theta\cosh\theta\big)\big(-\sinh^2\theta\big)\\ &= -n\sinh^{n+1}\theta\,\big(\cosh n\theta\cosh\theta + \sinh n\theta\sinh\theta\big)\\ &= -n\sinh^{n+1}\theta\,\cosh(n+1)\theta,\end{aligned}$$

that is,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh n\theta}{r^n}\right) = -\frac{n\cosh(n+1)\theta}{r^{n+1}}.$$

Now the induction. The formula holds for $n = 1$ (it is the first derivative found above). Suppose it holds for some $n = k \geq 1$. Then, using the calculation above with $n = k$,

$$\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = (-1)^{k-1}\,2\,(k-1)!\cdot\left(-\frac{k\cosh(k+1)\theta}{r^{k+1}}\right) = (-1)^k\,\frac{2\,k!\,\cosh(k+1)\theta}{r^{k+1}},$$

which is the formula for $n = k+1$. By induction,

$$\boxed{\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = (-1)^{n-1}\,\frac{2\,(n-1)!\,\cosh n\theta}{r^n} \quad \text{for all } n \geq 1.}$$

(The cases $n = 2$ and $n = 3$ agree with what we found directly.)

A second route

Partial fractions give an independent derivation. Since $y = \ln(x-1) + \ln(x+1)$,

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = (-1)^{n-1}(n-1)!\,\left[(x-1)^{-n} + (x+1)^{-n}\right].$$

But $x \pm 1 = \dfrac{\cosh\theta \pm \sinh\theta}{\sinh\theta} = r\,\mathrm{e}^{\pm\theta}$, so the bracket is $r^{-n}\left(\mathrm{e}^{-n\theta} + \mathrm{e}^{n\theta}\right) = 2\cosh n\theta / r^n$, the same formula. This also explains where the substitution comes from: θ is half of $\ln\dfrac{x+1}{x-1}$.

Checks

- At $x = 2$: partial fractions give $\mathrm{d}^3y/\mathrm{d}x^3 = 2\left(1 + \frac{1}{27}\right) = \frac{56}{27} \approx 2.0741$, and $4\cosh 3\theta / r^3$ with $\theta = \tanh^{-1}\frac{1}{2}$, $r = \sqrt{3}$ gives the same value.
- We checked the first three derivatives against finite differences of $\ln(x^2-1)$ at 200 random points with $x > 1$, and the general formula against the partial-fraction form for $n \leq 14$ (agreement to about 10^{-14}).
- Signs: $\ln(x^2-1)$ is increasing and concave for $x > 1$, matching $y' > 0$ and $y'' < 0$.

Common slips

- Writing $\mathrm{d}\theta/\mathrm{d}x = +\sinh^2\theta$: the derivative of $\coth\theta$ is $-\operatorname{cosech}^2\theta$, and the minus sign is what makes the derivatives alternate.
- Getting the factorial wrong by one: the nth derivative carries $(n-1)!$, since the first derivative has coefficient $2 = 2\cdot 0!$.
- Checking a few cases and stopping: "with proof" needs the general inductive step.

Beyond STEP

The neat closed form for every derivative is a glimpse of complex analysis: $\ln(x^2-1)$ has singularities at $x = \pm 1$, and its higher derivatives are governed by the distances to them, here $x \mp 1 = r\mathrm{e}^{\mp\theta}$. In a first-year analysis course, formulae like this are what you need to bound the remainder in a Taylor series and find its radius of convergence. Needham's *Visual Complex Analysis* explains beautifully why singularities control power series.

WORKED SOLUTION

STEP III 2007, Question 6

Pure • Complex Numbers and Circles

STEP III 2007, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q6

The key idea

On a circle of radius a centred at the origin, conjugation is the same as inversion: $pp^* = a^2$, so $p^* = a^2/p$. That one fact turns every conjugate in the problem into an ordinary fraction, and the identity drops out. For perpendicularity, the test is that the ratio of the two chord vectors is purely imaginary. The identity then turns this test into a simple product condition. For the polygon, that product condition is a recurrence: once B_1 is chosen, every other B_k is forced, and the only question is whether the chain closes up when it gets back round to B_1.

Solution

The identity

Since P lies on the circle, $|p| = a$, so $pp^* = a^2$ and $p^* = a^2/p$; similarly $q^* = a^2/q$. Hence

$$p^* - q^* = \frac{a^2}{p} - \frac{a^2}{q} = \frac{a^2(q-p)}{pq} = -\frac{a^2(p-q)}{pq}.$$

This is nonzero because $p \neq q$ (and $p, q \neq 0$ since $a > 0$), so we may rearrange:

$$pq = -a^2 \frac{p-q}{p^* - q^*},$$

as required.

Perpendicular chords

The chord PQ is in the direction of the complex number $p - q$, and RS in the direction of $r - s$. Two nonzero complex numbers are perpendicular exactly when their ratio

$$w = \frac{p-q}{r-s}$$

is purely imaginary, because the argument of the ratio is the angle between them.

Ask yourself How do we turn "w is purely imaginary" into an equation we can use?

A nonzero w is purely imaginary if and only if $w^* = -w$, that is, $w/w^* = -1$. Using the identity for both chords (the second with r, s in place of p, q),

$$\frac{w}{w^*} = \frac{p-q}{p^*-q^*} \cdot \frac{r^*-s^*}{r-s} = \frac{-pq/a^2}{-rs/a^2} = \frac{pq}{rs}.$$

So $PQ \perp RS$ if and only if $pq/rs = -1$. Since $rs \neq 0$, that means

$$pq + rs = 0.$$

Every step here was reversible, so we have shown slightly more than was asked: for chords of this circle, $PQ \perp RS \iff pq + rs = 0$. We need the converse direction below.

The polygon problem

Let the points A_k and B_k be represented by α_k and β_k. By the previous part (and its converse), the n perpendicularity conditions are exactly

$$\beta_k\beta_{k+1} = -\alpha_k\alpha_{k+1} \quad (k = 1, \ldots, n-1), \qquad \beta_n\beta_1 = -\alpha_n\alpha_1,$$

provided consecutive Bs are distinct, so that each B_kB_{k+1} really is a chord.

Choose β_1. The first $n-1$ conditions then force $\beta_2, \beta_3, \ldots, \beta_n$ in turn, since $\beta_{k+1} = -\alpha_k\alpha_{k+1}/\beta_k$, and each β_{k+1} automatically has modulus a. Working out the first few:

$$\beta_2 = -\frac{\alpha_1\alpha_2}{\beta_1}, \qquad \beta_3 = -\frac{\alpha_2\alpha_3}{\beta_2} = \frac{\alpha_3\beta_1}{\alpha_1}, \qquad \beta_4 = -\frac{\alpha_3\alpha_4}{\beta_3} = -\frac{\alpha_1\alpha_4}{\beta_1},$$

and an easy induction gives

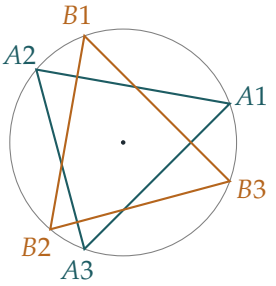
$$\beta_k = \frac{\alpha_k\beta_1}{\alpha_1} \ (k \text{ odd}), \qquad \beta_k = -\frac{\alpha_1\alpha_k}{\beta_1} \ (k \text{ even}).$$

(If it holds for k, then $\beta_{k+1} = -\alpha_k\alpha_{k+1}/\beta_k$ gives the other form for $k+1$.) Everything now depends on whether the last condition, $\beta_n\beta_1 = -\alpha_n\alpha_1$, holds.

$n = 3$. Here $\beta_3 = \alpha_3\beta_1/\alpha_1$, so the closing condition is

$$\frac{\alpha_3\beta_1^2}{\alpha_1} = -\alpha_3\alpha_1 \iff \beta_1^2 = -\alpha_1^2 \iff \beta_1 = \pm \mathrm{i}\,\alpha_1.$$

So there are only two choices of B_1. With $\beta_1 = \mathrm{i}\alpha_1$ the formulae give $\beta_2 = \mathrm{i}\alpha_2$ and $\beta_3 = \mathrm{i}\alpha_3$; with $\beta_1 = -\mathrm{i}\alpha_1$, every $\beta_k = -\mathrm{i}\alpha_k$. In each case the triangle $B_1B_2B_3$ is the triangle $A_1A_2A_3$ rotated through a right angle about the centre. Its vertices are distinct, so both choices genuinely work, and rotating a chord through 90° certainly makes it perpendicular to the original.



The figure shows the choice $\beta_1 = \mathrm{i}\alpha_1$: each side of the amber triangle is the corresponding teal side turned through $90°$.

$n = 4$. Now $\beta_4 = -\alpha_1\alpha_4/\beta_1$, so

$$\beta_4\beta_1 = -\alpha_1\alpha_4$$

holds automatically, whatever β_1 is. So B_1 can be chosen *anywhere* on the circle, giving infinitely many choices. The only exceptions are the finitely many positions in which two consecutive Bs would coincide. For instance, $\beta_1 = \beta_2$ needs $\beta_1^2 = -\alpha_1\alpha_2$, which rules out two points. Each of the four coincidences rules out at most two points, so at most eight positions are excluded.

$n > 4$. The same calculation works for every n, and only the parity of n matters:

- if n is odd, the closing condition is $\beta_1^2 = -\alpha_1^2$. There are exactly two choices, $\beta_1 = \pm\mathrm{i}\alpha_1$, and $B_1B_2\cdots B_n$ is the polygon $A_1A_2\cdots A_n$ rotated through $\pm 90°$ about the centre;
- if n is even, the closing condition holds automatically. B_1 may be any point of the circle apart from at most $2n$ positions that would make consecutive Bs coincide, so there are infinitely many choices.

Checks

- Try $p = a$, $q = \mathrm{i}a$: then $pq = \mathrm{i}a^2$, and $-a^2(a - \mathrm{i}a)/(a + \mathrm{i}a) = -a^2 \cdot (-\mathrm{i}) = \mathrm{i}a^2$, as it should be.
- Try the diameters $p = a$, $q = -a$ and $r = \mathrm{i}a$, $s = -\mathrm{i}a$, which are perpendicular: $pq + rs = -a^2 + a^2 = 0$.
- Numerically, we tested the identity and the criterion $PQ \perp RS \iff pq + rs = 0$ on 1000 random configurations. For $n = 3, \ldots, 9$ we ran the recurrence from random starting points. For odd n, scanning B_1 round the circle in $0.1°$ steps, the chain closed only near $\beta_1 = \pm\mathrm{i}\alpha_1$. For even n it closed every time, with all chords perpendicular to rounding error.

Common slips

- Proving only "perpendicular $\Rightarrow pq + rs = 0$". For the polygon we need the converse as well, to know that the βs we construct really do give perpendicular chords.
- Writing "perpendicular means w is imaginary" as $w + w^* = 0$ without noting $w \neq 0$. Here $w \neq 0$ is guaranteed because the points are distinct.
- For even n, saying "every B_1 works" without noticing that a few positions make two consecutive Bs coincide, so that the "chord" has no direction.

Beyond STEP

The map $z \mapsto a^2/z^*$ is *inversion* in the circle $|z| = a$, and the fact that points on the circle are fixed by it ($p = a^2/p^*$) is what powered this whole question. Inversion is one of the Möbius transformations, which are central to a first course in complex analysis and geometry. The polygon argument has the flavour of *monodromy*: follow a chain of forced choices round a loop and ask whether you return to where you started, and the answer depends on the parity of n. Needham's *Visual Complex Analysis* develops inversion and these geometric ideas beautifully.

WORKED SOLUTION

STEP III 2007, Question 7

Pure • Integration by Substitution

STEP III 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q7

The key idea

We are not allowed to say "arctan" or "π", so every fact has to come from changing variables inside the integrals. Each substitution is chosen so that the integrand $1/(1+u^2)$ (or $1/\sqrt{1-u^2}$) turns into itself or into the other one, with only the limits changing. Comparing limits then gives equations between values of t and s, and the symmetry $u \mapsto 1/u$ from part (i) is what turns those equations into multiples of p.

Solution

Part (i)

Take $x > 0$ and put $u = v^{-1}$, so $\mathrm{d}u = -v^{-2}\,\mathrm{d}v$. As u runs from 0 up to x, v runs from ∞ down to $1/x$. Also

$$\frac{1}{1+u^2} = \frac{1}{1+v^{-2}} = \frac{v^2}{v^2+1}.$$

Hence

$$t(x) = \int_{\infty}^{1/x} \frac{v^2}{1+v^2}\left(-\frac{1}{v^2}\right)\mathrm{d}v = \int_{1/x}^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v,$$

where the minus sign was used to swap the limits. This is the required result. (The improper integral converges because the integrand is at most $1/v^2$.)

Applying this with $1/x$ in place of x gives $t(1/x) = \int_x^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v$. Adding $t(x) = \int_0^x \frac{1}{1+v^2}\,\mathrm{d}v$, the two ranges join up to $[0,\infty)$:

$$t(x) + t(1/x) = \int_0^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v = \tfrac{1}{2}p \qquad (x > 0).$$

Putting $x = 1$ gives $2t(1) = \frac{1}{2}p$.

Part (ii)

For $u \geq 0$ we have $0 \leq y < 1$. Squaring, $y^2(1+u^2) = u^2$, so $u^2(1-y^2) = y^2$ and

$$u = \frac{y}{\sqrt{1-y^2}},$$

taking the positive root because u and y have the same sign. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{\sqrt{1-y^2} + \dfrac{y^2}{\sqrt{1-y^2}}}{1-y^2} = \frac{(1-y^2)+y^2}{(1-y^2)^{3/2}} = \frac{1}{\sqrt{(1-y^2)^3}},$$

as required.

Now substitute in $t(x)$. From $u^2 = y^2/(1-y^2)$ we get

$$1 + u^2 = \frac{1}{1-y^2},$$

so

$$\frac{1}{1+u^2}\,\frac{\mathrm{d}u}{\mathrm{d}y} = (1-y^2)\cdot\frac{1}{(1-y^2)^{3/2}} = \frac{1}{\sqrt{1-y^2}}.$$

The map $u \mapsto y$ is increasing (its inverse has positive derivative), with $u = 0 \mapsto y = 0$ and $u = x \mapsto y = x/\sqrt{1+x^2}$, which is less than 1. So

$$t(x) = \int_0^{x/\sqrt{1+x^2}} \frac{1}{\sqrt{1-y^2}}\,\mathrm{d}y = s\left(\frac{x}{\sqrt{1+x^2}}\right).$$

With $x = 1$ this reads $s\left(\frac{1}{\sqrt{2}}\right) = t(1) = \frac{1}{4}p$, using part (i).

Part (iii)

Write $a = \frac{1}{\sqrt{3}}$, so $z = \dfrac{u+a}{1-au}$. On $0 \leq u \leq a$ we have $au \leq \frac{1}{3}$, so the denominator stays positive and z is a smooth function of u there. Differentiating,

$$\frac{\mathrm{d}z}{\mathrm{d}u} = \frac{(1-au)+a(u+a)}{(1-au)^2} = \frac{1+a^2}{(1-au)^2}.$$

? **Ask yourself** Why should this substitution leave $1/(1+u^2)$ unchanged?

Compute $1+z^2$ over the common denominator. The cross terms $\mp 2au$ cancel:

$$1 + z^2 = \frac{(1-au)^2 + (u+a)^2}{(1-au)^2} = \frac{1 + a^2u^2 + u^2 + a^2}{(1-au)^2} = \frac{(1+a^2)(1+u^2)}{(1-au)^2}.$$

Dividing,

$$\frac{1}{1+z^2}\,\frac{\mathrm{d}z}{\mathrm{d}u} = \frac{1}{1+u^2},$$

so the integrand is unchanged and only the limits move. (Behind the scenes this is the tangent addition formula, but we don't need it.) When $u = 0$, $z = a = \frac{1}{\sqrt{3}}$. When $u = a$,

$$z = \frac{2/\sqrt{3}}{1 - \frac{1}{3}} = \frac{2}{\sqrt{3}} \cdot \frac{3}{2} = \sqrt{3}.$$

Therefore

$$t\left(\tfrac{1}{\sqrt{3}}\right) = \int_0^{1/\sqrt{3}} \frac{\mathrm{d}u}{1+u^2} = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2}.$$

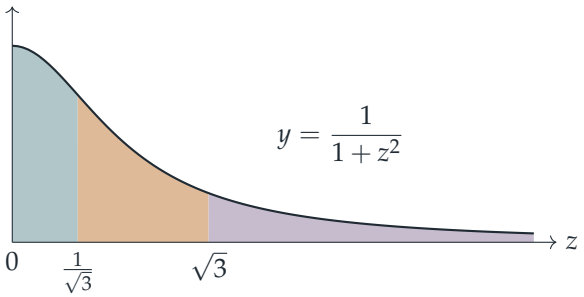
For the last step, split the range $[0, \infty)$ at $\frac{1}{\sqrt{3}}$ and $\sqrt{3}$:

$$\tfrac{1}{2}p = \int_0^{1/\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} + \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} + \int_{\sqrt{3}}^{\infty} \frac{\mathrm{d}z}{1+z^2}.$$

The first piece is $t(\frac{1}{\sqrt{3}})$ by definition, and the second is $t(\frac{1}{\sqrt{3}})$ by what we have just shown. By part (i) with $x = \frac{1}{\sqrt{3}}$, the third piece is $\int_{1/x}^{\infty} \frac{\mathrm{d}z}{1+z^2} = t(x) = t(\frac{1}{\sqrt{3}})$ as well. Hence

$$3\,t\left(\tfrac{1}{\sqrt{3}}\right) = \tfrac{1}{2}p.$$

The picture shows the three pieces: the area under $1/(1+z^2)$ from 0 to ∞ splits into three regions of equal area.



Checks

Numerically (Simpson's rule), $t(\frac{1}{\sqrt{3}}) \approx 0.5235988$ and $\int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} \approx 0.5235988$, while $p \approx 3.1415927$, so $3t(\frac{1}{\sqrt{3}}) \approx 1.5707963 \approx \frac{1}{2}p$. Likewise $s(\frac{1}{\sqrt{2}}) \approx t(1) \approx 0.7853982 \approx \frac{1}{4}p$. Of course p is really π, and these say $\arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$, $\arcsin\frac{1}{\sqrt{2}} = \arctan 1 = \frac{\pi}{4}$; you may use that to check, though not in the answer.

Common slips

- Forgetting that $u = 1/v$ reverses the direction of integration; the minus sign in $\mathrm{d}u$ is what puts the limits back in order.

- In part (iii), using the substitution and then stopping: the integral from $\frac{1}{\sqrt{3}}$ to $\sqrt{3}$ is only one of three equal pieces, and part (i) is needed for the piece beyond $\sqrt{3}$.
- Not checking that $1 - au \neq 0$ on the range; here $au \leq \frac{1}{3}$, so the substitution is valid.

Beyond STEP

This question is a glimpse of how analysts define the trigonometric functions rigorously: one defines $\arctan x$ as $\int_0^x \frac{\mathrm{d}u}{1+u^2}$, defines π as twice its limit at infinity, and derives every property from substitutions like these, without any geometry. The substitution in part (iii) is the addition formula for tan in disguise, and the same method proves $t(x) + t(y) = t\left(\frac{x+y}{1-xy}\right)$ for $xy < 1$. Abbott's *Understanding Analysis* shows how integrals and power series are used to put such definitions on a firm footing.

WORKED SOLUTION

STEP III 2007, Question 8

Pure • Riccati Equations

STEP III 2007, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q8

The key idea

The nonlinear equation $(*)$ is a Riccati equation: a y^2 term spoils linearity. Writing y as a multiple of u'/u turns $y' + y^2$ into u''/u, and that gets rid of the square, leaving a *linear* second-order equation for u. The first half of (i) hands us two solutions of exactly that linear equation, so everything else is fitting constants. For (ii) we need two solutions of a new linear equation, and the clue is that $1 - x$ is $1 + x$ with x replaced by $-x$.

Solution

Part (i)

Finding a **and** b**.** For $u = x$ we have $u' = 1$, $u'' = 0$, so we need

$$\mathrm{a}(x) + x\,\mathrm{b}(x) = 0.$$

For $u = \mathrm{e}^{-x}$ we have $u' = -\mathrm{e}^{-x}$, $u'' = \mathrm{e}^{-x}$; dividing by $\mathrm{e}^{-x} \neq 0$ gives

$$1 - \mathrm{a}(x) + \mathrm{b}(x) = 0.$$

The first equation gives $\mathrm{b} = -\mathrm{a}/x$ (for $x \neq 0$), and then the second gives $\mathrm{a}\left(1 + \frac{1}{x}\right) = 1$. So

$$\mathrm{a}(x) = \frac{x}{1+x}, \qquad \mathrm{b}(x) = -\frac{1}{1+x}.$$

(These also satisfy both conditions at $x = 0$.) The equation is linear and homogeneous, and x and e^{-x} are not multiples of each other, so the general solution is

$$u = Ax + B\mathrm{e}^{-x},$$

with A and B arbitrary constants.

The substitution. Put $y = \dfrac{u'}{3u}$, where $u \neq 0$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{u''}{3u} - \frac{(u')^2}{3u^2} = \frac{u''}{3u} - 3\left(\frac{u'}{3u}\right)^2 = \frac{u''}{3u} - 3y^2.$$

So $y' + 3y^2 = \dfrac{u''}{3u}$: the square cancels exactly, which is why the factor 3 was put in the substitution. Equation $(*)$ becomes

$$\frac{u''}{3u} + \frac{x}{1+x} \cdot \frac{u'}{3u} = \frac{1}{3(1+x)},$$

and multiplying by $3u$ gives

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \frac{x}{1+x}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{1}{1+x}u = 0,$$

as required. This is the equation from the start of the part, with the a and b we found.

The particular solution. So $u = Ax + B\mathrm{e}^{-x}$, and

$$y = \frac{u'}{3u} = \frac{A - B\mathrm{e}^{-x}}{3(Ax + B\mathrm{e}^{-x})}.$$

Ask yourself There are two constants but only one condition. Is that a problem?

No. Multiplying u by a non-zero constant doesn't change u'/u, so only the ratio $A : B$ matters, and a first-order equation should have just one free constant. The condition $y = 0$ at $x = 0$ gives $A - B = 0$. Then $B = A \neq 0$ (otherwise $u \equiv 0$), and cancelling A,

$$y = \frac{1 - \mathrm{e}^{-x}}{3(x + \mathrm{e}^{-x})}.$$

The denominator never vanishes, since $\mathrm{e}^{-x} \geq 1 - x$ gives $x + \mathrm{e}^{-x} \geq 1$. That completes part (i).

Part (ii)

Now the y^2 term has coefficient 1, so we use $y = \dfrac{u'}{u}$. Exactly as before,

$$y' = \frac{u''}{u} - \left(\frac{u'}{u}\right)^2 = \frac{u''}{u} - y^2,$$

so $y' + y^2 = u''/u$. Multiplying the equation by u gives the linear equation

$$u'' + \frac{x}{1-x}u' - \frac{1}{1-x}u = 0.$$

This is the equation of part (i) with x replaced by $-x$. Indeed, if $v(x) = w(-x)$ where w solves the part (i) equation, then $v'(x) = -w'(-x)$ and $v''(x) = w''(-x)$, and the part (i) equation at the point $-x$ reads

$$v''(x) + \frac{-x}{1-x}\big(-v'(x)\big) - \frac{1}{1-x}v(x) = 0,$$

which is the new equation. So from x and e^{-x} we get the solutions $-x$ (or just x) and e^{x}. Checking directly: for $u = x$, $0 + \frac{x}{1-x} - \frac{x}{1-x} = 0$; for $u = \mathrm{e}^{x}$, $\mathrm{e}^{x}\left(1 + \frac{x}{1-x} - \frac{1}{1-x}\right) = \mathrm{e}^{x}(1-1) = 0$. Hence

$$u = Ax + B\mathrm{e}^{x}, \qquad y = \frac{u'}{u} = \frac{A + B\mathrm{e}^{x}}{Ax + B\mathrm{e}^{x}}.$$

At $x = 0$ this is $(A+B)/B$, which must equal 2, so $A = B$ (and $B \neq 0$). Therefore

$$y = \frac{1 + \mathrm{e}^{x}}{x + \mathrm{e}^{x}}.$$

This is valid on the interval containing 0 on which $x + \mathrm{e}^{x} > 0$ and the equation is defined, namely $x_0 < x < 1$, where $x_0 \approx -0.567$ is the root of $x + \mathrm{e}^{x} = 0$.

Checks

- At $x = 0$ the part (ii) answer gives $2/1 = 2$, and differentiating it gives $y'(0) = \frac{1\cdot 1 - 2\cdot 2}{1} = -3$, which matches the equation: $1 - 4 - 0 = -3$.
- As $x \to \infty$ the part (i) answer tends to 0 like $\frac{1}{3x}$. Then $y' \approx -\frac{1}{3x^2}$ and $3y^2 \approx \frac{1}{3x^2}$ cancel, and $\frac{x}{1+x}y \approx \frac{1}{3(1+x)}$, consistent with $(*)$.
- We solved the part (ii) equation numerically (Runge–Kutta from $y(0) = 2$) and got $y(0.9) = 1.0297654\ldots$, agreeing with the formula to 10 decimal places.

Common slips

- Keeping the factor $\frac{1}{3}$ in part (ii): the multiplier must match the coefficient of y^2, so here it is $y = u'/u$.
- Trying to fix both A and B from one condition. Only their ratio affects y.
- Taking e^{-x} again in part (ii) without checking it: the sign change in $1 - x$ turns it into e^{x}.

Beyond STEP

An equation $y' + P(x)y^2 + Q(x)y = R(x)$ is a Riccati equation, and the substitution $y = u'/(Pu)$ always turns it into a linear second-order equation. Conversely, the ratio u'/u of any second-order linear equation satisfies a Riccati equation, which is why these nonlinear equations keep appearing in physics, from quantum mechanics to control theory. Boyce and DiPrima, *Elementary Differential Equations*, covers this and the theory of linear second-order equations (why two independent solutions give the general solution).

WORKED SOLUTION

STEP III 2007, Question 9

Mechanics • Simple Harmonic Motion

STEP III 2007, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q9

The key idea

The spring pulls or pushes the two beads equally and oppositely along the same line, and the hoop's reactions are radial, so the total angular momentum about the centre stays zero: the beads move as mirror images. That reduces everything to one coordinate, and conservation of energy then gives simple harmonic motion in θ. The catch is that θ is the *acute* half-angle: if the beads get far enough apart to pass through opposite ends of a diameter, the formula for the energy changes form, and the motion must be pieced together from two arcs of SHM.

Solution

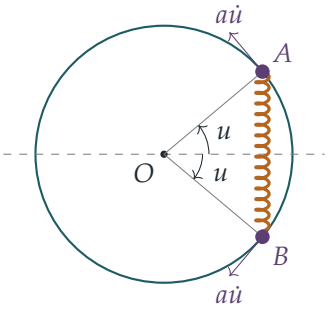
Setting up

Model the beads as particles on a fixed, smooth hoop, and the spring as light. The forces on the beads are gravity and the vertical part of the hoop's reaction (which cancel), the horizontal part of the reaction (along the radius, since the hoop is smooth) and the spring forces. The spring force on A and the spring force on B are equal and opposite and act along the same line AB, so their total moment about the vertical axis through O is zero; the radial forces have no moment about it at all. Hence the total angular momentum about that axis is conserved, and it is zero because the beads start at rest.

Measure angles round the hoop from a fixed radius, so that A and B are at angles ϕ_A and ϕ_B. Zero angular momentum means $ma^2(\dot{\phi}_A + \dot{\phi}_B) = 0$, so $\phi_A + \phi_B$ is constant. Choose the reference radius to bisect the initial angle AOB; then for all time

$$\phi_A = u, \qquad \phi_B = -u,$$

for a single coordinate u, with $u = \beta$ at the start. Each bead has speed $a\dot{u}$, so the kinetic energy is $2 \cdot \frac{1}{2}ma^2\dot{u}^2 = ma^2\dot{u}^2$.



The angle AOB is $2u$ while $0 < u \leq \frac{\pi}{2}$, but if u exceeds $\frac{\pi}{2}$ the beads have passed opposite ends of a diameter and the angle AOB is $2\pi - 2u$. So

$$\theta = u \quad (0 < u \leq \tfrac{\pi}{2}), \qquad \theta = \pi - u \quad (\tfrac{\pi}{2} \leq u < \pi).$$

Conservation of energy, with the beads released from rest at $u = \beta$, gives

$$ma^2\dot{u}^2 + mk^2a^2(\theta - \alpha)^2 = mk^2a^2(\beta - \alpha)^2, \qquad \text{so} \qquad \dot{u}^2 = k^2\left[(\alpha - \beta)^2 - (\theta - \alpha)^2\right].$$

"In compression" means the spring is shorter than its natural length. The energy is least at $\theta = \alpha$, so that is the natural length, and the length $2a\sin\theta$ increases with θ on $(0, \frac{\pi}{2})$; hence $0 < \beta < \alpha$.

The motion while $u \leq \frac{\pi}{2}$

Here $\theta = u$, and differentiating the energy equation with respect to t and dividing by $2\dot{u}$ (valid whenever the beads are moving, and then by continuity) gives

$$\ddot{\theta} = -k^2(\theta - \alpha).$$

This is SHM about $\theta = \alpha$ with angular frequency k. Starting from rest at $\theta = \beta$,

$$\theta - \alpha = -(\alpha - \beta)\cos kt,$$

so the beads move apart and, if nothing interrupts them, reach $\theta = 2\alpha - \beta$ at $t = \frac{\pi}{k}$. Everything depends on whether $2\alpha - \beta$ is below $\frac{\pi}{2}$, that is, on how β compares with $2\alpha - \frac{\pi}{2}$. This critical value lies in $(0, \alpha)$ precisely because $\frac{\pi}{4} < \alpha < \frac{\pi}{2}$, so all three cases below can occur.

Case 1: $2\alpha - \frac{\pi}{2} < \beta < \alpha$

Now $2\alpha - \beta < \frac{\pi}{2}$, so the beads never reach the ends of a diameter, $\theta = u$ throughout, and the motion is pure SHM between $\theta = \beta$ and $\theta = 2\alpha - \beta$. The period is

$$T = \frac{2\pi}{k}.$$

Case 2: $0 < \beta < 2\alpha - \frac{\pi}{2}$

Now θ reaches $\frac{\pi}{2}$ before the SHM turns round, at the time t_1 with

$$\tfrac{\pi}{2} - \alpha = -(\alpha - \beta)\cos kt_1, \qquad \text{so} \qquad \cos kt_1 = -\lambda, \quad \text{where } \lambda = \frac{\frac{\pi}{2} - \alpha}{\alpha - \beta}.$$

Since $\alpha < \frac{\pi}{2}$ and $\beta < 2\alpha - \frac{\pi}{2}$, we have $0 < \lambda < 1$. Hence

$$kt_1 = \pi - \arccos\lambda = \tfrac{\pi}{2} + \arcsin\lambda.$$

At this moment the energy equation gives $\dot{u}^2 = k^2[(\alpha - \beta)^2 - (\frac{\pi}{2} - \alpha)^2] > 0$, so the beads are still moving apart and carry on past the ends of the diameter.

Ask yourself What happens after the beads pass the ends of a diameter?

For $u \geq \frac{\pi}{2}$ we have $\theta = \pi - u$, so the energy equation reads

$$\dot{u}^2 = k^2\left[(\alpha - \beta)^2 - (u - (\pi - \alpha))^2\right],$$

which is SHM in u about $u = \pi - \alpha$, again with angular frequency k. This is the mirror image, in $u = \frac{\pi}{2}$, of the motion before the diameter. So the beads take another time t_1 to come to rest at $u = \pi - \beta$ (where $\theta = \beta$ again, with the beads now on the far side of O), then retrace the whole path back to $u = \beta$. The beads return to their starting positions after four such stages, so the period of the motion is

$$T = 4t_1 = \frac{4}{k}\left(\frac{\pi}{2} + \arcsin\frac{\frac{\pi}{2} - \alpha}{\alpha - \beta}\right) = \frac{2}{k}\left(\pi + 2\arcsin\frac{\pi - 2\alpha}{2(\alpha - \beta)}\right).$$

(The angle θ itself, which forgets which side of O the beads are on, repeats after $2t_1$.)

Case 3: $\beta = 2\alpha - \frac{\pi}{2}$

Now θ reaches $2\alpha - \beta = \frac{\pi}{2}$ at $t = \frac{\pi}{k}$ exactly when the SHM brings the beads to rest. They are then at opposite ends of a diameter, so the spring lies along AOB and its force on each bead is along the radius: it has no component along the hoop, and nothing else acts along the hoop. The beads are in equilibrium and at rest, so they stay there. So

$$\beta = 2\alpha - \frac{\pi}{2}$$

is the value for which no oscillations occur. This matches the two periods: as β approaches this value from below, $\lambda \to 1$ and the Case 2 period tends to $\frac{4\pi}{k}$, twice the Case 1 period, because the beads linger longer and longer near the diameter.

Checks

- Take $k = 1$, $\alpha = \frac{3\pi}{8}$, $\beta = \frac{\pi}{8}$, which is in Case 2 since $2\alpha - \frac{\pi}{2} = \frac{\pi}{4}$. Then $\lambda = \frac{\pi/8}{\pi/4} = \frac{1}{2}$, $\arcsin\lambda = \frac{\pi}{6}$, and $T = 2(\pi + \frac{\pi}{3}) = \frac{8\pi}{3} \approx 8.378$.

- We stepped the two beads' equations of motion numerically (with the energy as a function of both bead positions, and no symmetry assumed). With $k = 1$, $\alpha = 1.2$, $\beta = 0.5$ the beads returned to rest at their starting points after 8.5162, against the formula's 8.51622; with $\beta = 1.0$ (Case 1) the period was $6.2832 \approx 2\pi$.
- Dimensions: k has units of s^{-1} (the energy is ma^2k^2 times an angle squared), so $\frac{2\pi}{k}$ is a time.

Common slips

- Writing the kinetic energy as $\frac{1}{2}ma^2\dot{\theta}^2$: both beads move, so it is $ma^2\dot{\theta}^2$. Missing the factor of 2 gives the wrong angular frequency $\sqrt{2}\,k$.
- Assuming the beads move symmetrically without a reason: it follows from conservation of angular momentum about O, which needs the spring force on each bead to act along the same line AB.
- Forgetting that θ is acute, and so answering $\frac{2\pi}{k}$ in every case.

Beyond STEP

Reducing a two-particle system to one coordinate using a conserved quantity is the start of Lagrangian mechanics, where symmetries (here, rotating the whole configuration about O) give conservation laws, an idea made precise by Noether's theorem. The motion in Case 2 is an oscillation in a "double-well" potential, whose period depends on the amplitude and jumps where the energy equals the height of the central barrier. For more on conservation laws and oscillations, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2007, Question 10

Mechanics • Projectiles on Inclined Planes

STEP III 2007, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q10

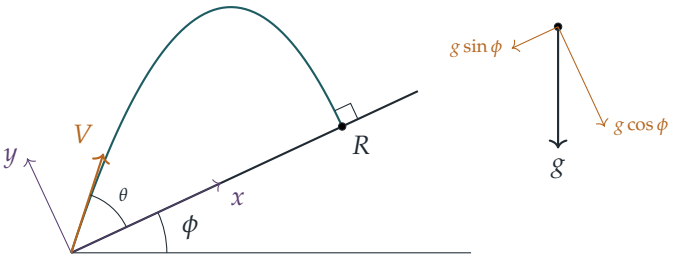
The key idea

Work in axes along and perpendicular to the slope, so that gravity becomes two constant accelerations, $g\sin\phi$ down the slope and $g\cos\phi$ into it. The striking condition is that the particle retraces its path. A smooth plane can only push perpendicular to itself, so it can reverse the velocity only if the particle arrives travelling straight into the plane. That single fact gives both results; the rest is algebra and an AM–GM minimisation.

Solution

The equations of motion

We model the particle as a point mass moving under gravity alone, with no air resistance. The weight mg acts vertically downwards. Resolving along the slope and perpendicular to it:



the acceleration is $-g\sin\phi$ in the x-direction and $-g\cos\phi$ in the y-direction, both constant. Integrating twice with the given initial velocity and starting point $(0,0)$,

$$x = Vt\cos\theta - \tfrac{1}{2}gt^2\sin\phi, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2\cos\phi.$$

The condition for retracing the path

The particle first returns to the plane when $y = 0$ with $t > 0$, that is at

$$t = T = \frac{2V \sin\theta}{g \cos\phi}.$$

Ask yourself What does "returns along the same path" tell us about the impact?

After the bounce, the particle again moves under constant gravity. If it is to run back along the same parabola to the origin, its velocity just after the bounce must be exactly the reverse of its velocity just before. (Think of running the film backwards.) We take the plane to be smooth, so the impulse it gives the particle is perpendicular to the plane: the component $\dot{x}$ along the slope is unchanged by the bounce. That component can only be reversed if it is zero. So the particle must strike the plane at right angles, with $\dot{x} = 0$ at $t = T$. (We also need the bounce to reverse $\dot{y}$ without loss of speed, so the impact must be perfectly elastic. The question takes this for granted.)

Now $\dot{x} = V\cos\theta - gt\sin\phi$, so $\dot{x}(T) = 0$ gives

$$V\cos\theta = gT\sin\phi = \frac{2V\sin\theta\sin\phi}{\cos\phi}.$$

Dividing by $V\cos\theta$, which is positive since $0 < \theta < \pi/2$,

$$2\tan\phi\tan\theta = 1,$$

as required.

The range

From $\dot{x}(T) = 0$ we have the neater form $T = \dfrac{V\cos\theta}{g\sin\phi}$. Then

$$\begin{aligned} R = x(T) &= V\cos\theta \cdot \frac{V\cos\theta}{g\sin\phi} - \frac{1}{2}g\sin\phi \cdot \frac{V^2\cos^2\theta}{g^2\sin^2\phi} \\ &= \frac{V^2\cos^2\theta}{g\sin\phi} - \frac{V^2\cos^2\theta}{2g\sin\phi} = \frac{V^2\cos^2\theta}{2g\sin\phi}, \end{aligned}$$

which is the required result. (Equivalently: x is greatest when $\dot{x} = 0$, so the landing point is the highest point up the slope that the motion would reach.)

The identity for $2V^2/(gR)$

From the range,

$$\frac{2V^2}{gR} = \frac{4\sin\phi}{\cos^2\theta} = 4\sin\phi\,(1 + \tan^2\theta).$$

Using $\tan\theta = \dfrac{1}{2\tan\phi}$, so $\tan^2\theta = \dfrac{\cos^2\phi}{4\sin^2\phi}$,

$$\frac{2V^2}{gR} = 4\sin\phi + \frac{\cos^2\phi}{\sin\phi} = 4\sin\phi + \frac{1-\sin^2\phi}{\sin\phi} = 3\sin\phi + \operatorname{cosec}\phi.$$

The largest range

For a given V, R is largest when $3\sin\phi + \operatorname{cosec}\phi$ is smallest. Write $s = \sin\phi$; since $0 < \phi < \pi/2$, $0 < s < 1$. By the AM–GM inequality for the positive numbers $3s$ and $1/s$,

$$3s + \frac{1}{s} \geq 2\sqrt{3s \cdot \frac{1}{s}} = 2\sqrt{3},$$

with equality exactly when $3s = 1/s$, that is $s = 1/\sqrt{3}$. This value does lie in $(0,1)$, so the bound is attained by a genuine slope. Hence

$$\frac{2V^2}{gR} \geq 2\sqrt{3} \quad\Longrightarrow\quad R \leq \frac{V^2}{\sqrt{3}\,g},$$

with equality when $\sin\phi = 1/\sqrt{3}$. The largest possible range is therefore $V^2/(\sqrt{3}\,g)$.

Ask yourself Is the condition $\phi + \theta < \pi/2$ satisfied at the maximum?

Yes, and in fact always. With $\tan\theta\tan\phi = \frac{1}{2}$ and both angles acute,

$$\tan(\phi+\theta) = \frac{\tan\phi + \tan\theta}{1 - \tan\phi\tan\theta} = 2(\tan\phi + \tan\theta) > 0,$$

and since $0 < \phi + \theta < \pi$, this forces $\phi + \theta < \pi/2$. At the maximum, $\sin\phi = 1/\sqrt{3}$ gives $\tan\phi = 1/\sqrt{2}$, so $\tan\theta = 1/\sqrt{2}$ too: the best case has $\theta = \phi \approx 35.3°$.

Checks

- Units: V^2/g is a length, as a range should be.
- Limits: as $\phi \to 0$, $\operatorname{cosec}\phi \to \infty$ and $R \to 0$. On a nearly flat plane, the only way to land at right angles is to fire almost straight up, which gives almost no range. As $\phi \to \pi/2$ the plane is a vertical wall, $\theta \to 0$, and $2V^2/(gR) \to 4$, so $R \to V^2/(2g)$: the height reached by a particle thrown straight up the wall. Both make sense.
- Numerically, with $g = 9.8$, $V = 10$ and $\phi = 0.7$, stepping the motion forward lands at $x \approx 5.8561$, matching the formula. The landing velocity has $\dot{x} \approx 0$, and reversing it brings the particle back to the origin. A grid search over ϕ gives maximum $R \approx 0.05891$ for $V = 1$, equal to $1/(9.8\sqrt{3})$.

Common slips

- Resolving gravity the wrong way round. The component *along* the slope is $g\sin\phi$ and the component *perpendicular* to it is $g\cos\phi$. Check with $\phi = 0$.
- Saying the particle must land "horizontally" or "at its highest point". The condition is that it hits the *plane* at right angles, $\dot{x} = 0$ in the tilted axes.
- Forgetting to check that $\sin\phi = 1/\sqrt{3}$ is attainable. AM–GM gives a bound, and it is only the maximum if equality can actually occur.

Beyond STEP

Retracing the path rests on the *time-reversibility* of Newtonian mechanics. For forces that depend only on position, reversing every velocity sends the system back along its history. Choosing axes adapted to the geometry, rather than the default horizontal and vertical, is a habit you will use constantly in university mechanics. Kleppner and Kolenkow's *An Introduction to Mechanics* develops both ideas.

WORKED SOLUTION

STEP III 2007, Question 11

Mechanics • Rolling Over an Edge

STEP III 2007, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q11

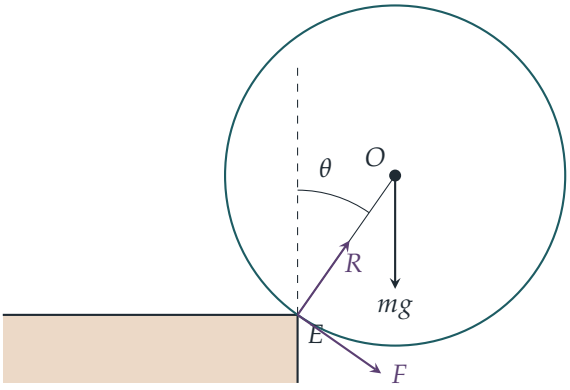
The key idea

Once the rim reaches the edge, the edge acts as a pivot: the point of the rim touching it is at rest, so the centre moves on a circle of radius a about the edge. We split the contact force at the edge into a component R along the radius (the normal reaction) and a component F along the rim (friction), and compare them with the motion of the centre of mass. The whole question turns on where the mass sits. In (i) all of it is at the centre, so the wheel has no moment of inertia about its centre. In (ii) the mass is out on the rim, and that is what produces a friction force.

Solution

Model and notation. Rim, spokes and hoop are light and rigid; the hub and the two particles are particles. Let E be the point of the edge in contact with the rim, O the centre, and θ the angle EO makes with the upward vertical, so $\theta = 0$ at the instant the wheel arrives at the edge. While the rim stays in contact with E without slipping, the material point of the rim at E is at rest. So the rigid body rotates about E with angular speed $\dot{\theta}$, and O moves on a circle of radius a about E with speed $a\dot{\theta}$.

The contact force at E has a component R along EO (away from E) and a component F perpendicular to EO, taken positive in the direction of increasing θ. Because E is at rest, the contact force does no work, so energy is conserved.



Here R acts along EO, and F is drawn in its positive sense.

For the centre of mass G at distance a from E, its acceleration has components $a\ddot{\theta}$ along the direction of motion and $a\dot{\theta}^2$ towards E. Resolving the external forces on the whole body (gravity mg and the contact force):

$$\text{along the motion:} \quad ma\ddot{\theta} = mg\sin\theta + F, \tag{1}$$

$$\text{towards } E: \quad ma\dot{\theta}^2 = mg\cos\theta - R. \tag{2}$$

In both parts the centre of mass is at O (the hub; the midpoint of the two particles), so (1) and (2) hold in both.

Part (i)

On the table the wheel moves at constant speed u, so at $\theta = 0$ we have $\dot{\theta} = u/a$. The only kinetic energy is that of the hub, $\frac{1}{2}ma^2\dot{\theta}^2$, and the hub has dropped $a(1-\cos\theta)$, so

$$a^2\dot{\theta}^2 = u^2 + 2ga(1-\cos\theta).$$

Differentiating with respect to time and cancelling $2a\dot{\theta}$ (which is non-zero, since $\dot{\theta} \geq u/a > 0$):

$$a\ddot{\theta} = g\sin\theta.$$

Substituting in (1) gives $F = 0$: **the frictional force is zero**, as required.

There is also a direct reason. The rim and spokes are light, so the total moment about O of the forces on them must vanish. Those forces are the contact force at E and the force from the hub at O, which has no moment about O. The moment of the contact force about O is Fa, so $F = 0$.

From (2), the normal reaction is

$$R = mg\cos\theta - \frac{m}{a}\left(u^2 + 2ga(1-\cos\theta)\right) = mg(3\cos\theta - 2) - \frac{mu^2}{a}.$$

At $\theta = 0$ this is $m(g - u^2/a) > 0$, because $u^2 < ag$, so the wheel does not leave the edge at once. R then decreases as θ increases, and the rim loses contact when $R = 0$, that is

when

$$\cos\theta = \frac{2ag + u^2}{3ag},$$

which is less than 1 because $u^2 < ag$. The hub has then fallen

$$a(1-\cos\theta) = a \cdot \frac{3ag - 2ag - u^2}{3ag} = \frac{ag - u^2}{3g},$$

as required. Since $F = 0$ throughout, the no-slip assumption needs no help from friction, and the analysis holds right up to this point.

Part (ii)

? **Ask yourself** Where are the two particles while the hoop pivots about E?

At $\theta = 0$ the diameter through the particles is vertical, so the lower particle is at the contact point E. The hoop then rotates rigidly about the material point at E, which is that particle, so the particle stays at E. The other particle stays diametrically opposite, at distance $2a$ from E along EO.

So the moment of inertia about E is $\frac{m}{2}(2a)^2 = 2ma^2$. On the table the hoop rolls at constant speed u (nothing does work), so $\dot\theta = u/a$ at $\theta = 0$. The moving particle is at height $2a\cos\theta$ above E, and energy conservation gives

$$\tfrac{1}{2}(2ma^2)\dot\theta^2 + \tfrac{m}{2}g(2a\cos\theta) = \tfrac{1}{2}(2ma^2)\frac{u^2}{a^2} + \tfrac{m}{2}g(2a), \qquad \text{so} \qquad a^2\dot\theta^2 = u^2 + ga(1-\cos\theta).$$

Differentiating and cancelling $2a\dot\theta \neq 0$ gives $a\ddot\theta = \frac{1}{2}g\sin\theta$. Then (1) gives

$$F = ma\ddot\theta - mg\sin\theta = -\tfrac{1}{2}mg\sin\theta.$$

So **the frictional force is non-zero** for every $\theta > 0$. (Its magnitude is $\frac{1}{2}mg\sin\theta$, acting back against the direction of motion of O.) Unlike in (i), the rim now carries mass, so the hoop needs a torque about O to make its rotation keep pace with the motion of O, and only friction can supply it.

From (2),

$$R = mg\cos\theta - \frac{m}{a}\left(u^2 + ga(1-\cos\theta)\right) = mg(2\cos\theta - 1) - \frac{mu^2}{a}.$$

At $\theta = 0$, $R = m(g - u^2/a) > 0$ and $F = 0$, so the motion over the edge does begin without slipping. If the hoop kept contact without slipping, R would fall to zero at

$$\cos\theta_1 = \frac{ag + u^2}{2ag} < 1,$$

using $u^2 < ag$. So $\theta_1 > 0$, and there $|F| = \frac{1}{2}mg\sin\theta_1 > 0$.

Ask yourself Why does this force a slip before contact is lost?

With coefficient of friction μ, no slipping requires $|F| \le \mu R$. As $\theta \to \theta_1$ from below, $R \to 0^+$ but $|F| \to \frac{1}{2}mg\sin\theta_1 > 0$, so $|F|/R \to \infty$. Whatever the (finite) value of μ, there is therefore an angle $\theta < \theta_1$, while R is still positive, at which $|F|$ would have to exceed μR. So the hoop **slips before it loses contact** with the table.

Checks

- In (i), if $u^2 \to ag$ the drop tends to 0: the hub is then on the point of flying off at $\theta = 0$, where $g = u^2/a$. If $u \to 0$ the drop is $a/3$, the familiar result for a particle sliding off a smooth sphere starting from rest at the top.
- Stepping the equations numerically with $a = 1\,\text{m}$, $u = 2\,\text{m s}^{-1}$ and $g = 9.81\,\text{m s}^{-2}$ gives a drop of $0.1974\,\text{m}$ in (i), matching $(ag - u^2)/(3g)$. In (ii) we found $\cos\theta_1 = 0.7039$, and $|F|/R$ exceeded 10^4 just before $R = 0$.
- Computing the contact force in (ii) directly from the two particles' accelerations ($m\ddot{\mathbf{r}}_G$ minus gravity) reproduces both F and R.

Common slips

- Using rolling kinetic energy $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ about O in (ii) but forgetting that, once pivoting, one particle is fixed at E. It's cleaner to work about E.
- Resolving horizontally and vertically at the edge. At a corner the "normal" direction is along the radius EO, not vertical.
- Concluding "it slips" just because $F \neq 0$. The argument needs $R \to 0$ while $|F|$ stays away from zero.

Beyond STEP

The contrast between the two parts is about the distribution of mass: the same centre-of-mass motion needs different contact forces depending on the moment of inertia, because the angular momentum equation and the linear momentum equation must both be satisfied. In a first mechanics course this becomes rigid-body dynamics, where a body's motion is governed by $\mathbf{F} = m\ddot{\mathbf{r}}_G$ together with the torque equation about G, and contact constraints (rolling, slipping, losing contact) are checked against the inequalities $R \ge 0$ and $|F| \le \mu R$. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both equations thoroughly.

WORKED SOLUTION

STEP III 2007, Question 12

Probability and Statistics • Random Sums

STEP III 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q12

The key idea

Y is a sum with a *random* number of terms, so we can't use the rules for sums of a fixed number of variables straight away. The trick is to condition on N: once we know $N = k$, Y is an ordinary sum of k independent variables, whose mean and variance we know. Then we average over the distribution of N, which is why the first part asks for $\mathrm{E}(N)$ and $\mathrm{E}(N^2)$.

Solution

The distribution of N

We take the number to be chosen uniformly at random, so $\mathrm{P}(N = k) = \frac{1}{2n-1}$ for $k = 1, 2, \ldots, 2n-1$. Using $\sum_{k=1}^{m} k = \frac{1}{2}m(m+1)$ and $\sum_{k=1}^{m} k^2 = \frac{1}{6}m(m+1)(2m+1)$ with $m = 2n-1$:

$$\mathrm{E}(N) = \frac{1}{2n-1} \cdot \frac{(2n-1)(2n)}{2} = n,$$
$$\mathrm{E}(N^2) = \frac{1}{2n-1} \cdot \frac{(2n-1)(2n)(4n-1)}{6} = \frac{n(4n-1)}{3}.$$

(The mean n is also clear by symmetry: n is the middle of $1, \ldots, 2n-1$.) For later use,

$$\mathrm{Var}(N) = \frac{n(4n-1)}{3} - n^2 = \frac{n^2 - n}{3} = \frac{n(n-1)}{3}.$$

Conditioning on N

Fix k and suppose $N = k$. Because the X_i are independent of N, conditioning on $N = k$ doesn't change their distributions, so given $N = k$ we have $Y = X_1 + \cdots + X_k$, a sum of k independent variables each with mean μ and variance σ^2. Hence

$$\mathrm{E}(Y \mid N = k) = k\mu, \qquad \mathrm{Var}(Y \mid N = k) = k\sigma^2,$$

the second using independence of the X_i (variances add). So

$$\mathrm{E}(Y^2 \mid N=k) = \mathrm{Var}(Y \mid N=k) + \big(\mathrm{E}(Y \mid N=k)\big)^2 = k\sigma^2 + k^2\mu^2.$$

Ask yourself How do we get from a conditional expectation back to an ordinary one?

By the law of total expectation (the partition theorem): $\mathrm{E}(g(Y,N)) = \sum_k \mathrm{P}(N=k)\,\mathrm{E}(g(Y,N) \mid N=k)$. Inside each term N is just the number k.

The mean of Y

$$\mathrm{E}(Y) = \sum_k \mathrm{P}(N=k)\, k\mu = \mu\,\mathrm{E}(N) = n\mu,$$

as required.

The covariance

Given $N=k$, $YN = kY$, so $\mathrm{E}(YN \mid N=k) = k\cdot k\mu = k^2\mu$. Therefore $\mathrm{E}(YN) = \mu\,\mathrm{E}(N^2)$ and

$$\begin{aligned}\mathrm{Cov}(Y,N) &= \mathrm{E}(YN) - \mathrm{E}(Y)\,\mathrm{E}(N) = \mu\,\mathrm{E}(N^2) - n\mu\cdot n \\ &= \mu\,\mathrm{Var}(N) = \tfrac{1}{3}n(n-1)\mu,\end{aligned}$$

which is the required result.

The variance of Y

Averaging $\mathrm{E}(Y^2 \mid N=k) = k\sigma^2 + k^2\mu^2$ over N:

$$\mathrm{E}(Y^2) = \sigma^2\,\mathrm{E}(N) + \mu^2\,\mathrm{E}(N^2) = n\sigma^2 + \frac{n(4n-1)}{3}\mu^2.$$

Hence

$$\mathrm{Var}(Y) = n\sigma^2 + \frac{n(4n-1)}{3}\mu^2 - n^2\mu^2 = n\sigma^2 + \frac{n(n-1)}{3}\mu^2.$$

The two terms have a natural meaning: $n\sigma^2$ is the average spread of Y for a fixed number of trials, and $\frac{1}{3}n(n-1)\mu^2 = \mu^2\,\mathrm{Var}(N)$ is the extra spread caused by not knowing how many trials there will be.

Checks

- If $n=1$ then $N=1$ always, and the formulas reduce to $\mathrm{E}(Y)=\mu$, $\mathrm{Cov}(Y,N)=0$ and $\mathrm{Var}(Y)=\sigma^2$, as they should for a single X_1.
- If $\mu=0$ then the extra term vanishes: the number of terms no longer affects the mean, so it can't add spread or correlate with Y.

- Take $n = 2$ and each X_i equal to 0 or 1 with probability $\frac{1}{2}$. Then $N \in \{1, 2, 3\}$ and $Y \mid N \sim \mathrm{B}(N, \frac{1}{2})$; the formula gives $\mathrm{Var}(Y) = 2 \cdot \frac{1}{4} + \frac{2}{3} \cdot \frac{1}{4} = \frac{2}{3}$. We confirmed this, and all the results, by exact enumeration of the joint distribution of (N, Y) for $n = 1, \ldots, 6$ and several distributions of X_i.

Common slips

- Writing $\mathrm{Var}(Y) = \mathrm{E}(N)\sigma^2$ only, forgetting that N itself varies. The second term is the whole point.
- Treating Y and N as independent and getting $\mathrm{Cov}(Y, N) = 0$. More trials means a bigger sum when $\mu > 0$, so they are positively correlated.

Beyond STEP

Y is a *random sum* (a compound distribution), and the variance result is a case of the law of total variance, $\mathrm{Var}(Y) = \mathrm{E}(\mathrm{Var}(Y \mid N)) + \mathrm{Var}(\mathrm{E}(Y \mid N))$, here $\sigma^2\mathrm{E}(N) + \mu^2\mathrm{Var}(N)$. Random sums model insurance claims and branching processes, and are handled elegantly with probability generating functions: $G_Y(s) = G_N(G_X(s))$. Grimmett and Welsh, *Probability: An Introduction*, treats random sums and branching processes this way.

WORKED SOLUTION

STEP III 2007, Question 13

Probability and Statistics • Random Walk Expectation

STEP III 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

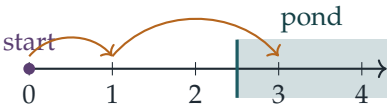
step.maths.org/questions/07-s3-q13

The key idea

The half-metre is there to remove any doubt about "landing on the edge": starting $(n - \frac{1}{2})$ m away, the frog is in the pond exactly when the total distance it has jumped is *at least* n. So everything is about the first time a running total of 1s and 2s reaches n. For small n we can simply list the possibilities; for the general formula, three values of u_n give three equations for the three constants.

Solution

Let S_j be the total distance (in metres) covered in the first j jumps, and let T be the number of jumps needed to reach the pond. Starting $(n - \frac{1}{2})$ m away, the frog is in the pond after j jumps if and only if $S_j > n - \frac{1}{2}$, that is $S_j \geq n$, since S_j is a whole number. So $p_n(j) = \mathrm{P}(T = j)$, where T is the first j with $S_j \geq n$.



The picture shows $n = 3$ (the edge is 2.5 m away) and the sequence "short, long", which reaches the pond in two jumps.

Part (i)

With $n = 2$ the edge is 1.5 m away. A long first jump (probability q) takes the frog straight in, so $p_2(1) = q$. Otherwise the first jump is short (probability p), the frog is 0.5 m from the edge, and *any* second jump takes it in. Jumps are independent, so

$$p_2(2) = p \times 1 = p,$$

as required. (Also $p_2(j) = 0$ for $j \geq 3$, and $p + q = 1$ confirms the distribution is complete.)

Part (ii)

From $\frac{1}{2}$ m away every jump lands in the pond, so $u_1 = 1$.

Using part (i),

$$u_2 = 1 \cdot p_2(1) + 2 \cdot p_2(2) = q + 2p = 2 - q.$$

For $n = 3$ we need $S_j \geq 3$.

- One jump is never enough, since $S_1 \leq 2$: $p_3(1) = 0$.
- Two jumps are not enough only if both are short ($S_2 = 2$); every other pair gives $S_2 \geq 3$. So $p_3(2) = 1 - p^2$.
- After two short jumps the frog is 0.5 m away and the third jump certainly lands in the pond: $p_3(3) = p^2$.

Therefore

$$\begin{aligned} u_3 &= 2(1 - p^2) + 3p^2 = 2 + p^2 \\ &= 2 + (1 - q)^2 = 3 - 2q + q^2, \end{aligned}$$

as required.

Ask yourself Is there a quicker way to see these values, and does it extend to all n?

Yes: condition on the first jump. After it the frog is in the same situation with n reduced by 1 or by 2, so for $n \geq 2$

$$u_n = 1 + p\,u_{n-1} + q\,u_{n-2}, \qquad u_0 = 0,$$

where $u_0 = 0$ covers a long first jump from 1.5 m. This gives $u_2 = 1 + p$ and $u_3 = 1 + p(1 + p) + q = 2 + p^2$ again. It is also why the form in part (iii) is right: the auxiliary equation $\lambda^2 - p\lambda - q = 0$ factorises as $(\lambda - 1)(\lambda + q) = 0$, giving the $(-q)^{n-1}$ term. The repeated root 1 (from the "+1") explains the $B + Cn$.

Part (iii)

Substituting $n = 1, 2, 3$ into $u_n = A(-q)^{n-1} + B + Cn$ and using part (ii):

$$A + B + C = 1, \tag{1}$$

$$-qA + B + 2C = 2 - q, \tag{2}$$

$$q^2A + B + 3C = 3 - 2q + q^2. \tag{3}$$

Subtracting to eliminate B:

$$(2) - (1): \quad -(1 + q)A + C = 1 - q, \tag{4}$$

$$(3) - (2): \quad q(1 + q)A + C = 1 - q + q^2. \tag{5}$$

Now $q \times (4) + (5)$ eliminates A:

$$(1+q)C = q - q^2 + 1 - q + q^2 = 1, \qquad \text{so} \qquad C = \frac{1}{1+q},$$

as required. Then (4) gives

$$(1+q)A = \frac{1}{1+q} - (1-q) = \frac{1-(1-q^2)}{1+q} = \frac{q^2}{1+q}, \qquad \text{so} \qquad A = \frac{q^2}{(1+q)^2},$$

and (1) gives

$$B = 1 - \frac{q^2}{(1+q)^2} - \frac{1}{1+q} = \frac{(1+q)^2 - q^2 - (1+q)}{(1+q)^2} = \frac{q}{(1+q)^2}.$$

So

$$u_n = \frac{q^2(-q)^{n-1} + q + (1+q)n}{(1+q)^2}.$$

Large n. Since $0 \le q \le 1$, the terms $A(-q)^{n-1}$ and B are bounded (each is at most $\frac{1}{4}$ in size), while Cn grows without limit. Hence for large n

$$u_n \approx Cn = \frac{n}{1+q} = \frac{n}{p+2q},$$

using $1 = p + q$, so $1 + q = p + 2q$.

Why this is expected. The mean length of one jump is $1 \cdot p + 2 \cdot q = p + 2q$ metres. The frog has to cover about n metres, and by the law of large numbers the total length of N jumps is close to $N(p+2q)$ when N is large. So the number of jumps needed is close to $n/(p+2q)$. The approximation is a good one because the half-metre start offset and the overshoot at the end (the frog lands at most 1.5 m past the edge) are bounded. They change the count by a bounded amount, which is negligible compared with n. That bounded correction is exactly the $A(-q)^{n-1} + B$ part.

Checks

- $n = 0$ in the formula gives $\left(q^2 \cdot (-1/q) + q\right)/(1+q)^2 = 0 = u_0$, as it should, and the formula satisfies the recurrence above.
- $q = 0$ (all jumps short): $u_n = n$, correct since the frog needs exactly n unit jumps.
- Exact enumeration of all jump sequences (with fractions) matched the formula for $n = 1, \ldots, 12$ and several values of p. A simulation with $q = 0.7$, $n = 200$ gave a mean of 117.90 jumps, against 117.89 from the formula and $n/(p+2q) \approx 117.65$.

Common slips

- Treating "lands in the pond" as $S_j = n$ exactly. Because of the half-metre the frog can overshoot, and every sequence with $S_j \geq n$ counts.
- Writing $p_3(2) = 2pq + q^2$ and forgetting the case where the long jump comes first. In fact $pq + qp + q^2 = 1 - p^2$ covers both orders.

Beyond STEP

The recurrence for u_n is a renewal equation. The result $u_n \sim n/\mu$, where μ is the mean step, is the elementary renewal theorem. An exact version follows from Wald's identity $\mathrm{E}(S_T) = \mu\,\mathrm{E}(T)$: here S_T is either n or $n+1$, so $(1+q)u_n = n + \mathrm{P}(\text{overshoot})$. Grimmett and Welsh, *Probability: An Introduction*, covers random walks and generating functions in this spirit.

WORKED SOLUTION

STEP III 2007, Question 14

Probability and Statistics • Geometrical Probability

STEP III 2007, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q14

The key idea

The smallest centred circle (or square) enclosing all the holes is fixed by the *furthest* hole, so we want the distribution of a maximum, and $P(\max \le r)$ is just the nth power of the single-throw probability. On the disc, the squared distance R^2 of one hole from O turns out to be uniform on $[0,1]$, which makes everything clean. On the square, the right "distance" is $\max(|x|,|y|)$, and it behaves exactly like the radius on the disc. That coincidence also settles part (iii) almost for free.

Solution

Part (i)

Let the n holes be at distances $R_1,\dots,R_n$ from O. These are independent, since the throws are. For a single throw and $0 \le r \le 1$, the hole lies within distance r of O exactly when it lands in the disc of radius r, so

$$P(R_i \le r) = \frac{\pi r^2}{\pi \cdot 1^2} = r^2.$$

Put $U_i = R_i^2$. Then $P(U_i \le u) = P(R_i \le \sqrt{u}) = u$ for $0 \le u \le 1$, so each U_i is *uniformly distributed on* $[0,1]$, and the U_i are independent.

The smallest circle centred at O enclosing every hole has radius $M = \max_i R_i$ and area $\pi M^2 = \pi \max_i U_i$. Write $V = \max_i U_i$. By independence,

$$P(V \le v) = P(U_1 \le v)\cdots P(U_n \le v) = v^n \qquad (0 \le v \le 1),$$

so V has density nv^{n-1} and

$$E(V) = \int_0^1 v \cdot nv^{n-1}\,\mathrm{d}v = \frac{n}{n+1}.$$

Hence the expected area of the smallest circle enclosing all n holes is

$$\boxed{\frac{n\pi}{n+1}}.$$

For the circle enclosing the $n-1$ holes nearest to O, the radius is the *second-largest* distance, so the area is πW where W is the second-largest of $U_1, \ldots, U_n$.

Ask yourself How do we get the distribution of the second-largest value?

We have $W \le w$ exactly when at most one of the U_i exceeds w. Either none does (probability w^n) or exactly one does: choose which one in n ways, giving probability $n(1-w)w^{n-1}$. So

$$P(W \le w) = w^n + n(1-w)w^{n-1} = nw^{n-1} - (n-1)w^n.$$

Rather than differentiating, we can use $E(W) = \int_0^1 P(W > w)\,\mathrm{d}w$, which holds for any random variable taking values in $[0,1]$:

$$E(W) = \int_0^1 \left(1 - nw^{n-1} + (n-1)w^n\right)\mathrm{d}w = 1 - 1 + \frac{n-1}{n+1} = \frac{n-1}{n+1}.$$

So the expected area of the circle enclosing the $n-1$ nearest holes is

$$\boxed{\frac{(n-1)\pi}{n+1}}.$$

(This fits a pleasant pattern: n independent uniform points split $[0,1]$ into $n+1$ gaps of equal expected length $\frac{1}{n+1}$, so the kth smallest has mean $\frac{k}{n+1}$.)

Part (ii)

Take axes through Q parallel to the sides of the board, so the board is $-1 \le x, y \le 1$. We take the enclosing squares to have sides parallel to the board's. A square centred at Q with half-side s is $\{|x| \le s,\ |y| \le s\}$, so it contains the hole (x,y) exactly when $\max(|x|,|y|) \le s$.

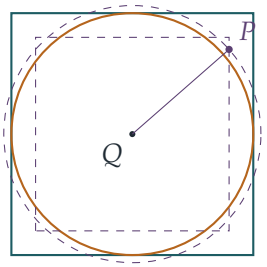
Let $S_i = \max(|x_i|, |y_i|)$ for the ith hole. For $0 \le s \le 1$,

$$P(S_i \le s) = \frac{(2s)^2}{2^2} = s^2,$$

the same as $P(R_i \le r) = r^2$ in part (i). The smallest enclosing square has half-side $\max_i S_i$ and area $(2\max_i S_i)^2 = 4\max_i S_i^2$. Exactly as in part (i), the S_i^2 are independent and uniform on $[0,1]$, so their maximum has expectation $\frac{n}{n+1}$. The expected area of the smallest enclosing square is

$$\boxed{\frac{4n}{n+1}}.$$

Part (iii)



Now the board is the square, and we want the smallest circle centred at Q enclosing every hole. A hole $P = (x, y)$ is at distance $\sqrt{x^2 + y^2}$ from Q, and

$$\sqrt{x^2 + y^2} \;\geq\; \max(|x|, |y|) = S,$$

with equality only when P lies on an axis. (In the figure, the dashed circle through P is never smaller than the inscribed circle of the dashed square through P.) So if D is the largest distance of a hole from Q, then $D \geq \max_i S_i$, and the area of the enclosing circle satisfies

$$\pi D^2 \;\geq\; \pi \max_i S_i^2.$$

From part (ii), $\max_i S_i^2$ has exactly the same distribution as $\max_i R_i^2$ on the circular board, so

$$E(\pi D^2) \;\geq\; \pi\, E\big(\max_i S_i^2\big) = \frac{n\pi}{n+1},$$

which is the circular-board answer from part (i). The inequality is strict: with positive probability no hole lies on an axis, and then $D > \max_i S_i$. So the expected area of the circle on the square board is **larger**.

Intuitively: the square board contains the circular one (the disc is its inscribed circle), and the darts spread out into the corners, so the furthest hole tends to be further away.

Checks

- For $n = 1$ the formulae give $\frac{\pi}{2}$ and 2: the mean of a uniform R^2 is $\frac{1}{2}$, so that is right. As $n \to \infty$ the areas tend to π and 4, the whole boards, as they should.
- A Monte Carlo simulation with 2×10^5 trials for $n = 2, 3, 5$ matched $\frac{n\pi}{n+1}$, $\frac{(n-1)\pi}{n+1}$ and $\frac{4n}{n+1}$ to about three significant figures; for example, $n = 3$ gave 2.3561 against $\frac{3\pi}{4} \approx 2.3562$.
- For part (iii), a quadrature of the exact distribution gave, for $n = 3$, an expected area of about 3.257 on the square board against 2.356 on the disc: larger, as claimed.

Common slips

- Working with $E(M)$ and then squaring: the expected area is $\pi E(M^2)$, not $\pi\big(E(M)\big)^2$.
- Taking the distance from the centre to be uniform on $[0,1]$. Area-proportional hitting makes R^2, not R, uniform; the outer rings are bigger.
- For the second-largest, forgetting the factor n for which hole is the one outside the circle.

Beyond STEP

The maximum and second-largest of a sample are *order statistics*. For n independent uniform variables on $[0,1]$, the kth smallest has a Beta distribution with mean $\frac{k}{n+1}$. Part (iii) is an example of *stochastic dominance* (coupling): if one random variable is always at least another, so is its expectation, and the comparison needs no detailed calculation. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both treat order statistics and transformations of random variables.

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